

P24 — Multi-Seed Hardening Pass (canonical 5 seeds {1,2,3,5,42})

Purpose. Promote the P24 *thin arms* from single-seed (42 only) to the canonical 5-seed protocol, so their numbers can carry error bars in the thesis instead of being labelled "indicative". Nothing here invents new science — it is the validation pass flagged in `Thesis_Final_Structure_and_Remaining_Work.md` §2.

Non-destructive by design. This notebook and its two helper scripts (`p24/seed_report_5.py` , `p24/edge_budget_multiseed.py`) are *new*. No original P24 file is edited or deleted. New seeds land as *new sibling* checkpoint dirs (`full_nc-accel_only_s1` , ...), exactly as `train_save.py` already names them; the seed-42 dirs and the original `Run_P24_S0_Compression_Colab.ipynb` are untouched.

What gets computed

1. The 4 thin arms at the 4 missing seeds {1,2,3,5}: `accel_only` , `gyro_only` , `aw40` , `aw30` .
2. `full` at seeds {1,2,3,5} — needed *only* to make the RED/GS edge-budget table 5-seed.
3. `no_gz` & `tiny` topped up to seeds {3,5} (they were 3-seed) so every headline P24 number is 5-seed.
4. The two aggregators → `results/p24_seed_report_5.json` and `results/p24_edge_budget_multiseed.json` , with paired Wilcoxon tests and GenPerByte + CIs.

Resumable. `train_save.py` skips any fold/seed whose checkpoint already exists, so a Colab disconnect is harmless — just re-run. Seed 42 for every arm already exists and will be skipped instantly. Estimated GPU time for the *missing* work: ~6–8 h, resumable.

1. Mount & paths

Same layout as `Run_P24_S0_Compression_Colab.ipynb` . The frozen ConvNet probs for **all 5** seeds already exist under P21 `results/sweep/base_s{1,2,3,5,42}` — this notebook only regenerates the *GNN* side.

```
In [ ]: from google.colab import drive
drive.mount('/content/drive')
import os

PROJECT_PATH = "/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P2
0_GraphRevival_DG/P24_S0_Compression"
DATA_PATH    = "/content/drive/MyDrive/HAR-Datasets/DAGHAR/standardize
d_view"
P21_SWEEP    = "/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P2
0_GraphRevival_DG/P21_PearsonGraph/results/sweep"
os.environ["DATA_PATH"] = DATA_PATH
assert os.path.isdir(PROJECT_PATH) and os.path.isdir(DATA_PATH)
assert os.path.isdir(P21_SWEEP), "fix P21_SWEEP (per-seed frozen ConvN
et base_s* live here)"
for s in (1, 2, 3, 5, 42):
    d = os.path.join(P21_SWEEP, f"base_s{s}")
    assert os.path.isdir(d), f"missing {d} - the 5-seed fusion needs e
very base_s*"
os.chdir(PROJECT_PATH); print("cwd:", os.getcwd())
```

Mounted at /content/drive

cwd: /content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphRevi
val_DG/P24_S0_Compression

2. Setup (deps + data check)

```
In [ ]: import setup_colab
setup_colab.install_deps(); assert setup_colab.verify(), "fix DATA_PATH"
# scipy is used by the aggregators for the paired Wilcoxon tests (sign
# -test fallback otherwise)
import importlib
try:
    import scipy # noqa
except ModuleNotFoundError:
    import subprocess, sys; subprocess.run([sys.executable, '-m', 'pip',
    'install', '-q', 'scipy'])
```

Installing PyTorch Geometric ...
Done.

Verifying imports ...
OK torch 2.11.0+cu128 CUDA=True
GPU Tesla T4
OK torch_geometric 2.8.0

Checking data at: /content/drive/MyDrive/HAR-Datasets/DAGHAR/standardi
zed_view

OK KuHar	3 CSVs
OK MotionSense	3 CSVs
OK RealWorld-Thigh	3 CSVs
OK RealWorld-Waist	3 CSVs
OK UCI	3 CSVs
OK WISDM	3 CSVs

3. Smoke test (~2 min)

One short arm/seed at 2 epochs to confirm the pipeline, then delete the smoke checkpoint so the real run retrains it.

```
In [ ]: python -m p24.train_save --data_path "$DATA_PATH" --targets KuHar --e
pochs 2 --log_every 1 --nc accel_only --seed 1
import torch
ck = torch.load("results/checkpoints/full_nc-acccl_only_s1/ckpt_KuHar.
pt", map_location="cpu", weights_only=False)
print("smoke OK: accel_only params =", ck["n_params"])
import shutil, os
shutil.rmtree("results/checkpoints/full_nc-acccl_only_s1"); print("smo
ke checkpoint removed")
```

```
[P24] train+save | arm=full_nc-acccl_only_s1 {'cnn_embed_dim': 64, 'gn
n_hidden': 64, 'cnn_channels': [32, 64]} | nc=acccl_only aw=- | seed=1
| targets=['KuHar'] | out=/content/drive/MyDrive/Thesis/S0_Experiment
s/DAGHAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/fu
ll_nc-acccl_only_s1
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (0, 1):100%, (1, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-acccl_only_s1:KuHar] trainable params: 51,339
[full_nc-acccl_only_s1:KuHar] epoch 1 | act_loss 0.7852 | dom_lo
ss 1.4593 | lambda 0.987 | tr_acc 67.1% | val_loss 0.8140 | val_acc
70.3% *
[full_nc-acccl_only_s1:KuHar] epoch 2 | act_loss 0.6487 | dom_lo
ss 1.4442 | lambda 1.000 | tr_acc 73.4% | val_loss 0.7145 | val_acc
70.7% *
[full_nc-acccl_only_s1:KuHar] src-val 70.66% | TARGET-TEST 59.03%
| saved ckpt + npz
smoke OK: accel_only params = 51339
smoke checkpoint removed
```

4. Thin arms × 4 missing seeds (the core of the ask)

Each cell = one arm across seeds {1,2,3,5} (seed 42 already exists, skipped automatically). Cells are independent and resumable — run them in any order, re-run freely after a disconnect. Per arm ≈ 4 seeds × 6 LODO folds of a small SAGE+DANN model.

Tip for parallel wall-clock: these arms don't collide, so you can open this notebook on a second Colab runtime and run different cells there to finish faster. The skip-logic makes it safe.

4a. accel_only (3-node graph — accelerometer axes only)

```
In [ ]: for s in (1, 2, 3, 5):  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant full  
        --nc accel_only --seed {s} --epochs 100 --log_every 20
```

```

[P24] train+save | arm=full_nc-accel_only_s1 {'cnn_embed_dim': 64, 'gn
n_hidden': 64, 'cnn_channels': [32, 64]} | nc=accel_only aw=- | seed=1
| targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Wais
t', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiment
s/DAGHAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/fu
ll_nc-accel_only_s1
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (0, 1):100%, (1, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-accel_only_s1:KuHar] trainable params: 51,339
[full_nc-accel_only_s1:KuHar] epoch 1 | act_loss 0.7642 | dom_lo
ss 1.4208 | lambda 0.050 | tr_acc 68.4% | val_loss 0.8625 | val_acc
64.5% *
[full_nc-accel_only_s1:KuHar] epoch 2 | act_loss 0.6338 | dom_lo
ss 1.3606 | lambda 0.100 | tr_acc 73.8% | val_loss 0.7450 | val_acc
74.2% *
[full_nc-accel_only_s1:KuHar] epoch 12 | act_loss 0.5338 | dom_lo
ss 1.3914 | lambda 0.537 | tr_acc 77.8% | val_loss 0.6351 | val_acc
75.7% *
[full_nc-accel_only_s1:KuHar] epoch 20 | act_loss 0.5205 | dom_lo
ss 1.4060 | lambda 0.762 | tr_acc 78.2% | val_loss 0.6429 | val_acc
75.0%
[full_nc-accel_only_s1:KuHar] epoch 21 | act_loss 0.5198 | dom_lo
ss 1.4082 | lambda 0.782 | tr_acc 78.4% | val_loss 0.6447 | val_acc
76.1% *
[full_nc-accel_only_s1:KuHar] epoch 34 | act_loss 0.5066 | dom_lo
ss 1.4172 | lambda 0.935 | tr_acc 78.9% | val_loss 0.6219 | val_acc
76.3% *
[full_nc-accel_only_s1:KuHar] epoch 40 | act_loss 0.5027 | dom_lo
ss 1.4240 | lambda 0.964 | tr_acc 79.1% | val_loss 0.6880 | val_acc
74.6%
[full_nc-accel_only_s1:KuHar] early stop at epoch 54
[full_nc-accel_only_s1:KuHar] src-val 76.26% | TARGET-TEST 52.08%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC

```

```

I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 33230 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
[full_nc-accel_only_s1:MotionSense] trainable params: 51,339
[full_nc-accel_only_s1:MotionSense] epoch 1 | act_loss 0.7750 |
dom_loss 1.3464 | lambda 0.050 | tr_acc 67.5% | val_loss 0.8697 | val
_acc 63.3% *
[full_nc-accel_only_s1:MotionSense] epoch 2 | act_loss 0.6483 |
dom_loss 1.2907 | lambda 0.100 | tr_acc 72.7% | val_loss 0.8517 | val
_acc 67.1% *
[full_nc-accel_only_s1:MotionSense] epoch 3 | act_loss 0.6178 |
dom_loss 1.2727 | lambda 0.149 | tr_acc 73.9% | val_loss 0.6493 | val
_acc 75.3% *
[full_nc-accel_only_s1:MotionSense] epoch 20 | act_loss 0.5277 |
dom_loss 1.3192 | lambda 0.762 | tr_acc 77.5% | val_loss 0.8535 | val
_acc 69.2%
[full_nc-accel_only_s1:MotionSense] early stop at epoch 23
[full_nc-accel_only_s1:MotionSense] src-val 75.33% | TARGET-TEST 8
0.98% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs

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    built 25000/26450 graphs
    done: 26450 graphs built
[full_nc-accel_only_s1:RealWorld-Thigh] trainable params: 51,339
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 1 | act_loss 0.775
3 | dom_loss 1.3107 | lambda 0.050 | tr_acc 66.4% | val_loss 0.7264 |
val_acc 67.1% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 2 | act_loss 0.650
7 | dom_loss 1.2499 | lambda 0.100 | tr_acc 71.7% | val_loss 0.6992 |
val_acc 68.7% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 3 | act_loss 0.611
4 | dom_loss 1.2248 | lambda 0.149 | tr_acc 73.1% | val_loss 0.7286 |
val_acc 70.2% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 4 | act_loss 0.582
5 | dom_loss 1.2190 | lambda 0.197 | tr_acc 74.4% | val_loss 0.6742 |
val_acc 72.3% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 8 | act_loss 0.538
0 | dom_loss 1.2379 | lambda 0.380 | tr_acc 76.2% | val_loss 0.6343 |
val_acc 73.8% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 19 | act_loss 0.515
3 | dom_loss 1.2749 | lambda 0.740 | tr_acc 76.6% | val_loss 0.6609 |
val_acc 73.8% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 20 | act_loss 0.513
9 | dom_loss 1.2768 | lambda 0.762 | tr_acc 76.9% | val_loss 0.6758 |
val_acc 73.6%
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 25 | act_loss 0.506
9 | dom_loss 1.2855 | lambda 0.848 | tr_acc 77.0% | val_loss 0.6219 |
val_acc 75.3% *
[full_nc-accel_only_s1:RealWorld-Thigh] epoch 40 | act_loss 0.502
7 | dom_loss 1.3105 | lambda 0.964 | tr_acc 76.9% | val_loss 0.6578 |
val_acc 70.3%
[full_nc-accel_only_s1:RealWorld-Thigh] early stop at epoch 45
[full_nc-accel_only_s1:RealWorld-Thigh] src-val 75.26% | TARGET-TE
ST 70.70% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_nc-accel_only_s1:RealWorld-Waist] trainable params: 51,339
[full_nc-accel_only_s1:RealWorld-Waist] epoch 1 | act_loss 0.774

```

```

5 | dom_loss 1.3238 | lambda 0.050 | tr_acc 66.9% | val_loss 0.6996 |
val_acc 69.3% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 3 | act_loss 0.600
5 | dom_loss 1.2612 | lambda 0.149 | tr_acc 74.6% | val_loss 0.6757 |
val_acc 73.3% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 4 | act_loss 0.577
8 | dom_loss 1.2629 | lambda 0.197 | tr_acc 75.5% | val_loss 0.6285 |
val_acc 75.6% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 7 | act_loss 0.544
0 | dom_loss 1.2568 | lambda 0.336 | tr_acc 77.2% | val_loss 0.6193 |
val_acc 77.3% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 20 | act_loss 0.495
5 | dom_loss 1.2850 | lambda 0.762 | tr_acc 79.0% | val_loss 0.8123 |
val_acc 70.3%
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 25 | act_loss 0.489
8 | dom_loss 1.2910 | lambda 0.848 | tr_acc 79.5% | val_loss 0.6362 |
val_acc 77.3% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 32 | act_loss 0.485
8 | dom_loss 1.3093 | lambda 0.922 | tr_acc 79.9% | val_loss 0.6020 |
val_acc 77.6% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 40 | act_loss 0.475
2 | dom_loss 1.3148 | lambda 0.964 | tr_acc 80.7% | val_loss 0.6977 |
val_acc 73.4%
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 45 | act_loss 0.465
8 | dom_loss 1.3146 | lambda 0.978 | tr_acc 81.1% | val_loss 0.6496 |
val_acc 77.7% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 46 | act_loss 0.459
6 | dom_loss 1.3119 | lambda 0.980 | tr_acc 81.3% | val_loss 0.6157 |
val_acc 78.8% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 59 | act_loss 0.443
6 | dom_loss 1.3119 | lambda 0.995 | tr_acc 82.4% | val_loss 0.6225 |
val_acc 79.3% *
    [full_nc-accel_only_s1:RealWorld-Waist] epoch 60 | act_loss 0.439
1 | dom_loss 1.3128 | lambda 0.995 | tr_acc 82.0% | val_loss 0.6024 |
val_acc 77.0%
    [full_nc-accel_only_s1:RealWorld-Waist] early stop at epoch 79
    [full_nc-accel_only_s1:RealWorld-Waist] src-val 79.32% | TARGET-TE
ST 66.40% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs

```

```

built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_nc-accel_only_s1:UCI] trainable params: 51,339
[full_nc-accel_only_s1:UCI] epoch 1 | act_loss 0.7592 | dom_loss
1.3925 | lambda 0.050 | tr_acc 68.6% | val_loss 0.7123 | val_acc 71.
6% *
[full_nc-accel_only_s1:UCI] epoch 3 | act_loss 0.6078 | dom_loss
1.3316 | lambda 0.149 | tr_acc 74.9% | val_loss 0.7322 | val_acc 72.
0% *
[full_nc-accel_only_s1:UCI] epoch 5 | act_loss 0.5805 | dom_loss
1.3302 | lambda 0.245 | tr_acc 76.0% | val_loss 0.7924 | val_acc 72.
5% *
[full_nc-accel_only_s1:UCI] epoch 11 | act_loss 0.5401 | dom_loss
1.3446 | lambda 0.501 | tr_acc 77.5% | val_loss 0.6636 | val_acc 75.
1% *
[full_nc-accel_only_s1:UCI] epoch 20 | act_loss 0.5228 | dom_loss
1.3587 | lambda 0.762 | tr_acc 78.2% | val_loss 0.7231 | val_acc 71.
9%
[full_nc-accel_only_s1:UCI] early stop at epoch 31
[full_nc-accel_only_s1:UCI] src-val 75.09% | TARGET-TEST 63.91% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-accel_only_s1:WISDM] trainable params: 51,339
[full_nc-accel_only_s1:WISDM] epoch 1 | act_loss 0.7686 | dom_lo
ss 1.3287 | lambda 0.050 | tr_acc 68.2% | val_loss 0.8122 | val_acc
70.6% *
[full_nc-accel_only_s1:WISDM] epoch 2 | act_loss 0.6312 | dom_lo
ss 1.2964 | lambda 0.100 | tr_acc 74.1% | val_loss 0.7260 | val_acc
72.8% *
[full_nc-accel_only_s1:WISDM] epoch 4 | act_loss 0.5828 | dom_lo
ss 1.2973 | lambda 0.197 | tr_acc 76.4% | val_loss 0.7137 | val_acc
73.0% *
[full_nc-accel_only_s1:WISDM] epoch 6 | act_loss 0.5557 | dom_lo
ss 1.3025 | lambda 0.291 | tr_acc 77.3% | val_loss 0.6815 | val_acc

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75.0% *
[full_nc-accel_only_s1:WISDM] epoch 15 | act_loss 0.5151 | dom_lo
ss 1.3248 | lambda 0.635 | tr_acc 79.0% | val_loss 0.6774 | val_acc
75.7% *
[full_nc-accel_only_s1:WISDM] epoch 20 | act_loss 0.5015 | dom_lo
ss 1.3292 | lambda 0.762 | tr_acc 79.6% | val_loss 0.9909 | val_acc
67.6%
[full_nc-accel_only_s1:WISDM] early stop at epoch 35
[full_nc-accel_only_s1:WISDM] src-val 75.67% | TARGET-TEST 69.32%
| saved ckpt + npz
[P24] train+save | arm=full_nc-accel_only_s2 {'cnn_embed_dim': 64, 'gn
n_hidden': 64, 'cnn_channels': [32, 64]} | nc=accel_only aw=- | seed=2
| targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Wais
t', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiment
s/DAGHAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/fu
ll_nc-accel_only_s2
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (0, 1):100%, (1, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-accel_only_s2:KuHar] trainable params: 51,339
[full_nc-accel_only_s2:KuHar] epoch 1 | act_loss 0.7673 | dom_lo
ss 1.4151 | lambda 0.050 | tr_acc 68.1% | val_loss 0.8213 | val_acc
66.6% *
[full_nc-accel_only_s2:KuHar] epoch 2 | act_loss 0.6396 | dom_lo
ss 1.3664 | lambda 0.100 | tr_acc 73.6% | val_loss 0.8923 | val_acc
68.8% *
[full_nc-accel_only_s2:KuHar] epoch 3 | act_loss 0.6009 | dom_lo
ss 1.3673 | lambda 0.149 | tr_acc 75.2% | val_loss 0.6691 | val_acc
74.8% *
[full_nc-accel_only_s2:KuHar] epoch 4 | act_loss 0.5840 | dom_lo
ss 1.3737 | lambda 0.197 | tr_acc 75.6% | val_loss 0.6517 | val_acc
75.4% *
[full_nc-accel_only_s2:KuHar] epoch 13 | act_loss 0.5283 | dom_lo
ss 1.3969 | lambda 0.572 | tr_acc 78.1% | val_loss 0.6656 | val_acc
75.7% *
[full_nc-accel_only_s2:KuHar] epoch 20 | act_loss 0.5187 | dom_lo

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ss 1.4058 | lambda 0.762 | tr_acc 78.4% | val_loss 0.7367 | val_acc
73.2%
[full_nc-accel_only_s2:KuHar] early stop at epoch 33
[full_nc-accel_only_s2:KuHar] src-val 75.72% | TARGET-TEST 53.47%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_nc-accel_only_s2:MotionSense] trainable params: 51,339
[full_nc-accel_only_s2:MotionSense] epoch 1 | act_loss 0.7872 |
dom_loss 1.3354 | lambda 0.050 | tr_acc 66.7% | val_loss 0.7522 | val
_acc 71.2% *
[full_nc-accel_only_s2:MotionSense] epoch 4 | act_loss 0.5943 |
dom_loss 1.2697 | lambda 0.197 | tr_acc 74.9% | val_loss 0.7758 | val
_acc 71.3% *
[full_nc-accel_only_s2:MotionSense] epoch 5 | act_loss 0.5803 |
dom_loss 1.2730 | lambda 0.245 | tr_acc 75.7% | val_loss 0.6882 | val
_acc 74.1% *
[full_nc-accel_only_s2:MotionSense] epoch 11 | act_loss 0.5443 |
dom_loss 1.2980 | lambda 0.501 | tr_acc 77.0% | val_loss 0.6959 | val
_acc 75.4% *
[full_nc-accel_only_s2:MotionSense] epoch 20 | act_loss 0.5284 |
dom_loss 1.3212 | lambda 0.762 | tr_acc 77.5% | val_loss 0.7529 | val
_acc 71.4%
[full_nc-accel_only_s2:MotionSense] early stop at epoch 31
[full_nc-accel_only_s2:MotionSense] src-val 75.44% | TARGET-TEST 8
0.32% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa

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irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
  built 5000/26450 graphs
  built 10000/26450 graphs
  built 15000/26450 graphs
  built 20000/26450 graphs
  built 25000/26450 graphs
  done: 26450 graphs built
  [full_nc-accel_only_s2:RealWorld-Thigh] trainable params: 51,339
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 1 | act_loss 0.785
4 | dom_loss 1.3091 | lambda 0.050 | tr_acc 66.3% | val_loss 0.7909 |
val_acc 64.8% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 2 | act_loss 0.651
9 | dom_loss 1.2380 | lambda 0.100 | tr_acc 71.5% | val_loss 0.8062 |
val_acc 66.6% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 3 | act_loss 0.612
8 | dom_loss 1.2181 | lambda 0.149 | tr_acc 73.1% | val_loss 0.7404 |
val_acc 69.5% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 6 | act_loss 0.553
2 | dom_loss 1.2390 | lambda 0.291 | tr_acc 75.5% | val_loss 0.6533 |
val_acc 72.3% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 8 | act_loss 0.537
4 | dom_loss 1.2478 | lambda 0.380 | tr_acc 76.4% | val_loss 0.6650 |
val_acc 73.3% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 12 | act_loss 0.521
0 | dom_loss 1.2628 | lambda 0.537 | tr_acc 76.8% | val_loss 0.6685 |
val_acc 73.4% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 16 | act_loss 0.508
9 | dom_loss 1.2696 | lambda 0.664 | tr_acc 77.0% | val_loss 0.6221 |
val_acc 75.2% *
  [full_nc-accel_only_s2:RealWorld-Thigh] epoch 20 | act_loss 0.504
4 | dom_loss 1.2767 | lambda 0.762 | tr_acc 77.4% | val_loss 0.6897 |
val_acc 69.3%
  [full_nc-accel_only_s2:RealWorld-Thigh] early stop at epoch 36
  [full_nc-accel_only_s2:RealWorld-Thigh] src-val 75.24% | TARGET-TE
ST 70.74% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
  built 5000/26456 graphs

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    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_nc-accel_only_s2:RealWorld-Waist] trainable params: 51,339
[full_nc-accel_only_s2:RealWorld-Waist] epoch 1 | act_loss 0.780
3 | dom_loss 1.3172 | lambda 0.050 | tr_acc 67.0% | val_loss 1.1695 |
val_acc 57.1% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 2 | act_loss 0.654
0 | dom_loss 1.2626 | lambda 0.100 | tr_acc 72.6% | val_loss 0.8025 |
val_acc 68.1% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 6 | act_loss 0.547
4 | dom_loss 1.2505 | lambda 0.291 | tr_acc 76.9% | val_loss 0.8098 |
val_acc 68.8% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 7 | act_loss 0.538
1 | dom_loss 1.2514 | lambda 0.336 | tr_acc 77.4% | val_loss 0.6608 |
val_acc 74.9% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 12 | act_loss 0.513
4 | dom_loss 1.2689 | lambda 0.537 | tr_acc 78.3% | val_loss 0.6586 |
val_acc 75.8% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 20 | act_loss 0.497
7 | dom_loss 1.2878 | lambda 0.762 | tr_acc 79.2% | val_loss 0.6317 |
val_acc 75.7%
[full_nc-accel_only_s2:RealWorld-Waist] epoch 27 | act_loss 0.486
5 | dom_loss 1.3047 | lambda 0.874 | tr_acc 80.0% | val_loss 0.6072 |
val_acc 77.0% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 38 | act_loss 0.475
4 | dom_loss 1.3118 | lambda 0.956 | tr_acc 80.7% | val_loss 0.6224 |
val_acc 78.6% *
[full_nc-accel_only_s2:RealWorld-Waist] epoch 40 | act_loss 0.470
3 | dom_loss 1.3099 | lambda 0.964 | tr_acc 80.8% | val_loss 0.6245 |
val_acc 76.5%
[full_nc-accel_only_s2:RealWorld-Waist] early stop at epoch 58
[full_nc-accel_only_s2:RealWorld-Waist] src-val 78.57% | TARGET-TE
ST 69.17% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs

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built 30000/34368 graphs
done: 34368 graphs built
[full_nc-accel_only_s2:UCI] trainable params: 51,339
[full_nc-accel_only_s2:UCI] epoch 1 | act_loss 0.7652 | dom_loss
1.4011 | lambda 0.050 | tr_acc 68.0% | val_loss 0.8769 | val_acc 65.
6% *
[full_nc-accel_only_s2:UCI] epoch 3 | act_loss 0.6053 | dom_loss
1.3285 | lambda 0.149 | tr_acc 75.1% | val_loss 0.8831 | val_acc 68.
0% *
[full_nc-accel_only_s2:UCI] epoch 5 | act_loss 0.5665 | dom_loss
1.3222 | lambda 0.245 | tr_acc 76.5% | val_loss 0.6896 | val_acc 74.
2% *
[full_nc-accel_only_s2:UCI] epoch 8 | act_loss 0.5482 | dom_loss
1.3330 | lambda 0.380 | tr_acc 77.3% | val_loss 0.6862 | val_acc 75.
0% *
[full_nc-accel_only_s2:UCI] epoch 20 | act_loss 0.5202 | dom_loss
1.3641 | lambda 0.762 | tr_acc 78.2% | val_loss 0.9755 | val_acc 66.
2%
[full_nc-accel_only_s2:UCI] early stop at epoch 28
[full_nc-accel_only_s2:UCI] src-val 75.02% | TARGET-TEST 65.07% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-accel_only_s2:WISDM] trainable params: 51,339
[full_nc-accel_only_s2:WISDM] epoch 1 | act_loss 0.7913 | dom_lo
ss 1.3273 | lambda 0.050 | tr_acc 67.5% | val_loss 0.8755 | val_acc
66.6% *
[full_nc-accel_only_s2:WISDM] epoch 4 | act_loss 0.5805 | dom_lo
ss 1.3046 | lambda 0.197 | tr_acc 76.3% | val_loss 0.7641 | val_acc
70.6% *
[full_nc-accel_only_s2:WISDM] epoch 7 | act_loss 0.5373 | dom_lo
ss 1.3042 | lambda 0.336 | tr_acc 78.1% | val_loss 0.7381 | val_acc
73.7% *
[full_nc-accel_only_s2:WISDM] epoch 9 | act_loss 0.5277 | dom_lo
ss 1.3133 | lambda 0.422 | tr_acc 78.4% | val_loss 0.6762 | val_acc
74.7% *
[full_nc-accel_only_s2:WISDM] epoch 20 | act_loss 0.4921 | dom_lo

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ss 1.3312 | lambda 0.762 | tr_acc 80.1% | val_loss 0.8614 | val_acc
69.3%
[full_nc-accel_only_s2:WISDM] early stop at epoch 29
[full_nc-accel_only_s2:WISDM] src-val 74.65% | TARGET-TEST 70.85%
| saved ckpt + npz
[P24] train+save | arm=full_nc-accel_only_s3 {'cnn_embed_dim': 64, 'gn
n_hidden': 64, 'cnn_channels': [32, 64]} | nc=accel_only aw=- | seed=3
| targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Wais
t', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiment
s/DAGHAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/fu
ll_nc-accel_only_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (0, 1):100%, (1, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-accel_only_s3:KuHar] trainable params: 51,339
[full_nc-accel_only_s3:KuHar] epoch 1 | act_loss 0.7648 | dom_lo
ss 1.4258 | lambda 0.050 | tr_acc 68.0% | val_loss 0.8502 | val_acc
68.1% *
[full_nc-accel_only_s3:KuHar] epoch 3 | act_loss 0.6000 | dom_lo
ss 1.3698 | lambda 0.149 | tr_acc 75.2% | val_loss 0.6856 | val_acc
75.6% *
[full_nc-accel_only_s3:KuHar] epoch 20 | act_loss 0.5157 | dom_lo
ss 1.4063 | lambda 0.762 | tr_acc 78.5% | val_loss 0.7192 | val_acc
69.1%
[full_nc-accel_only_s3:KuHar] early stop at epoch 23
[full_nc-accel_only_s3:KuHar] src-val 75.57% | TARGET-TEST 54.86%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%

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[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (3 nodes, 6 edges, 90 features each)...
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built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_nc-accel_only_s3:MotionSense] trainable params: 51,339
[full_nc-accel_only_s3:MotionSense] epoch 1 | act_loss 0.7880 | dom_loss 1.3449 | lambda 0.050 | tr_acc 66.5% | val_loss 0.8647 | val_acc 65.0% *
[full_nc-accel_only_s3:MotionSense] epoch 2 | act_loss 0.6504 | dom_loss 1.2956 | lambda 0.100 | tr_acc 72.6% | val_loss 0.7608 | val_acc 70.3% *
[full_nc-accel_only_s3:MotionSense] epoch 5 | act_loss 0.5758 | dom_loss 1.2704 | lambda 0.245 | tr_acc 75.9% | val_loss 0.7739 | val_acc 70.5% *
[full_nc-accel_only_s3:MotionSense] epoch 8 | act_loss 0.5592 | dom_loss 1.2912 | lambda 0.380 | tr_acc 76.5% | val_loss 0.6853 | val_acc 74.3% *
[full_nc-accel_only_s3:MotionSense] epoch 20 | act_loss 0.5293 | dom_loss 1.3193 | lambda 0.762 | tr_acc 77.5% | val_loss 0.8351 | val_acc 69.8%
[full_nc-accel_only_s3:MotionSense] early stop at epoch 28
[full_nc-accel_only_s3:MotionSense] src-val 74.28% | TARGET-TEST 79.19% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (3 nodes, 6 edges, 90 features each)...
```

```

built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_nc-accel_only_s3:RealWorld-Thigh] trainable params: 51,339
[full_nc-accel_only_s3:RealWorld-Thigh] epoch 1 | act_loss 0.786
```

```

5 | dom_loss 1.3078 | lambda 0.050 | tr_acc 66.1% | val_loss 0.7387 |
val_acc 70.8% *
    [full_nc-accel_only_s3:RealWorld-Thigh] epoch 4 | act_loss 0.582
7 | dom_loss 1.2183 | lambda 0.197 | tr_acc 74.4% | val_loss 0.5962 |
val_acc 74.9% *
    [full_nc-accel_only_s3:RealWorld-Thigh] epoch 9 | act_loss 0.529
8 | dom_loss 1.2491 | lambda 0.422 | tr_acc 76.3% | val_loss 0.6357 |
val_acc 75.0% *
    [full_nc-accel_only_s3:RealWorld-Thigh] epoch 18 | act_loss 0.509
5 | dom_loss 1.2733 | lambda 0.716 | tr_acc 76.8% | val_loss 0.6198 |
val_acc 75.3% *
    [full_nc-accel_only_s3:RealWorld-Thigh] epoch 20 | act_loss 0.506
6 | dom_loss 1.2795 | lambda 0.762 | tr_acc 76.8% | val_loss 0.7199 |
val_acc 69.4%
    [full_nc-accel_only_s3:RealWorld-Thigh] early stop at epoch 38
    [full_nc-accel_only_s3:RealWorld-Thigh] src-val 75.29% | TARGET-TE
ST 69.81% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
    [full_nc-accel_only_s3:RealWorld-Waist] trainable params: 51,339
    [full_nc-accel_only_s3:RealWorld-Waist] epoch 1 | act_loss 0.783
2 | dom_loss 1.3281 | lambda 0.050 | tr_acc 66.5% | val_loss 0.6896 |
val_acc 71.3% *
    [full_nc-accel_only_s3:RealWorld-Waist] epoch 2 | act_loss 0.641
4 | dom_loss 1.2832 | lambda 0.100 | tr_acc 73.2% | val_loss 0.6725 |
val_acc 74.2% *
    [full_nc-accel_only_s3:RealWorld-Waist] epoch 3 | act_loss 0.599
3 | dom_loss 1.2715 | lambda 0.149 | tr_acc 74.8% | val_loss 0.6536 |
val_acc 74.7% *
    [full_nc-accel_only_s3:RealWorld-Waist] epoch 10 | act_loss 0.517
2 | dom_loss 1.2599 | lambda 0.462 | tr_acc 78.6% | val_loss 0.6138 |
val_acc 77.1% *
    [full_nc-accel_only_s3:RealWorld-Waist] epoch 17 | act_loss 0.495
9 | dom_loss 1.2789 | lambda 0.691 | tr_acc 79.1% | val_loss 0.5902 |
val_acc 77.3% *
    [full_nc-accel_only_s3:RealWorld-Waist] epoch 20 | act_loss 0.484
9 | dom_loss 1.2855 | lambda 0.762 | tr_acc 79.9% | val_loss 0.7963 |

```

```

val_acc 69.9%
[full_nc-accel_only_s3:RealWorld-Waist] early stop at epoch 37
[full_nc-accel_only_s3:RealWorld-Waist] src-val 77.27% | TARGET-TE
ST 69.60% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_nc-accel_only_s3:UCI] trainable params: 51,339
[full_nc-accel_only_s3:UCI] epoch 1 | act_loss 0.7660 | dom_loss
1.3995 | lambda 0.050 | tr_acc 67.9% | val_loss 0.8208 | val_acc 67.
2% *
[full_nc-accel_only_s3:UCI] epoch 3 | act_loss 0.6038 | dom_loss
1.3276 | lambda 0.149 | tr_acc 75.1% | val_loss 0.7597 | val_acc 71.
9% *
[full_nc-accel_only_s3:UCI] epoch 9 | act_loss 0.5424 | dom_loss
1.3404 | lambda 0.422 | tr_acc 77.6% | val_loss 0.6906 | val_acc 74.
4% *
[full_nc-accel_only_s3:UCI] epoch 20 | act_loss 0.5185 | dom_loss
1.3604 | lambda 0.762 | tr_acc 78.3% | val_loss 0.7428 | val_acc 71.
1%
[full_nc-accel_only_s3:UCI] epoch 23 | act_loss 0.5160 | dom_loss
1.3611 | lambda 0.818 | tr_acc 78.1% | val_loss 0.6829 | val_acc 74.
9% *
[full_nc-accel_only_s3:UCI] epoch 40 | act_loss 0.5032 | dom_loss
1.3908 | lambda 0.964 | tr_acc 79.0% | val_loss 0.8620 | val_acc 65.
3%
[full_nc-accel_only_s3:UCI] early stop at epoch 43
[full_nc-accel_only_s3:UCI] src-val 74.87% | TARGET-TEST 67.54% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept

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pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 28040 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
  built 5000/28040 graphs
  built 10000/28040 graphs
  built 15000/28040 graphs
  built 20000/28040 graphs
  built 25000/28040 graphs
  done: 28040 graphs built
  [full_nc-accel_only_s3:WISDM] trainable params: 51,339
  [full_nc-accel_only_s3:WISDM] epoch 1 | act_loss 0.7847 | dom_lo
ss 1.3308 | lambda 0.050 | tr_acc 67.6% | val_loss 1.0387 | val_acc
60.3% *
  [full_nc-accel_only_s3:WISDM] epoch 2 | act_loss 0.6384 | dom_lo
ss 1.3008 | lambda 0.100 | tr_acc 73.7% | val_loss 0.7727 | val_acc
67.6% *
  [full_nc-accel_only_s3:WISDM] epoch 3 | act_loss 0.6049 | dom_lo
ss 1.2958 | lambda 0.149 | tr_acc 75.4% | val_loss 0.7750 | val_acc
71.5% *
  [full_nc-accel_only_s3:WISDM] epoch 11 | act_loss 0.5182 | dom_lo
ss 1.3186 | lambda 0.501 | tr_acc 79.0% | val_loss 0.7078 | val_acc
73.6% *
  [full_nc-accel_only_s3:WISDM] epoch 17 | act_loss 0.4961 | dom_lo
ss 1.3257 | lambda 0.691 | tr_acc 79.7% | val_loss 0.7084 | val_acc
74.8% *
  [full_nc-accel_only_s3:WISDM] epoch 20 | act_loss 0.4917 | dom_lo
ss 1.3302 | lambda 0.762 | tr_acc 80.1% | val_loss 0.7812 | val_acc
72.0%
  [full_nc-accel_only_s3:WISDM] early stop at epoch 37
  [full_nc-accel_only_s3:WISDM] src-val 74.76% | TARGET-TEST 70.58%
  | saved ckpt + npz
[P24] train+save | arm=full_nc-accel_only_s5 {'cnn_embed_dim': 64, 'gn
n_hidden': 64, 'cnn_channels': [32, 64]} | nc=accel_only aw=- | seed=5
| targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Wais
t', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiment
s/DAGHAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/fu
ll_nc-accel_only_s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (0, 1):100%, (1, 2):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 35396 sensor graphs (3 nodes, 6 edges, 90 features eac

```

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h)...
  built 5000/35396 graphs
  built 10000/35396 graphs
  built 15000/35396 graphs
  built 20000/35396 graphs
  built 25000/35396 graphs
  built 30000/35396 graphs
  built 35000/35396 graphs
  done: 35396 graphs built
[full_nc-accel_only_s5:KuHar] trainable params: 51,339
[full_nc-accel_only_s5:KuHar] epoch 1 | act_loss 0.7667 | dom_lo
ss 1.4210 | lambda 0.050 | tr_acc 67.9% | val_loss 0.9221 | val_acc
61.4% *
[full_nc-accel_only_s5:KuHar] epoch 2 | act_loss 0.6311 | dom_lo
ss 1.3796 | lambda 0.100 | tr_acc 74.0% | val_loss 0.9690 | val_acc
62.1% *
[full_nc-accel_only_s5:KuHar] epoch 3 | act_loss 0.5985 | dom_lo
ss 1.3647 | lambda 0.149 | tr_acc 75.2% | val_loss 0.7198 | val_acc
74.0% *
[full_nc-accel_only_s5:KuHar] epoch 11 | act_loss 0.5357 | dom_lo
ss 1.3909 | lambda 0.501 | tr_acc 77.7% | val_loss 0.6623 | val_acc
75.4% *
[full_nc-accel_only_s5:KuHar] epoch 20 | act_loss 0.5160 | dom_lo
ss 1.4055 | lambda 0.762 | tr_acc 78.4% | val_loss 0.6794 | val_acc
74.9%
[full_nc-accel_only_s5:KuHar] epoch 22 | act_loss 0.5111 | dom_lo
ss 1.4065 | lambda 0.800 | tr_acc 78.5% | val_loss 0.6576 | val_acc
75.9% *
[full_nc-accel_only_s5:KuHar] epoch 25 | act_loss 0.5105 | dom_lo
ss 1.4084 | lambda 0.848 | tr_acc 78.7% | val_loss 0.6705 | val_acc
75.9% *
[full_nc-accel_only_s5:KuHar] epoch 40 | act_loss 0.4999 | dom_lo
ss 1.4259 | lambda 0.964 | tr_acc 79.1% | val_loss 0.7158 | val_acc
72.3%
[full_nc-accel_only_s5:KuHar] early stop at epoch 45
[full_nc-accel_only_s5:KuHar] src-val 75.89% | TARGET-TEST 50.00%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
  built 5000/33230 graphs
  built 10000/33230 graphs
  built 15000/33230 graphs
  built 20000/33230 graphs

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    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
[full_nc-accel_only_s5:MotionSense] trainable params: 51,339
[full_nc-accel_only_s5:MotionSense] epoch 1 | act_loss 0.7841 |
dom_loss 1.3374 | lambda 0.050 | tr_acc 67.0% | val_loss 0.8156 | val
_acc 67.8% *
[full_nc-accel_only_s5:MotionSense] epoch 4 | act_loss 0.5974 |
dom_loss 1.2807 | lambda 0.197 | tr_acc 74.9% | val_loss 0.7442 | val
_acc 72.4% *
[full_nc-accel_only_s5:MotionSense] epoch 5 | act_loss 0.5846 |
dom_loss 1.2717 | lambda 0.245 | tr_acc 75.5% | val_loss 0.6698 | val
_acc 74.9% *
[full_nc-accel_only_s5:MotionSense] epoch 20 | act_loss 0.5244 |
dom_loss 1.3209 | lambda 0.762 | tr_acc 77.5% | val_loss 0.6994 | val
_acc 71.1%
[full_nc-accel_only_s5:MotionSense] early stop at epoch 25
[full_nc-accel_only_s5:MotionSense] src-val 74.94% | TARGET-TEST 8
1.54% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
[full_nc-accel_only_s5:RealWorld-Thigh] trainable params: 51,339
[full_nc-accel_only_s5:RealWorld-Thigh] epoch 1 | act_loss 0.788
1 | dom_loss 1.3105 | lambda 0.050 | tr_acc 65.6% | val_loss 0.8372 |
val_acc 65.9% *
[full_nc-accel_only_s5:RealWorld-Thigh] epoch 3 | act_loss 0.607
7 | dom_loss 1.2346 | lambda 0.149 | tr_acc 73.3% | val_loss 0.6236 |
val_acc 73.6% *
[full_nc-accel_only_s5:RealWorld-Thigh] epoch 6 | act_loss 0.548
6 | dom_loss 1.2389 | lambda 0.291 | tr_acc 75.7% | val_loss 0.6441 |
val_acc 74.5% *
[full_nc-accel_only_s5:RealWorld-Thigh] epoch 7 | act_loss 0.545
6 | dom_loss 1.2457 | lambda 0.336 | tr_acc 75.6% | val_loss 0.6387 |
val_acc 75.5% *
[full_nc-accel_only_s5:RealWorld-Thigh] epoch 20 | act_loss 0.510
9 | dom_loss 1.2786 | lambda 0.762 | tr_acc 76.7% | val_loss 0.6740 |
val_acc 73.0%

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[full_nc-accel_only_s5:RealWorld-Thigh] early stop at epoch 27
[full_nc-accel_only_s5:RealWorld-Thigh] src-val 75.49% | TARGET-TE
ST 69.81% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_nc-accel_only_s5:RealWorld-Waist] trainable params: 51,339
[full_nc-accel_only_s5:RealWorld-Waist] epoch 1 | act_loss 0.786
2 | dom_loss 1.3267 | lambda 0.050 | tr_acc 66.4% | val_loss 0.7255 |
val_acc 68.6% *
[full_nc-accel_only_s5:RealWorld-Waist] epoch 2 | act_loss 0.656
7 | dom_loss 1.2886 | lambda 0.100 | tr_acc 72.3% | val_loss 0.6617 |
val_acc 73.0% *
[full_nc-accel_only_s5:RealWorld-Waist] epoch 5 | act_loss 0.554
1 | dom_loss 1.2641 | lambda 0.245 | tr_acc 76.5% | val_loss 0.6561 |
val_acc 75.6% *
[full_nc-accel_only_s5:RealWorld-Waist] epoch 16 | act_loss 0.493
3 | dom_loss 1.2800 | lambda 0.664 | tr_acc 79.2% | val_loss 0.6676 |
val_acc 75.9% *
[full_nc-accel_only_s5:RealWorld-Waist] epoch 20 | act_loss 0.491
1 | dom_loss 1.2910 | lambda 0.762 | tr_acc 79.5% | val_loss 0.8492 |
val_acc 71.6%
[full_nc-accel_only_s5:RealWorld-Waist] epoch 34 | act_loss 0.477
2 | dom_loss 1.3106 | lambda 0.935 | tr_acc 80.7% | val_loss 0.7044 |
val_acc 76.2% *
[full_nc-accel_only_s5:RealWorld-Waist] epoch 35 | act_loss 0.474
6 | dom_loss 1.3120 | lambda 0.941 | tr_acc 80.7% | val_loss 0.6348 |
val_acc 77.5% *
[full_nc-accel_only_s5:RealWorld-Waist] epoch 40 | act_loss 0.465
2 | dom_loss 1.3105 | lambda 0.964 | tr_acc 81.1% | val_loss 0.7302 |
val_acc 74.9%
[full_nc-accel_only_s5:RealWorld-Waist] early stop at epoch 55
[full_nc-accel_only_s5:RealWorld-Waist] src-val 77.47% | TARGET-TE
ST 69.02% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel

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[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs
    built 30000/34368 graphs
    done: 34368 graphs built
  [full_nc-accel_only_s5:UCI] trainable params: 51,339
  [full_nc-accel_only_s5:UCI] epoch 1 | act_loss 0.7637 | dom_loss
1.3942 | lambda 0.050 | tr_acc 68.0% | val_loss 0.8354 | val_acc 69.
6% *
  [full_nc-accel_only_s5:UCI] epoch 2 | act_loss 0.6319 | dom_loss
1.3269 | lambda 0.100 | tr_acc 74.0% | val_loss 0.7212 | val_acc 71.
3% *
  [full_nc-accel_only_s5:UCI] epoch 5 | act_loss 0.5648 | dom_loss
1.3222 | lambda 0.245 | tr_acc 76.5% | val_loss 0.8153 | val_acc 71.
8% *
  [full_nc-accel_only_s5:UCI] epoch 7 | act_loss 0.5560 | dom_loss
1.3330 | lambda 0.336 | tr_acc 77.0% | val_loss 0.6833 | val_acc 74.
6% *
  [full_nc-accel_only_s5:UCI] epoch 8 | act_loss 0.5478 | dom_loss
1.3332 | lambda 0.380 | tr_acc 77.2% | val_loss 0.6740 | val_acc 75.
2% *
  [full_nc-accel_only_s5:UCI] epoch 20 | act_loss 0.5185 | dom_loss
1.3629 | lambda 0.762 | tr_acc 78.3% | val_loss 0.7084 | val_acc 72.
5%
  [full_nc-accel_only_s5:UCI] early stop at epoch 28
  [full_nc-accel_only_s5:UCI] src-val 75.24% | TARGET-TEST 68.41% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [0, 1, 2] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (1, 2):100%, (0, 1):100%, (0, 2):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 28040 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...

```

```

built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-accel_only_s5:WISDM] trainable params: 51,339
[full_nc-accel_only_s5:WISDM] epoch 1 | act_loss 0.7799 | dom_lo
ss 1.3182 | lambda 0.050 | tr_acc 67.7% | val_loss 0.7424 | val_acc
72.9% *
[full_nc-accel_only_s5:WISDM] epoch 10 | act_loss 0.5276 | dom_lo
ss 1.3119 | lambda 0.462 | tr_acc 78.4% | val_loss 0.7196 | val_acc
73.2% *
[full_nc-accel_only_s5:WISDM] epoch 13 | act_loss 0.5180 | dom_lo
ss 1.3206 | lambda 0.572 | tr_acc 78.9% | val_loss 0.7206 | val_acc
73.9% *
[full_nc-accel_only_s5:WISDM] epoch 20 | act_loss 0.4943 | dom_lo
ss 1.3290 | lambda 0.762 | tr_acc 80.0% | val_loss 1.0586 | val_acc
66.9%
[full_nc-accel_only_s5:WISDM] epoch 22 | act_loss 0.4935 | dom_lo
ss 1.3301 | lambda 0.800 | tr_acc 80.1% | val_loss 0.7519 | val_acc
73.9% *
[full_nc-accel_only_s5:WISDM] epoch 34 | act_loss 0.4694 | dom_lo
ss 1.3347 | lambda 0.935 | tr_acc 80.8% | val_loss 0.7789 | val_acc
74.0% *
[full_nc-accel_only_s5:WISDM] epoch 40 | act_loss 0.4634 | dom_lo
ss 1.3367 | lambda 0.964 | tr_acc 81.2% | val_loss 1.0272 | val_acc
66.7%
[full_nc-accel_only_s5:WISDM] early stop at epoch 54
[full_nc-accel_only_s5:WISDM] src-val 73.98% | TARGET-TEST 74.62%
| saved ckpt + npz

```

4b. gyro_only (3-node graph — gyroscope axes only)

```
In [ ]: for s in (1, 2, 3, 5):  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant full  
        --nc gyro_only --seed {s} --epochs 100 --log_every 20
```

```

[P24] train+save | arm=full_nc-gyro_only_s1 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=gyro_only aw=- | seed=1 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiments/DAG HAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/full_nc-gyro_only_s1
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 1):100%, (0, 2):100%, (1, 2):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-gyro_only_s1:KuHar] trainable params: 51,339
[full_nc-gyro_only_s1:KuHar] epoch 1 | act_loss 0.8373 | dom_loss 1.3564 | lambda 0.050 | tr_acc 65.3% | val_loss 0.7449 | val_acc 65.6% *
[full_nc-gyro_only_s1:KuHar] epoch 2 | act_loss 0.6928 | dom_loss 1.3328 | lambda 0.100 | tr_acc 71.8% | val_loss 0.7499 | val_acc 72.0% *
[full_nc-gyro_only_s1:KuHar] epoch 9 | act_loss 0.5914 | dom_loss 1.4062 | lambda 0.422 | tr_acc 76.9% | val_loss 0.7007 | val_acc 75.5% *
[full_nc-gyro_only_s1:KuHar] epoch 20 | act_loss 0.5602 | dom_loss 1.4149 | lambda 0.762 | tr_acc 78.2% | val_loss 1.2952 | val_acc 67.1%
[full_nc-gyro_only_s1:KuHar] early stop at epoch 29
[full_nc-gyro_only_s1:KuHar] src-val 75.50% | TARGET-TEST 58.33%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%

```

```

[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (0, 1):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_nc-gyro_only_s1:MotionSense] trainable params: 51,339
[full_nc-gyro_only_s1:MotionSense] epoch 1 | act_loss 0.8628 | d
om_loss 1.2617 | lambda 0.050 | tr_acc 64.4% | val_loss 0.7766 | val_
acc 68.5% *
[full_nc-gyro_only_s1:MotionSense] epoch 2 | act_loss 0.7130 | d
om_loss 1.2077 | lambda 0.100 | tr_acc 71.0% | val_loss 0.7166 | val_
acc 71.6% *
[full_nc-gyro_only_s1:MotionSense] epoch 4 | act_loss 0.6562 | d
om_loss 1.2892 | lambda 0.197 | tr_acc 73.7% | val_loss 0.6892 | val_
acc 72.8% *
[full_nc-gyro_only_s1:MotionSense] epoch 10 | act_loss 0.5896 | d
om_loss 1.3159 | lambda 0.462 | tr_acc 76.4% | val_loss 0.7030 | val_
acc 73.5% *
[full_nc-gyro_only_s1:MotionSense] epoch 20 | act_loss 0.5646 | d
om_loss 1.3253 | lambda 0.762 | tr_acc 77.5% | val_loss 0.7071 | val_
acc 69.9%
[full_nc-gyro_only_s1:MotionSense] epoch 23 | act_loss 0.5578 | d
om_loss 1.3262 | lambda 0.818 | tr_acc 77.9% | val_loss 0.6924 | val_
acc 74.9% *
[full_nc-gyro_only_s1:MotionSense] epoch 28 | act_loss 0.5503 | d
om_loss 1.3285 | lambda 0.885 | tr_acc 78.4% | val_loss 0.6841 | val_
acc 75.6% *
[full_nc-gyro_only_s1:MotionSense] epoch 40 | act_loss 0.5330 | d
om_loss 1.3406 | lambda 0.964 | tr_acc 79.4% | val_loss 0.8072 | val_
acc 65.5%
[full_nc-gyro_only_s1:MotionSense] epoch 42 | act_loss 0.5283 | d
om_loss 1.3388 | lambda 0.970 | tr_acc 79.5% | val_loss 0.6810 | val_
acc 75.9% *
[full_nc-gyro_only_s1:MotionSense] epoch 50 | act_loss 0.5169 | d
om_loss 1.3421 | lambda 0.987 | tr_acc 80.1% | val_loss 0.7135 | val_
acc 76.1% *
[full_nc-gyro_only_s1:MotionSense] epoch 60 | act_loss 0.4952 | d
om_loss 1.3431 | lambda 0.995 | tr_acc 81.3% | val_loss 0.8491 | val_
acc 69.9%
[full_nc-gyro_only_s1:MotionSense] epoch 70 | act_loss 0.4782 | d
om_loss 1.3445 | lambda 0.998 | tr_acc 81.8% | val_loss 0.7032 | val_
acc 76.1% *
[full_nc-gyro_only_s1:MotionSense] epoch 80 | act_loss 0.4609 | d
om_loss 1.3430 | lambda 0.999 | tr_acc 82.7% | val_loss 0.7338 | val_
acc 73.5%
[full_nc-gyro_only_s1:MotionSense] epoch 89 | act_loss 0.4429 | d
om_loss 1.3397 | lambda 1.000 | tr_acc 83.4% | val_loss 0.6933 | val_
acc 76.9% *
[full_nc-gyro_only_s1:MotionSense] epoch 100 | act_loss 0.4386 | d
om_loss 1.3404 | lambda 1.000 | tr_acc 83.6% | val_loss 0.7537 | val_

```

```

acc 72.0%
[full_nc-gyro_only_s1:MotionSense] src-val 76.90% | TARGET-TEST 6
6.76% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_nc-gyro_only_s1:RealWorld-Thigh] trainable params: 51,339
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 1 | act_loss 0.8128
| dom_loss 1.2364 | lambda 0.050 | tr_acc 65.5% | val_loss 0.7054 | v
al_acc 70.6% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 4 | act_loss 0.6162
| dom_loss 1.2274 | lambda 0.197 | tr_acc 74.2% | val_loss 0.6956 | v
al_acc 72.6% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 7 | act_loss 0.5846
| dom_loss 1.2533 | lambda 0.336 | tr_acc 75.8% | val_loss 0.6007 | v
al_acc 74.5% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 8 | act_loss 0.5798
| dom_loss 1.2624 | lambda 0.380 | tr_acc 76.0% | val_loss 0.6164 | v
al_acc 76.8% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 13 | act_loss 0.5538
| dom_loss 1.2786 | lambda 0.572 | tr_acc 77.5% | val_loss 0.5858 | v
al_acc 77.7% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 16 | act_loss 0.5468
| dom_loss 1.2790 | lambda 0.664 | tr_acc 77.9% | val_loss 0.6302 | v
al_acc 77.7% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 19 | act_loss 0.5462
| dom_loss 1.2833 | lambda 0.740 | tr_acc 78.0% | val_loss 0.6080 | v
al_acc 79.6% *
[full_nc-gyro_only_s1:RealWorld-Thigh] epoch 20 | act_loss 0.5379
| dom_loss 1.2866 | lambda 0.762 | tr_acc 78.4% | val_loss 0.5875 | v
al_acc 78.9%
[full_nc-gyro_only_s1:RealWorld-Thigh] early stop at epoch 39
[full_nc-gyro_only_s1:RealWorld-Thigh] src-val 79.60% | TARGET-TEST
65.77% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel

```

```

[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
    [full_nc-gyro_only_s1:RealWorld-Waist] trainable params: 51,339
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 1 | act_loss 0.8378
| dom_loss 1.3143 | lambda 0.050 | tr_acc 64.8% | val_loss 0.7122 | v
al_acc 65.5% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 2 | act_loss 0.6967
| dom_loss 1.2486 | lambda 0.100 | tr_acc 71.1% | val_loss 0.7944 | v
al_acc 69.0% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 3 | act_loss 0.6579
| dom_loss 1.2253 | lambda 0.149 | tr_acc 72.9% | val_loss 0.7415 | v
al_acc 71.6% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 6 | act_loss 0.5975
| dom_loss 1.2637 | lambda 0.291 | tr_acc 75.7% | val_loss 0.6812 | v
al_acc 73.0% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 9 | act_loss 0.5710
| dom_loss 1.2775 | lambda 0.422 | tr_acc 77.1% | val_loss 0.6940 | v
al_acc 74.4% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 10 | act_loss 0.5653
| dom_loss 1.2791 | lambda 0.462 | tr_acc 77.1% | val_loss 0.6129 | v
al_acc 77.5% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 20 | act_loss 0.5311
| dom_loss 1.2927 | lambda 0.762 | tr_acc 79.0% | val_loss 0.7585 | v
al_acc 66.2%
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 24 | act_loss 0.5203
| dom_loss 1.2947 | lambda 0.834 | tr_acc 79.5% | val_loss 0.6472 | v
al_acc 77.6% *
    [full_nc-gyro_only_s1:RealWorld-Waist] epoch 40 | act_loss 0.4905
| dom_loss 1.3121 | lambda 0.964 | tr_acc 81.2% | val_loss 0.9762 | v
al_acc 65.2%
    [full_nc-gyro_only_s1:RealWorld-Waist] early stop at epoch 44
    [full_nc-gyro_only_s1:RealWorld-Waist] src-val 77.62% | TARGET-TES
T 63.12% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept

```

```

pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_nc-gyro_only_s1:UCI] trainable params: 51,339
[full_nc-gyro_only_s1:UCI] epoch 1 | act_loss 0.8473 | dom_loss
1.3101 | lambda 0.050 | tr_acc 65.0% | val_loss 0.7994 | val_acc 69.
0% *
[full_nc-gyro_only_s1:UCI] epoch 3 | act_loss 0.6632 | dom_loss
1.3289 | lambda 0.149 | tr_acc 73.4% | val_loss 0.6720 | val_acc 76.
2% *
[full_nc-gyro_only_s1:UCI] epoch 17 | act_loss 0.5613 | dom_loss
1.3727 | lambda 0.691 | tr_acc 78.0% | val_loss 0.6680 | val_acc 77.
5% *
[full_nc-gyro_only_s1:UCI] epoch 20 | act_loss 0.5567 | dom_loss
1.3742 | lambda 0.762 | tr_acc 78.4% | val_loss 0.8780 | val_acc 62.
8%
[full_nc-gyro_only_s1:UCI] early stop at epoch 37
[full_nc-gyro_only_s1:UCI] src-val 77.49% | TARGET-TEST 74.64% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-gyro_only_s1:WISDM] trainable params: 51,339
[full_nc-gyro_only_s1:WISDM] epoch 1 | act_loss 0.8447 | dom_lo
s 1.2479 | lambda 0.050 | tr_acc 65.8% | val_loss 0.9022 | val_acc 6
2.8% *

```

```

[full_nc-gyro_only_s1:WISDM] epoch 4 | act_loss 0.6162 | dom_loss 1.3031 | lambda 0.197 | tr_acc 76.1% | val_loss 0.7075 | val_acc 69.3% *
[full_nc-gyro_only_s1:WISDM] epoch 5 | act_loss 0.6082 | dom_loss 1.3169 | lambda 0.245 | tr_acc 76.1% | val_loss 0.7798 | val_acc 71.9% *
[full_nc-gyro_only_s1:WISDM] epoch 6 | act_loss 0.5878 | dom_loss 1.3231 | lambda 0.291 | tr_acc 77.5% | val_loss 0.7819 | val_acc 73.2% *
[full_nc-gyro_only_s1:WISDM] epoch 7 | act_loss 0.5920 | dom_loss 1.3249 | lambda 0.336 | tr_acc 77.3% | val_loss 0.7239 | val_acc 74.7% *
[full_nc-gyro_only_s1:WISDM] epoch 8 | act_loss 0.5729 | dom_loss 1.3241 | lambda 0.380 | tr_acc 77.8% | val_loss 0.7840 | val_acc 75.2% *
[full_nc-gyro_only_s1:WISDM] epoch 9 | act_loss 0.5659 | dom_loss 1.3243 | lambda 0.422 | tr_acc 78.2% | val_loss 0.6807 | val_acc 75.8% *
[full_nc-gyro_only_s1:WISDM] epoch 16 | act_loss 0.5407 | dom_loss 1.3297 | lambda 0.664 | tr_acc 79.6% | val_loss 0.6831 | val_acc 76.7% *
[full_nc-gyro_only_s1:WISDM] epoch 20 | act_loss 0.5269 | dom_loss 1.3288 | lambda 0.762 | tr_acc 79.9% | val_loss 0.7535 | val_acc 72.4%
[full_nc-gyro_only_s1:WISDM] epoch 25 | act_loss 0.5158 | dom_loss 1.3321 | lambda 0.848 | tr_acc 80.7% | val_loss 0.6930 | val_acc 77.8% *
[full_nc-gyro_only_s1:WISDM] epoch 38 | act_loss 0.4928 | dom_loss 1.3362 | lambda 0.956 | tr_acc 81.4% | val_loss 0.7223 | val_acc 78.8% *
[full_nc-gyro_only_s1:WISDM] epoch 40 | act_loss 0.4884 | dom_loss 1.3353 | lambda 0.964 | tr_acc 81.7% | val_loss 1.2169 | val_acc 61.0%
[full_nc-gyro_only_s1:WISDM] early stop at epoch 58
[full_nc-gyro_only_s1:WISDM] src-val 78.82% | TARGET-TEST 58.56%
| saved ckpt + npz
[P24] train+save | arm=full_nc-gyro_only_s2 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=gyro_only aw=- | seed=2 |
targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiments/DAG
HAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/full_nc-gyro_only_s2
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 1):100%, (0, 2):100%, (1, 2):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features eac

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h)...
  built 5000/35396 graphs
  built 10000/35396 graphs
  built 15000/35396 graphs
  built 20000/35396 graphs
  built 25000/35396 graphs
  built 30000/35396 graphs
  built 35000/35396 graphs
  done: 35396 graphs built
[full_nc-gyro_only_s2:KuHar] trainable params: 51,339
[full_nc-gyro_only_s2:KuHar] epoch 1 | act_loss 0.8177 | dom_loss 1.3704 | lambda 0.050 | tr_acc 66.3% | val_loss 0.9600 | val_acc 63.4% *
[full_nc-gyro_only_s2:KuHar] epoch 2 | act_loss 0.6948 | dom_loss 1.3484 | lambda 0.100 | tr_acc 72.1% | val_loss 0.7492 | val_acc 68.7% *
[full_nc-gyro_only_s2:KuHar] epoch 8 | act_loss 0.5987 | dom_loss 1.4019 | lambda 0.380 | tr_acc 76.4% | val_loss 0.6741 | val_acc 75.9% *
[full_nc-gyro_only_s2:KuHar] epoch 16 | act_loss 0.5699 | dom_loss 1.4124 | lambda 0.664 | tr_acc 77.7% | val_loss 0.6562 | val_acc 76.3% *
[full_nc-gyro_only_s2:KuHar] epoch 20 | act_loss 0.5566 | dom_loss 1.4135 | lambda 0.762 | tr_acc 78.4% | val_loss 0.8475 | val_acc 67.7%
[full_nc-gyro_only_s2:KuHar] epoch 26 | act_loss 0.5441 | dom_loss 1.4172 | lambda 0.862 | tr_acc 79.0% | val_loss 0.6573 | val_acc 77.0% *
[full_nc-gyro_only_s2:KuHar] epoch 40 | act_loss 0.5208 | dom_loss 1.4232 | lambda 0.964 | tr_acc 80.0% | val_loss 0.9579 | val_acc 64.7%
[full_nc-gyro_only_s2:KuHar] early stop at epoch 46
[full_nc-gyro_only_s2:KuHar] src-val 77.03% | TARGET-TEST 58.33%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (0, 1):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (3 nodes, 6 edges, 90 features each)
h)...
  built 5000/33230 graphs
  built 10000/33230 graphs
  built 15000/33230 graphs
  built 20000/33230 graphs
  built 25000/33230 graphs
  built 30000/33230 graphs
  done: 33230 graphs built

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[full_nc-gyro_only_s2:MotionSense] trainable params: 51,339
[full_nc-gyro_only_s2:MotionSense] epoch 1 | act_loss 0.8347 | d
om_loss 1.2852 | lambda 0.050 | tr_acc 65.3% | val_loss 0.7665 | val_
acc 70.1% *
[full_nc-gyro_only_s2:MotionSense] epoch 7 | act_loss 0.6095 | d
om_loss 1.3167 | lambda 0.336 | tr_acc 75.6% | val_loss 0.7049 | val_
acc 71.4% *
[full_nc-gyro_only_s2:MotionSense] epoch 10 | act_loss 0.5882 | d
om_loss 1.3206 | lambda 0.462 | tr_acc 76.6% | val_loss 0.6702 | val_
acc 76.2% *
[full_nc-gyro_only_s2:MotionSense] epoch 20 | act_loss 0.5592 | d
om_loss 1.3270 | lambda 0.762 | tr_acc 77.9% | val_loss 0.8368 | val_
acc 73.5%
[full_nc-gyro_only_s2:MotionSense] early stop at epoch 30
[full_nc-gyro_only_s2:MotionSense] src-val 76.24% | TARGET-TEST 7
3.07% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_nc-gyro_only_s2:RealWorld-Thigh] trainable params: 51,339
[full_nc-gyro_only_s2:RealWorld-Thigh] epoch 1 | act_loss 0.7978
| dom_loss 1.2510 | lambda 0.050 | tr_acc 66.4% | val_loss 0.8945 | v
al_acc 59.3% *
[full_nc-gyro_only_s2:RealWorld-Thigh] epoch 2 | act_loss 0.6598
| dom_loss 1.1654 | lambda 0.100 | tr_acc 71.8% | val_loss 0.6431 | v
al_acc 73.4% *
[full_nc-gyro_only_s2:RealWorld-Thigh] epoch 7 | act_loss 0.5846
| dom_loss 1.2601 | lambda 0.336 | tr_acc 76.0% | val_loss 0.6738 | v
al_acc 76.3% *
[full_nc-gyro_only_s2:RealWorld-Thigh] epoch 12 | act_loss 0.5565
| dom_loss 1.2672 | lambda 0.537 | tr_acc 77.5% | val_loss 0.6582 | v
al_acc 77.9% *
[full_nc-gyro_only_s2:RealWorld-Thigh] epoch 20 | act_loss 0.5382
| dom_loss 1.2816 | lambda 0.762 | tr_acc 78.3% | val_loss 0.7324 | v
al_acc 71.2%
[full_nc-gyro_only_s2:RealWorld-Thigh] early stop at epoch 32
[full_nc-gyro_only_s2:RealWorld-Thigh] src-val 77.89% | TARGET-TES
T 66.56% | saved ckpt + npz

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[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (3 nodes, 6 edges, 90 features each)...)
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_nc-gyro_only_s2:RealWorld-Waist] trainable params: 51,339
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 1 | act_loss 0.8189 | dom_loss 1.3196 | lambda 0.050 | tr_acc 65.9% | val_loss 0.8440 | val_acc 59.7% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 2 | act_loss 0.6903 | dom_loss 1.2517 | lambda 0.100 | tr_acc 71.0% | val_loss 0.7633 | val_acc 69.3% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 4 | act_loss 0.6258 | dom_loss 1.2668 | lambda 0.197 | tr_acc 74.4% | val_loss 0.7135 | val_acc 71.9% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 9 | act_loss 0.5681 | dom_loss 1.2763 | lambda 0.422 | tr_acc 77.3% | val_loss 0.7468 | val_acc 73.2% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 10 | act_loss 0.5584 | dom_loss 1.2765 | lambda 0.462 | tr_acc 77.8% | val_loss 0.6312 | val_acc 77.1% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 20 | act_loss 0.5312 | dom_loss 1.2893 | lambda 0.762 | tr_acc 79.3% | val_loss 0.9960 | val_acc 67.9%
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 29 | act_loss 0.5101 | dom_loss 1.2955 | lambda 0.896 | tr_acc 80.2% | val_loss 0.6513 | val_acc 77.5% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 36 | act_loss 0.4957 | dom_loss 1.3000 | lambda 0.947 | tr_acc 80.8% | val_loss 0.6997 | val_acc 77.6% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 40 | act_loss 0.5004 | dom_loss 1.3089 | lambda 0.964 | tr_acc 80.6% | val_loss 0.7529 | val_acc 76.2%
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 44 | act_loss 0.4870 | dom_loss 1.3069 | lambda 0.976 | tr_acc 81.1% | val_loss 0.7369 | val_acc 78.4% *
[full_nc-gyro_only_s2:RealWorld-Waist] epoch 60 | act_loss 0.4531 | dom_loss 1.3155 | lambda 0.995 | tr_acc 83.0% | val_loss 0.7250 | val_acc 75.8%
[full_nc-gyro_only_s2:RealWorld-Waist] early stop at epoch 64

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[full_nc-gyro_only_s2:RealWorld-Waist] src-val 78.37% | TARGET-TEST 61.07% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 34368 sensor graphs (3 nodes, 6 edges, 90 features each)...)
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs
    built 30000/34368 graphs
    done: 34368 graphs built
[full_nc-gyro_only_s2:UCI] trainable params: 51,339
[full_nc-gyro_only_s2:UCI] epoch 1 | act_loss 0.8320 | dom_loss 1.3249 | lambda 0.050 | tr_acc 65.5% | val_loss 0.7564 | val_acc 68.3% *
[full_nc-gyro_only_s2:UCI] epoch 3 | act_loss 0.6567 | dom_loss 1.3191 | lambda 0.149 | tr_acc 73.7% | val_loss 0.7298 | val_acc 74.0% *
[full_nc-gyro_only_s2:UCI] epoch 13 | act_loss 0.5735 | dom_loss 1.3701 | lambda 0.572 | tr_acc 77.8% | val_loss 0.7042 | val_acc 74.8% *
[full_nc-gyro_only_s2:UCI] epoch 20 | act_loss 0.5559 | dom_loss 1.3757 | lambda 0.762 | tr_acc 78.2% | val_loss 0.7911 | val_acc 72.1%
[full_nc-gyro_only_s2:UCI] early stop at epoch 33
[full_nc-gyro_only_s2:UCI] src-val 74.80% | TARGET-TEST 73.91% | saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 28040 sensor graphs (3 nodes, 6 edges, 90 features each)...)

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built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-gyro_only_s2:WISDM] trainable params: 51,339
[full_nc-gyro_only_s2:WISDM] epoch 1 | act_loss 0.8255 | dom_loss 1.2841 | lambda 0.050 | tr_acc 66.8% | val_loss 1.0620 | val_acc 58.1% *
[full_nc-gyro_only_s2:WISDM] epoch 2 | act_loss 0.6781 | dom_loss 1.2611 | lambda 0.100 | tr_acc 73.3% | val_loss 0.7920 | val_acc 65.7% *
[full_nc-gyro_only_s2:WISDM] epoch 3 | act_loss 0.6339 | dom_loss 1.2909 | lambda 0.149 | tr_acc 75.3% | val_loss 0.8334 | val_acc 69.7% *
[full_nc-gyro_only_s2:WISDM] epoch 6 | act_loss 0.5938 | dom_loss 1.3266 | lambda 0.291 | tr_acc 76.9% | val_loss 0.7547 | val_acc 74.6% *
[full_nc-gyro_only_s2:WISDM] epoch 9 | act_loss 0.5746 | dom_loss 1.3256 | lambda 0.422 | tr_acc 77.8% | val_loss 0.6703 | val_acc 77.2% *
[full_nc-gyro_only_s2:WISDM] epoch 20 | act_loss 0.5256 | dom_loss 1.3293 | lambda 0.762 | tr_acc 80.0% | val_loss 1.5343 | val_acc 64.4%
[full_nc-gyro_only_s2:WISDM] epoch 29 | act_loss 0.5145 | dom_loss 1.3353 | lambda 0.896 | tr_acc 80.3% | val_loss 0.7186 | val_acc 77.3% *
[full_nc-gyro_only_s2:WISDM] epoch 40 | act_loss 0.4910 | dom_loss 1.3363 | lambda 0.964 | tr_acc 81.6% | val_loss 0.7123 | val_acc 74.4%
[full_nc-gyro_only_s2:WISDM] early stop at epoch 49
[full_nc-gyro_only_s2:WISDM] src-val 77.29% | TARGET-TEST 63.25%
| saved ckpt + npz
[P24] train+save | arm=full_nc-gyro_only_s3 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=gyro_only aw=- | seed=3 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiments/DAG HAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/full_nc-gyro_only_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 1):100%, (0, 2):100%, (1, 2):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features each)...)
built 5000/35396 graphs

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```

built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-gyro_only_s3:KuHar] trainable params: 51,339
[full_nc-gyro_only_s3:KuHar] epoch 1 | act_loss 0.8365 | dom_loss 1.3752 | lambda 0.050 | tr_acc 65.7% | val_loss 0.7940 | val_acc 72.2% *
[full_nc-gyro_only_s3:KuHar] epoch 9 | act_loss 0.5941 | dom_loss 1.4064 | lambda 0.422 | tr_acc 76.8% | val_loss 0.6861 | val_acc 73.5% *
[full_nc-gyro_only_s3:KuHar] epoch 10 | act_loss 0.5857 | dom_loss 1.4040 | lambda 0.462 | tr_acc 77.2% | val_loss 0.7201 | val_acc 74.9% *
[full_nc-gyro_only_s3:KuHar] epoch 12 | act_loss 0.5765 | dom_loss 1.4059 | lambda 0.537 | tr_acc 77.6% | val_loss 0.6798 | val_acc 75.8% *
[full_nc-gyro_only_s3:KuHar] epoch 17 | act_loss 0.5612 | dom_loss 1.4108 | lambda 0.691 | tr_acc 78.3% | val_loss 0.6705 | val_acc 77.4% *
[full_nc-gyro_only_s3:KuHar] epoch 20 | act_loss 0.5573 | dom_loss 1.4119 | lambda 0.762 | tr_acc 78.6% | val_loss 0.9925 | val_acc 60.7%
[full_nc-gyro_only_s3:KuHar] epoch 29 | act_loss 0.5386 | dom_loss 1.4163 | lambda 0.896 | tr_acc 79.1% | val_loss 0.6641 | val_acc 77.8% *
[full_nc-gyro_only_s3:KuHar] epoch 40 | act_loss 0.5205 | dom_loss 1.4182 | lambda 0.964 | tr_acc 80.2% | val_loss 0.6876 | val_acc 75.1%
[full_nc-gyro_only_s3:KuHar] epoch 48 | act_loss 0.5043 | dom_loss 1.4222 | lambda 0.984 | tr_acc 81.0% | val_loss 0.6018 | val_acc 80.0% *
[full_nc-gyro_only_s3:KuHar] epoch 60 | act_loss 0.4872 | dom_loss 1.4240 | lambda 0.995 | tr_acc 81.6% | val_loss 0.7971 | val_acc 71.7%
[full_nc-gyro_only_s3:KuHar] epoch 66 | act_loss 0.4716 | dom_loss 1.4228 | lambda 0.997 | tr_acc 82.4% | val_loss 0.5864 | val_acc 80.0% *
[full_nc-gyro_only_s3:KuHar] epoch 80 | act_loss 0.4470 | dom_loss 1.4214 | lambda 0.999 | tr_acc 83.5% | val_loss 0.6664 | val_acc 73.2%
[full_nc-gyro_only_s3:KuHar] epoch 81 | act_loss 0.4412 | dom_loss 1.4213 | lambda 0.999 | tr_acc 83.8% | val_loss 0.5534 | val_acc 81.3% *
[full_nc-gyro_only_s3:KuHar] epoch 100 | act_loss 0.4250 | dom_loss 1.4206 | lambda 1.000 | tr_acc 84.5% | val_loss 0.6103 | val_acc 78.4%
[full_nc-gyro_only_s3:KuHar] src-val 81.28% | TARGET-TEST 62.50%
| saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062

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[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (0, 1):100%, (1, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 33230 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
  [full_nc-gyro_only_s3:MotionSense] trainable params: 51,339
  [full_nc-gyro_only_s3:MotionSense] epoch 1 | act_loss 0.8559 | d
om_loss 1.3029 | lambda 0.050 | tr_acc 64.2% | val_loss 0.7638 | val_
acc 69.2% *
  [full_nc-gyro_only_s3:MotionSense] epoch 5 | act_loss 0.6446 | d
om_loss 1.3002 | lambda 0.245 | tr_acc 74.1% | val_loss 0.6886 | val_
acc 74.0% *
  [full_nc-gyro_only_s3:MotionSense] epoch 10 | act_loss 0.5992 | d
om_loss 1.3145 | lambda 0.462 | tr_acc 76.2% | val_loss 0.6476 | val_
acc 76.9% *
  [full_nc-gyro_only_s3:MotionSense] epoch 20 | act_loss 0.5660 | d
om_loss 1.3272 | lambda 0.762 | tr_acc 77.5% | val_loss 0.8767 | val_
acc 64.3%
  [full_nc-gyro_only_s3:MotionSense] early stop at epoch 30
  [full_nc-gyro_only_s3:MotionSense] src-val 76.90% | TARGET-TEST 7
1.19% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
  [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
  [full_nc-gyro_only_s3:RealWorld-Thigh] trainable params: 51,339

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[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 1 | act_loss 0.8175
| dom_loss 1.2690 | lambda 0.050 | tr_acc 65.4% | val_loss 0.7862 | v
al_acc 64.5% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 2 | act_loss 0.6735
| dom_loss 1.1906 | lambda 0.100 | tr_acc 71.5% | val_loss 0.7167 | v
al_acc 71.4% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 4 | act_loss 0.6210
| dom_loss 1.2349 | lambda 0.197 | tr_acc 74.2% | val_loss 0.6521 | v
al_acc 76.0% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 7 | act_loss 0.5797
| dom_loss 1.2581 | lambda 0.336 | tr_acc 76.2% | val_loss 0.6703 | v
al_acc 76.1% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 20 | act_loss 0.5414
| dom_loss 1.2844 | lambda 0.762 | tr_acc 78.1% | val_loss 0.7710 | v
al_acc 68.5%
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 21 | act_loss 0.5369
| dom_loss 1.2870 | lambda 0.782 | tr_acc 78.2% | val_loss 0.6687 | v
al_acc 76.5% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 23 | act_loss 0.5335
| dom_loss 1.2870 | lambda 0.818 | tr_acc 78.3% | val_loss 0.6263 | v
al_acc 78.0% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 31 | act_loss 0.5273
| dom_loss 1.3002 | lambda 0.914 | tr_acc 78.8% | val_loss 0.6658 | v
al_acc 78.5% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 40 | act_loss 0.5178
| dom_loss 1.3119 | lambda 0.964 | tr_acc 79.2% | val_loss 0.5559 | v
al_acc 78.1%
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 42 | act_loss 0.5110
| dom_loss 1.3122 | lambda 0.970 | tr_acc 79.7% | val_loss 0.6922 | v
al_acc 78.6% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 52 | act_loss 0.4910
| dom_loss 1.3170 | lambda 0.989 | tr_acc 80.7% | val_loss 0.6712 | v
al_acc 81.1% *
[full_nc-gyro_only_s3:RealWorld-Thigh] epoch 60 | act_loss 0.4774
| dom_loss 1.3144 | lambda 0.995 | tr_acc 81.3% | val_loss 0.9172 | v
al_acc 66.0%
[full_nc-gyro_only_s3:RealWorld-Thigh] early stop at epoch 72
[full_nc-gyro_only_s3:RealWorld-Thigh] src-val 81.10% | TARGET-TES
T 61.01% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs

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    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_nc-gyro_only_s3:RealWorld-Waist] trainable params: 51,339
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 1 | act_loss 0.8422
| dom_loss 1.3242 | lambda 0.050 | tr_acc 64.5% | val_loss 0.8283 | v
al_acc 62.7% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 2 | act_loss 0.6905
| dom_loss 1.2527 | lambda 0.100 | tr_acc 71.2% | val_loss 0.7047 | v
al_acc 72.8% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 5 | act_loss 0.6174
| dom_loss 1.2746 | lambda 0.245 | tr_acc 74.7% | val_loss 0.6792 | v
al_acc 74.3% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 11 | act_loss 0.5639
| dom_loss 1.2826 | lambda 0.501 | tr_acc 77.4% | val_loss 0.6598 | v
al_acc 76.1% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 20 | act_loss 0.5312
| dom_loss 1.2885 | lambda 0.762 | tr_acc 79.0% | val_loss 0.6657 | v
al_acc 76.9% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 24 | act_loss 0.5281
| dom_loss 1.2963 | lambda 0.834 | tr_acc 79.2% | val_loss 0.6098 | v
al_acc 78.1% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 28 | act_loss 0.5167
| dom_loss 1.2971 | lambda 0.885 | tr_acc 79.8% | val_loss 0.5978 | v
al_acc 78.4% *
[full_nc-gyro_only_s3:RealWorld-Waist] epoch 40 | act_loss 0.4976
| dom_loss 1.3072 | lambda 0.964 | tr_acc 80.9% | val_loss 0.6322 | v
al_acc 75.9%
[full_nc-gyro_only_s3:RealWorld-Waist] early stop at epoch 48
[full_nc-gyro_only_s3:RealWorld-Waist] src-val 78.37% | TARGET-TES
T 60.26% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs
    built 30000/34368 graphs
    done: 34368 graphs built
[full_nc-gyro_only_s3:UCI] trainable params: 51,339
[full_nc-gyro_only_s3:UCI] epoch 1 | act_loss 0.8427 | dom_loss

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1.3258 | lambda 0.050 | tr_acc 64.9% | val_loss 0.8783 | val_acc 62.5% *
[full_nc-gyro_only_s3:UCI] epoch 2 | act_loss 0.7095 | dom_loss
1.2656 | lambda 0.100 | tr_acc 71.1% | val_loss 0.7068 | val_acc 72.5% *
[full_nc-gyro_only_s3:UCI] epoch 3 | act_loss 0.6745 | dom_loss
1.3134 | lambda 0.149 | tr_acc 72.8% | val_loss 0.7412 | val_acc 72.6% *
[full_nc-gyro_only_s3:UCI] epoch 6 | act_loss 0.6187 | dom_loss
1.3583 | lambda 0.291 | tr_acc 75.8% | val_loss 0.7112 | val_acc 74.9% *
[full_nc-gyro_only_s3:UCI] epoch 9 | act_loss 0.5963 | dom_loss
1.3672 | lambda 0.422 | tr_acc 76.6% | val_loss 0.6814 | val_acc 76.0% *
[full_nc-gyro_only_s3:UCI] epoch 20 | act_loss 0.5613 | dom_loss
1.3772 | lambda 0.762 | tr_acc 78.1% | val_loss 1.0814 | val_acc 61.7%
[full_nc-gyro_only_s3:UCI] early stop at epoch 29
[full_nc-gyro_only_s3:UCI] src-val 76.02% | TARGET-TEST 75.07% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (3 nodes, 6 edges, 90 features each)
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-gyro_only_s3:WISDM] trainable params: 51,339
[full_nc-gyro_only_s3:WISDM] epoch 1 | act_loss 0.8536 | dom_loss 1.2628 | lambda 0.050 | tr_acc 64.7% | val_loss 0.9003 | val_acc 66.2% *
[full_nc-gyro_only_s3:WISDM] epoch 2 | act_loss 0.6893 | dom_loss 1.2364 | lambda 0.100 | tr_acc 72.5% | val_loss 0.7103 | val_acc 68.8% *
[full_nc-gyro_only_s3:WISDM] epoch 3 | act_loss 0.6443 | dom_loss 1.2855 | lambda 0.149 | tr_acc 74.6% | val_loss 0.7342 | val_acc 73.1% *
[full_nc-gyro_only_s3:WISDM] epoch 6 | act_loss 0.5971 | dom_loss 1.3159 | lambda 0.291 | tr_acc 76.8% | val_loss 0.6852 | val_acc 73.3% *
[full_nc-gyro_only_s3:WISDM] epoch 7 | act_loss 0.5846 | dom_loss 1.3210 | lambda 0.336 | tr_acc 77.5% | val_loss 0.6744 | val_acc 7

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5.7% *
[full_nc-gyro_only_s3:WISDM] epoch 10 | act_loss 0.5653 | dom_loss 1.3210 | lambda 0.462 | tr_acc 78.3% | val_loss 0.6493 | val_acc 78.7% *
[full_nc-gyro_only_s3:WISDM] epoch 20 | act_loss 0.5307 | dom_loss 1.3287 | lambda 0.762 | tr_acc 79.7% | val_loss 0.7226 | val_acc 74.2%
[full_nc-gyro_only_s3:WISDM] early stop at epoch 30
[full_nc-gyro_only_s3:WISDM] src-val 78.69% | TARGET-TEST 71.20%
| saved ckpt + npz
[P24] train+save | arm=full_nc-gyro_only_s5 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=gyro_only aw=- | seed=5 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiments/DAG HAR/P20_GraphRevival_DG/P24_S0_Compression/results/checkpoints/full_nc-gyro_only_s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 1):100%, (0, 2):100%, (1, 2):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (1, 2):100%, (0, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (3 nodes, 6 edges, 90 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-gyro_only_s5:KuHar] trainable params: 51,339
[full_nc-gyro_only_s5:KuHar] epoch 1 | act_loss 0.8348 | dom_loss 1.3637 | lambda 0.050 | tr_acc 65.6% | val_loss 0.8951 | val_acc 67.2% *
[full_nc-gyro_only_s5:KuHar] epoch 3 | act_loss 0.6683 | dom_loss 1.3809 | lambda 0.149 | tr_acc 73.2% | val_loss 0.7216 | val_acc 72.4% *
[full_nc-gyro_only_s5:KuHar] epoch 8 | act_loss 0.6066 | dom_loss 1.4090 | lambda 0.380 | tr_acc 75.9% | val_loss 0.6360 | val_acc 77.1% *
[full_nc-gyro_only_s5:KuHar] epoch 20 | act_loss 0.5597 | dom_loss 1.4156 | lambda 0.762 | tr_acc 78.2% | val_loss 0.8093 | val_acc 72.2%
[full_nc-gyro_only_s5:KuHar] early stop at epoch 28
[full_nc-gyro_only_s5:KuHar] src-val 77.14% | TARGET-TEST 66.67%
| saved ckpt + npz
[LOD0] target = MotionSense

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[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (0, 1):100%, (1, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 33230 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
[full_nc-gyro_only_s5:MotionSense] trainable params: 51,339
[full_nc-gyro_only_s5:MotionSense] epoch 1 | act_loss 0.8586 | d
om_loss 1.2801 | lambda 0.050 | tr_acc 63.9% | val_loss 0.7582 | val_
acc 66.3% *
[full_nc-gyro_only_s5:MotionSense] epoch 2 | act_loss 0.7216 | d
om_loss 1.2482 | lambda 0.100 | tr_acc 70.1% | val_loss 0.7216 | val_
acc 72.8% *
[full_nc-gyro_only_s5:MotionSense] epoch 4 | act_loss 0.6568 | d
om_loss 1.2831 | lambda 0.197 | tr_acc 73.5% | val_loss 0.6788 | val_
acc 75.3% *
[full_nc-gyro_only_s5:MotionSense] epoch 20 | act_loss 0.5601 | d
om_loss 1.3244 | lambda 0.762 | tr_acc 77.7% | val_loss 1.0131 | val_
acc 62.6%
[full_nc-gyro_only_s5:MotionSense] early stop at epoch 24
[full_nc-gyro_only_s5:MotionSense] src-val 75.33% | TARGET-TEST 7
2.79% | saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 1):100%, (0, 2):100%, (1, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26450 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs

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    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
[full_nc-gyro_only_s5:RealWorld-Thigh] trainable params: 51,339
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 1 | act_loss 0.7994
| dom_loss 1.2636 | lambda 0.050 | tr_acc 66.2% | val_loss 0.8704 | v
al_acc 65.6% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 2 | act_loss 0.6715
| dom_loss 1.1767 | lambda 0.100 | tr_acc 71.4% | val_loss 0.7320 | v
al_acc 68.8% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 6 | act_loss 0.5943
| dom_loss 1.2568 | lambda 0.291 | tr_acc 75.8% | val_loss 0.6899 | v
al_acc 73.7% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 7 | act_loss 0.5882
| dom_loss 1.2614 | lambda 0.336 | tr_acc 75.8% | val_loss 0.6049 | v
al_acc 75.6% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 19 | act_loss 0.5441
| dom_loss 1.2833 | lambda 0.740 | tr_acc 77.6% | val_loss 0.6415 | v
al_acc 77.2% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 20 | act_loss 0.5424
| dom_loss 1.2837 | lambda 0.762 | tr_acc 78.1% | val_loss 0.7791 | v
al_acc 72.4%
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 27 | act_loss 0.5247
| dom_loss 1.2927 | lambda 0.874 | tr_acc 78.8% | val_loss 0.6474 | v
al_acc 77.5% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 33 | act_loss 0.5213
| dom_loss 1.3025 | lambda 0.929 | tr_acc 79.2% | val_loss 0.5976 | v
al_acc 79.8% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 34 | act_loss 0.5149
| dom_loss 1.3024 | lambda 0.935 | tr_acc 79.1% | val_loss 0.6309 | v
al_acc 80.2% *
[full_nc-gyro_only_s5:RealWorld-Thigh] epoch 40 | act_loss 0.5105
| dom_loss 1.3136 | lambda 0.964 | tr_acc 79.4% | val_loss 0.6173 | v
al_acc 78.6%
[full_nc-gyro_only_s5:RealWorld-Thigh] early stop at epoch 54
[full_nc-gyro_only_s5:RealWorld-Thigh] src-val 80.15% | TARGET-TEST
65.87% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs

```

```

    built 25000/26456 graphs
    done: 26456 graphs built
    [full_nc-gyro_only_s5:RealWorld-Waist] trainable params: 51,339
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 1 | act_loss 0.8371
    | dom_loss 1.3255 | lambda 0.050 | tr_acc 64.9% | val_loss 0.7380 | v
    al_acc 64.5% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 2 | act_loss 0.7100
    | dom_loss 1.2653 | lambda 0.100 | tr_acc 70.6% | val_loss 0.7815 | v
    al_acc 67.3% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 5 | act_loss 0.6142
    | dom_loss 1.2619 | lambda 0.245 | tr_acc 75.2% | val_loss 0.6576 | v
    al_acc 75.3% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 13 | act_loss 0.5490
    | dom_loss 1.2857 | lambda 0.572 | tr_acc 78.0% | val_loss 0.6777 | v
    al_acc 75.8% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 17 | act_loss 0.5349
    | dom_loss 1.2855 | lambda 0.691 | tr_acc 78.9% | val_loss 0.6462 | v
    al_acc 75.9% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 19 | act_loss 0.5322
    | dom_loss 1.2913 | lambda 0.740 | tr_acc 78.8% | val_loss 0.6431 | v
    al_acc 76.4% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 20 | act_loss 0.5307
    | dom_loss 1.2936 | lambda 0.762 | tr_acc 79.0% | val_loss 0.6749 | v
    al_acc 76.0%
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 22 | act_loss 0.5294
    | dom_loss 1.2904 | lambda 0.800 | tr_acc 79.0% | val_loss 0.6563 | v
    al_acc 76.8% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 33 | act_loss 0.5056
    | dom_loss 1.2987 | lambda 0.929 | tr_acc 80.2% | val_loss 0.6610 | v
    al_acc 77.6% *
    [full_nc-gyro_only_s5:RealWorld-Waist] epoch 40 | act_loss 0.4942
    | dom_loss 1.3024 | lambda 0.964 | tr_acc 80.9% | val_loss 0.7398 | v
    al_acc 74.4%
    [full_nc-gyro_only_s5:RealWorld-Waist] early stop at epoch 53
    [full_nc-gyro_only_s5:RealWorld-Waist] src-val 77.57% | TARGET-TEST
    60.57% | saved ckpt + npz
    [LOD0] target = UCI
    [LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-
    d-Waist', 'WISDM']
    [LOD0] feature_mode = hybrid normalize_mode = per_channel
    [LOD0] samples: train=34368 val=5504 test=690
    [LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
    [LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
    pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
    irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
    [pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
    airs: (1, 2):100%, (0, 1):100%, (0, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 34368 sensor graphs (3 nodes, 6 edges, 90 features eac
    h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs

```

```

built 30000/34368 graphs
done: 34368 graphs built
[full_nc-gyro_only_s5:UCI] trainable params: 51,339
[full_nc-gyro_only_s5:UCI] epoch 1 | act_loss 0.8489 | dom_loss
1.3423 | lambda 0.050 | tr_acc 64.8% | val_loss 0.7310 | val_acc 65.
9% *
[full_nc-gyro_only_s5:UCI] epoch 5 | act_loss 0.6366 | dom_loss
1.3523 | lambda 0.245 | tr_acc 74.8% | val_loss 0.7100 | val_acc 73.
5% *
[full_nc-gyro_only_s5:UCI] epoch 7 | act_loss 0.6161 | dom_loss
1.3592 | lambda 0.336 | tr_acc 75.7% | val_loss 0.6972 | val_acc 74.
6% *
[full_nc-gyro_only_s5:UCI] epoch 14 | act_loss 0.5773 | dom_loss
1.3714 | lambda 0.604 | tr_acc 77.5% | val_loss 0.6686 | val_acc 77.
0% *
[full_nc-gyro_only_s5:UCI] epoch 20 | act_loss 0.5652 | dom_loss
1.3768 | lambda 0.762 | tr_acc 78.0% | val_loss 0.7917 | val_acc 66.
0%
[full_nc-gyro_only_s5:UCI] epoch 21 | act_loss 0.5560 | dom_loss
1.3761 | lambda 0.782 | tr_acc 78.3% | val_loss 0.6582 | val_acc 78.
2% *
[full_nc-gyro_only_s5:UCI] epoch 40 | act_loss 0.5278 | dom_loss
1.3870 | lambda 0.964 | tr_acc 79.7% | val_loss 0.7488 | val_acc 74.
1%
[full_nc-gyro_only_s5:UCI] early stop at epoch 41
[full_nc-gyro_only_s5:UCI] src-val 78.16% | TARGET-TEST 71.16% |
saved ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [3, 4, 5] -> 3 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=3 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 3.0 min 3 max 3 | most-kept
pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 3.0 min 3 max 3 | most-kept pa
irs: (0, 2):100%, (1, 2):100%, (0, 1):100%
[pearson test] edges/window: mean 3.0 min 3 max 3 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (3 nodes, 6 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-gyro_only_s5:WISDM] trainable params: 51,339
[full_nc-gyro_only_s5:WISDM] epoch 1 | act_loss 0.8451 | dom_loss
1.2294 | lambda 0.050 | tr_acc 65.5% | val_loss 0.7644 | val_acc 7
1.2% *
[full_nc-gyro_only_s5:WISDM] epoch 3 | act_loss 0.6525 | dom_loss
1.2894 | lambda 0.149 | tr_acc 74.4% | val_loss 0.7341 | val_acc 7
3.0% *
[full_nc-gyro_only_s5:WISDM] epoch 5 | act_loss 0.6159 | dom_loss

```

```
s 1.3118 | lambda 0.245 | tr_acc 76.0% | val_loss 0.7040 | val_acc 74.7% *  
    [full_nc-gyro_only_s5:WISDM] epoch 9 | act_loss 0.5742 | dom_loss 1.3253 | lambda 0.422 | tr_acc 78.0% | val_loss 0.6931 | val_acc 74.7% *  
    [full_nc-gyro_only_s5:WISDM] epoch 15 | act_loss 0.5529 | dom_loss 1.3295 | lambda 0.635 | tr_acc 78.9% | val_loss 0.6730 | val_acc 76.8% *  
    [full_nc-gyro_only_s5:WISDM] epoch 20 | act_loss 0.5359 | dom_loss 1.3301 | lambda 0.762 | tr_acc 79.4% | val_loss 0.8017 | val_acc 73.1%  
    [full_nc-gyro_only_s5:WISDM] epoch 29 | act_loss 0.5191 | dom_loss 1.3328 | lambda 0.896 | tr_acc 80.3% | val_loss 0.6842 | val_acc 77.3% *  
    [full_nc-gyro_only_s5:WISDM] epoch 40 | act_loss 0.4874 | dom_loss 1.3365 | lambda 0.964 | tr_acc 81.7% | val_loss 0.8549 | val_acc 71.4%  
    [full_nc-gyro_only_s5:WISDM] early stop at epoch 49  
    [full_nc-gyro_only_s5:WISDM] src-val 77.27% | TARGET-TEST 66.13%  
| saved ckpt + npz
```

4c. aw40 (Adaptive Windowing → 40 timesteps/window)

```
In [ ]: for s in (1, 2, 3, 5):  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant full  
        --aw 40 --seed {s} --epochs 100 --log_every 20
```

```

[P24] train+save | arm=full_aw40_s1 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=40 | seed=1 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw40_s1
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 1):100%, (0, 2):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 4):100%, (3, 4):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (1, 4):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
    building 35396 sensor graphs (6 nodes, 18 edges, 60 features each)...)
    built 5000/35396 graphs
    built 10000/35396 graphs
    built 15000/35396 graphs
    built 20000/35396 graphs
    built 25000/35396 graphs
    built 30000/35396 graphs
    built 35000/35396 graphs
    done: 35396 graphs built
[full_aw40_s1:KuHar] trainable params: 51,531
[full_aw40_s1:KuHar] epoch 1 | act_loss 0.7766 | dom_loss 1.4058
| lambda 0.050 | tr_acc 67.2% | val_loss 0.6632 | val_acc 74.8% *
[full_aw40_s1:KuHar] epoch 3 | act_loss 0.5685 | dom_loss 1.3497
| lambda 0.149 | tr_acc 77.0% | val_loss 0.6676 | val_acc 75.7% *
[full_aw40_s1:KuHar] epoch 5 | act_loss 0.5343 | dom_loss 1.3690
| lambda 0.245 | tr_acc 78.7% | val_loss 0.6379 | val_acc 76.8% *
[full_aw40_s1:KuHar] epoch 8 | act_loss 0.5077 | dom_loss 1.3868
| lambda 0.380 | tr_acc 79.7% | val_loss 0.6682 | val_acc 77.6% *
[full_aw40_s1:KuHar] epoch 11 | act_loss 0.4905 | dom_loss 1.3917
| lambda 0.501 | tr_acc 80.3% | val_loss 0.6130 | val_acc 79.4% *
[full_aw40_s1:KuHar] epoch 17 | act_loss 0.4708 | dom_loss 1.4013
| lambda 0.691 | tr_acc 81.7% | val_loss 0.5693 | val_acc 81.0% *
[full_aw40_s1:KuHar] epoch 20 | act_loss 0.4649 | dom_loss 1.4035
| lambda 0.762 | tr_acc 82.2% | val_loss 0.5564 | val_acc 82.2% *
[full_aw40_s1:KuHar] epoch 26 | act_loss 0.4578 | dom_loss 1.4083
| lambda 0.862 | tr_acc 82.7% | val_loss 0.5237 | val_acc 83.7% *
[full_aw40_s1:KuHar] epoch 40 | act_loss 0.4409 | dom_loss 1.4226
| lambda 0.964 | tr_acc 83.6% | val_loss 0.5931 | val_acc 77.6%
[full_aw40_s1:KuHar] early stop at epoch 46
[full_aw40_s1:KuHar] src-val 83.67% | TARGET-TEST 62.50% | saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)

```

```

[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (1, 4):100%
  feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
  building 33230 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
  [full_aw40_s1:MotionSense] trainable params: 51,531
  [full_aw40_s1:MotionSense] epoch 1 | act_loss 0.8043 | dom_loss
1.3263 | lambda 0.050 | tr_acc 65.5% | val_loss 0.8393 | val_acc 66.
3% *
    [full_aw40_s1:MotionSense] epoch 2 | act_loss 0.6343 | dom_loss
1.2558 | lambda 0.100 | tr_acc 73.8% | val_loss 0.8202 | val_acc 67.
7% *
    [full_aw40_s1:MotionSense] epoch 3 | act_loss 0.5864 | dom_loss
1.2512 | lambda 0.149 | tr_acc 76.0% | val_loss 0.7446 | val_acc 72.
0% *
    [full_aw40_s1:MotionSense] epoch 6 | act_loss 0.5348 | dom_loss
1.2853 | lambda 0.291 | tr_acc 78.1% | val_loss 0.6801 | val_acc 76.
1% *
    [full_aw40_s1:MotionSense] epoch 9 | act_loss 0.5118 | dom_loss
1.3006 | lambda 0.422 | tr_acc 79.6% | val_loss 0.6635 | val_acc 76.
3% *
    [full_aw40_s1:MotionSense] epoch 12 | act_loss 0.4960 | dom_loss
1.3074 | lambda 0.537 | tr_acc 79.8% | val_loss 0.6110 | val_acc 77.
7% *
    [full_aw40_s1:MotionSense] epoch 15 | act_loss 0.4891 | dom_loss
1.3135 | lambda 0.635 | tr_acc 80.4% | val_loss 0.6056 | val_acc 80.
3% *
    [full_aw40_s1:MotionSense] epoch 20 | act_loss 0.4780 | dom_loss
1.3161 | lambda 0.762 | tr_acc 81.0% | val_loss 0.6490 | val_acc 76.
0%
    [full_aw40_s1:MotionSense] early stop at epoch 35
  [full_aw40_s1:MotionSense] src-val 80.29% | TARGET-TEST 75.80% |
saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p

```

```

airs: (3, 4):100%, (0, 1):100%, (0, 2):100%, (4, 5):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 60 features each)...)
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_aw40_s1:RealWorld-Thigh] trainable params: 51,531
[full_aw40_s1:RealWorld-Thigh] epoch 1 | act_loss 0.7810 | dom_loss 1.2783 | lambda 0.050 | tr_acc 66.2% | val_loss 0.7700 | val_acc 68.4% *
[full_aw40_s1:RealWorld-Thigh] epoch 2 | act_loss 0.5925 | dom_loss 1.1853 | lambda 0.100 | tr_acc 75.1% | val_loss 0.6933 | val_acc 73.3% *
[full_aw40_s1:RealWorld-Thigh] epoch 3 | act_loss 0.5530 | dom_loss 1.1871 | lambda 0.149 | tr_acc 77.1% | val_loss 0.6429 | val_acc 75.7% *
[full_aw40_s1:RealWorld-Thigh] epoch 7 | act_loss 0.4971 | dom_loss 1.2399 | lambda 0.336 | tr_acc 79.5% | val_loss 0.6106 | val_acc 78.1% *
[full_aw40_s1:RealWorld-Thigh] epoch 8 | act_loss 0.4899 | dom_loss 1.2443 | lambda 0.380 | tr_acc 80.0% | val_loss 0.5616 | val_acc 79.6% *
[full_aw40_s1:RealWorld-Thigh] epoch 14 | act_loss 0.4514 | dom_loss 1.2628 | lambda 0.604 | tr_acc 81.6% | val_loss 0.5598 | val_acc 80.5% *
[full_aw40_s1:RealWorld-Thigh] epoch 16 | act_loss 0.4443 | dom_loss 1.2654 | lambda 0.664 | tr_acc 81.9% | val_loss 0.5570 | val_acc 80.7% *
[full_aw40_s1:RealWorld-Thigh] epoch 19 | act_loss 0.4409 | dom_loss 1.2688 | lambda 0.740 | tr_acc 82.2% | val_loss 0.5415 | val_acc 81.1% *
[full_aw40_s1:RealWorld-Thigh] epoch 20 | act_loss 0.4322 | dom_loss 1.2690 | lambda 0.762 | tr_acc 82.4% | val_loss 0.7913 | val_acc 74.8%
[full_aw40_s1:RealWorld-Thigh] epoch 22 | act_loss 0.4332 | dom_loss 1.2713 | lambda 0.800 | tr_acc 82.5% | val_loss 0.5762 | val_acc 81.4% *
[full_aw40_s1:RealWorld-Thigh] epoch 25 | act_loss 0.4334 | dom_loss 1.2745 | lambda 0.848 | tr_acc 82.5% | val_loss 0.5284 | val_acc 82.8% *
[full_aw40_s1:RealWorld-Thigh] epoch 36 | act_loss 0.4283 | dom_loss 1.3096 | lambda 0.947 | tr_acc 83.0% | val_loss 0.5621 | val_acc 82.9% *
[full_aw40_s1:RealWorld-Thigh] epoch 40 | act_loss 0.4256 | dom_loss 1.3125 | lambda 0.964 | tr_acc 83.4% | val_loss 0.5238 | val_acc 84.1% *
[full_aw40_s1:RealWorld-Thigh] epoch 44 | act_loss 0.4172 | dom_loss 1.3084 | lambda 0.976 | tr_acc 83.7% | val_loss 0.4795 | val_acc 84.2% *
[full_aw40_s1:RealWorld-Thigh] epoch 60 | act_loss 0.3861 | dom_loss 1.3096 | lambda 0.995 | tr_acc 85.0% | val_loss 0.5819 | val_acc 78.9%
[full_aw40_s1:RealWorld-Thigh] early stop at epoch 64

```

```

[full_aw40_s1:RealWorld-Thigh] src-val 84.24% | TARGET-TEST 71.01%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (0, 1):100%, (3, 5):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_aw40_s1:RealWorld-Waist] trainable params: 51,531
[full_aw40_s1:RealWorld-Waist] epoch 1 | act_loss 0.7885 | dom_l
oss 1.3276 | lambda 0.050 | tr_acc 66.0% | val_loss 0.7592 | val_acc
70.6% *
[full_aw40_s1:RealWorld-Waist] epoch 3 | act_loss 0.5788 | dom_l
oss 1.2549 | lambda 0.149 | tr_acc 75.9% | val_loss 0.6125 | val_acc
76.2% *
[full_aw40_s1:RealWorld-Waist] epoch 9 | act_loss 0.4904 | dom_l
oss 1.2607 | lambda 0.422 | tr_acc 79.8% | val_loss 0.5748 | val_acc
78.9% *
[full_aw40_s1:RealWorld-Waist] epoch 20 | act_loss 0.4489 | dom_l
oss 1.2738 | lambda 0.762 | tr_acc 81.9% | val_loss 0.6668 | val_acc
70.4%
[full_aw40_s1:RealWorld-Waist] early stop at epoch 29
[full_aw40_s1:RealWorld-Waist] src-val 78.95% | TARGET-TEST 73.65%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/34368 graphs

```

```

built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw40_s1:UCI] trainable params: 51,531
[full_aw40_s1:UCI] epoch 1 | act_loss 0.7812 | dom_loss 1.3766 |
lambda 0.050 | tr_acc 66.8% | val_loss 0.7759 | val_acc 71.1% *
[full_aw40_s1:UCI] epoch 5 | act_loss 0.5445 | dom_loss 1.3347 |
lambda 0.245 | tr_acc 78.1% | val_loss 0.7000 | val_acc 75.0% *
[full_aw40_s1:UCI] epoch 7 | act_loss 0.5206 | dom_loss 1.3434 |
lambda 0.336 | tr_acc 79.0% | val_loss 0.7383 | val_acc 75.4% *
[full_aw40_s1:UCI] epoch 8 | act_loss 0.5183 | dom_loss 1.3466 |
lambda 0.380 | tr_acc 79.7% | val_loss 0.6580 | val_acc 77.1% *
[full_aw40_s1:UCI] epoch 10 | act_loss 0.5026 | dom_loss 1.3527 |
lambda 0.462 | tr_acc 80.2% | val_loss 0.6415 | val_acc 78.5% *
[full_aw40_s1:UCI] epoch 18 | act_loss 0.4807 | dom_loss 1.3642 |
lambda 0.716 | tr_acc 81.2% | val_loss 0.5544 | val_acc 81.4% *
[full_aw40_s1:UCI] epoch 20 | act_loss 0.4772 | dom_loss 1.3655 |
lambda 0.762 | tr_acc 81.3% | val_loss 0.5954 | val_acc 80.6%
[full_aw40_s1:UCI] epoch 32 | act_loss 0.4589 | dom_loss 1.3759 |
lambda 0.922 | tr_acc 82.5% | val_loss 0.5431 | val_acc 82.3% *
[full_aw40_s1:UCI] epoch 40 | act_loss 0.4502 | dom_loss 1.3852 |
lambda 0.964 | tr_acc 83.1% | val_loss 0.6388 | val_acc 77.7%
[full_aw40_s1:UCI] early stop at epoch 52
[full_aw40_s1:UCI] src-val 82.34% | TARGET-TEST 86.23% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (4, 5):100%, (0, 1):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw40_s1:WISDM] trainable params: 51,531
[full_aw40_s1:WISDM] epoch 1 | act_loss 0.8031 | dom_loss 1.3310
| lambda 0.050 | tr_acc 66.6% | val_loss 0.9337 | val_acc 61.2% *
[full_aw40_s1:WISDM] epoch 2 | act_loss 0.6057 | dom_loss 1.2793
| lambda 0.100 | tr_acc 75.8% | val_loss 0.7606 | val_acc 71.1% *
[full_aw40_s1:WISDM] epoch 4 | act_loss 0.5430 | dom_loss 1.3078
| lambda 0.197 | tr_acc 78.7% | val_loss 0.7788 | val_acc 71.7% *

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[full_aw40_s1:WISDM] epoch 5 | act_loss 0.5207 | dom_loss 1.3125
| lambda 0.245 | tr_acc 79.7% | val_loss 0.6520 | val_acc 73.7% *
[full_aw40_s1:WISDM] epoch 7 | act_loss 0.4988 | dom_loss 1.3198
| lambda 0.336 | tr_acc 80.7% | val_loss 0.7144 | val_acc 74.4% *
[full_aw40_s1:WISDM] epoch 9 | act_loss 0.4836 | dom_loss 1.3257
| lambda 0.422 | tr_acc 81.1% | val_loss 0.5999 | val_acc 79.0% *
[full_aw40_s1:WISDM] epoch 12 | act_loss 0.4673 | dom_loss 1.3278
| lambda 0.537 | tr_acc 82.1% | val_loss 0.6016 | val_acc 81.5% *
[full_aw40_s1:WISDM] epoch 20 | act_loss 0.4380 | dom_loss 1.3319
| lambda 0.762 | tr_acc 83.7% | val_loss 0.7265 | val_acc 76.5%
[full_aw40_s1:WISDM] early stop at epoch 32
[full_aw40_s1:WISDM] src-val 81.49% | TARGET-TEST 72.20% | saved
ckpt + npz
[P24] train+save | arm=full_aw40_s2 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=40 | seed=2 | targets=
['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI',
'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20
_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw40_s2
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 4):100%, (3, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (1, 2):100%, (4, 5):100%, (1, 4):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 60 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_aw40_s2:KuHar] trainable params: 51,531
[full_aw40_s2:KuHar] epoch 1 | act_loss 0.7635 | dom_loss 1.4245
| lambda 0.050 | tr_acc 68.2% | val_loss 0.7752 | val_acc 69.5% *
[full_aw40_s2:KuHar] epoch 2 | act_loss 0.6092 | dom_loss 1.3580
| lambda 0.100 | tr_acc 75.0% | val_loss 0.6725 | val_acc 70.3% *
[full_aw40_s2:KuHar] epoch 3 | act_loss 0.5692 | dom_loss 1.3530
| lambda 0.149 | tr_acc 77.0% | val_loss 0.6959 | val_acc 75.4% *
[full_aw40_s2:KuHar] epoch 10 | act_loss 0.4971 | dom_loss 1.3893
| lambda 0.462 | tr_acc 80.4% | val_loss 0.6708 | val_acc 77.0% *
[full_aw40_s2:KuHar] epoch 16 | act_loss 0.4694 | dom_loss 1.3957
| lambda 0.664 | tr_acc 81.8% | val_loss 0.7053 | val_acc 77.2% *
[full_aw40_s2:KuHar] epoch 20 | act_loss 0.4612 | dom_loss 1.4027
| lambda 0.762 | tr_acc 82.2% | val_loss 0.7295 | val_acc 74.9%
[full_aw40_s2:KuHar] epoch 22 | act_loss 0.4576 | dom_loss 1.4042
| lambda 0.800 | tr_acc 82.6% | val_loss 0.6221 | val_acc 78.3% *

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[full_aw40_s2:KuHar] epoch 24 | act_loss 0.4552 | dom_loss 1.4058
| lambda 0.834 | tr_acc 82.7% | val_loss 0.6160 | val_acc 81.6% *
[full_aw40_s2:KuHar] epoch 40 | act_loss 0.4389 | dom_loss 1.4203
| lambda 0.964 | tr_acc 83.7% | val_loss 0.5729 | val_acc 80.7%
[full_aw40_s2:KuHar] epoch 43 | act_loss 0.4337 | dom_loss 1.4220
| lambda 0.973 | tr_acc 84.0% | val_loss 0.5730 | val_acc 82.3% *
[full_aw40_s2:KuHar] epoch 53 | act_loss 0.4105 | dom_loss 1.4230
| lambda 0.990 | tr_acc 85.0% | val_loss 0.5277 | val_acc 83.1% *
[full_aw40_s2:KuHar] epoch 60 | act_loss 0.3999 | dom_loss 1.4245
| lambda 0.995 | tr_acc 85.4% | val_loss 0.5872 | val_acc 80.5%
[full_aw40_s2:KuHar] early stop at epoch 73
[full_aw40_s2:KuHar] src-val 83.06% | TARGET-TEST 64.58% | saved
ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_aw40_s2:MotionSense] trainable params: 51,531
[full_aw40_s2:MotionSense] epoch 1 | act_loss 0.7872 | dom_loss
1.3374 | lambda 0.050 | tr_acc 66.5% | val_loss 0.7597 | val_acc 69.
9% *
[full_aw40_s2:MotionSense] epoch 2 | act_loss 0.6200 | dom_loss
1.2603 | lambda 0.100 | tr_acc 74.2% | val_loss 0.6679 | val_acc 74.
5% *
[full_aw40_s2:MotionSense] epoch 7 | act_loss 0.5210 | dom_loss
1.2994 | lambda 0.336 | tr_acc 78.8% | val_loss 0.6374 | val_acc 76.
8% *
[full_aw40_s2:MotionSense] epoch 8 | act_loss 0.5099 | dom_loss
1.3023 | lambda 0.380 | tr_acc 79.4% | val_loss 0.6388 | val_acc 77.
9% *
[full_aw40_s2:MotionSense] epoch 16 | act_loss 0.4835 | dom_loss
1.3124 | lambda 0.664 | tr_acc 80.7% | val_loss 0.5985 | val_acc 78.
1% *
[full_aw40_s2:MotionSense] epoch 18 | act_loss 0.4770 | dom_loss
1.3154 | lambda 0.716 | tr_acc 80.9% | val_loss 0.6141 | val_acc 78.
8% *
[full_aw40_s2:MotionSense] epoch 19 | act_loss 0.4749 | dom_loss
1.3167 | lambda 0.740 | tr_acc 81.3% | val_loss 0.6142 | val_acc 79.

```

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6% *
[full_aw40_s2:MotionSense] epoch 20 | act_loss 0.4755 | dom_loss
1.3170 | lambda 0.762 | tr_acc 81.3% | val_loss 0.7017 | val_acc 74.
2%
[full_aw40_s2:MotionSense] epoch 24 | act_loss 0.4707 | dom_loss
1.3205 | lambda 0.834 | tr_acc 81.4% | val_loss 0.6047 | val_acc 80.
5% *
[full_aw40_s2:MotionSense] epoch 40 | act_loss 0.4545 | dom_loss
1.3401 | lambda 0.964 | tr_acc 82.7% | val_loss 0.5778 | val_acc 79.
7%
[full_aw40_s2:MotionSense] early stop at epoch 44
[full_aw40_s2:MotionSense] src-val 80.46% | TARGET-TEST 78.91% |
saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (0, 1):100%, (0, 2):100%, (4, 5):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_aw40_s2:RealWorld-Thigh] trainable params: 51,531
[full_aw40_s2:RealWorld-Thigh] epoch 1 | act_loss 0.7634 | dom_l
oss 1.2836 | lambda 0.050 | tr_acc 66.9% | val_loss 0.7132 | val_acc
70.3% *
[full_aw40_s2:RealWorld-Thigh] epoch 2 | act_loss 0.5932 | dom_l
oss 1.1829 | lambda 0.100 | tr_acc 74.6% | val_loss 0.7053 | val_acc
71.2% *
[full_aw40_s2:RealWorld-Thigh] epoch 3 | act_loss 0.5575 | dom_l
oss 1.1806 | lambda 0.149 | tr_acc 76.4% | val_loss 0.5907 | val_acc
74.0% *
[full_aw40_s2:RealWorld-Thigh] epoch 4 | act_loss 0.5417 | dom_l
oss 1.2018 | lambda 0.197 | tr_acc 77.4% | val_loss 0.5710 | val_acc
78.2% *
[full_aw40_s2:RealWorld-Thigh] epoch 9 | act_loss 0.4788 | dom_l
oss 1.2537 | lambda 0.422 | tr_acc 80.6% | val_loss 0.5690 | val_acc
81.2% *
[full_aw40_s2:RealWorld-Thigh] epoch 12 | act_loss 0.4552 | dom_l
oss 1.2621 | lambda 0.537 | tr_acc 81.5% | val_loss 0.5400 | val_acc
81.9% *
[full_aw40_s2:RealWorld-Thigh] epoch 20 | act_loss 0.4406 | dom_l
oss 1.2740 | lambda 0.762 | tr_acc 82.0% | val_loss 0.6998 | val_acc
70.4%

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[full_aw40_s2:RealWorld-Thigh] epoch 27 | act_loss 0.4351 | dom_l
oss 1.2825 | lambda 0.874 | tr_acc 82.7% | val_loss 0.5014 | val_acc
83.6% *
[full_aw40_s2:RealWorld-Thigh] epoch 40 | act_loss 0.4189 | dom_l
oss 1.3090 | lambda 0.964 | tr_acc 83.2% | val_loss 0.5068 | val_acc
79.4%
[full_aw40_s2:RealWorld-Thigh] early stop at epoch 47
[full_aw40_s2:RealWorld-Thigh] src-val 83.56% | TARGET-TEST 70.29%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (0, 1):100%, (3, 5):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_aw40_s2:RealWorld-Waist] trainable params: 51,531
[full_aw40_s2:RealWorld-Waist] epoch 1 | act_loss 0.7732 | dom_l
oss 1.3249 | lambda 0.050 | tr_acc 66.9% | val_loss 0.6725 | val_acc
72.3% *
[full_aw40_s2:RealWorld-Waist] epoch 4 | act_loss 0.5377 | dom_l
oss 1.2508 | lambda 0.197 | tr_acc 77.6% | val_loss 0.7175 | val_acc
73.3% *
[full_aw40_s2:RealWorld-Waist] epoch 6 | act_loss 0.5111 | dom_l
oss 1.2506 | lambda 0.291 | tr_acc 79.0% | val_loss 0.6594 | val_acc
75.0% *
[full_aw40_s2:RealWorld-Waist] epoch 8 | act_loss 0.4917 | dom_l
oss 1.2513 | lambda 0.380 | tr_acc 80.0% | val_loss 0.6538 | val_acc
76.4% *
[full_aw40_s2:RealWorld-Waist] epoch 10 | act_loss 0.4823 | dom_l
oss 1.2618 | lambda 0.462 | tr_acc 80.2% | val_loss 0.5669 | val_acc
79.0% *
[full_aw40_s2:RealWorld-Waist] epoch 17 | act_loss 0.4555 | dom_l
oss 1.2746 | lambda 0.691 | tr_acc 81.8% | val_loss 0.5886 | val_acc
79.1% *
[full_aw40_s2:RealWorld-Waist] epoch 20 | act_loss 0.4462 | dom_l
oss 1.2765 | lambda 0.762 | tr_acc 82.3% | val_loss 0.6326 | val_acc
72.1%
[full_aw40_s2:RealWorld-Waist] epoch 29 | act_loss 0.4350 | dom_l
oss 1.2946 | lambda 0.896 | tr_acc 83.0% | val_loss 0.5603 | val_acc
80.2% *
[full_aw40_s2:RealWorld-Waist] epoch 40 | act_loss 0.4232 | dom_l

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oss 1.3064 | lambda 0.964 | tr_acc 83.9% | val_loss 0.7583 | val_acc
72.5%
[full_aw40_s2:RealWorld-Waist] epoch 47 | act_loss 0.4077 | dom_l
oss 1.3069 | lambda 0.982 | tr_acc 84.7% | val_loss 0.5362 | val_acc
82.1% *
[full_aw40_s2:RealWorld-Waist] epoch 60 | act_loss 0.3856 | dom_l
oss 1.3100 | lambda 0.995 | tr_acc 85.9% | val_loss 0.5930 | val_acc
78.6%
[full_aw40_s2:RealWorld-Waist] early stop at epoch 67
[full_aw40_s2:RealWorld-Waist] src-val 82.06% | TARGET-TEST 67.63%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw40_s2:UCI] trainable params: 51,531
[full_aw40_s2:UCI] epoch 1 | act_loss 0.7714 | dom_loss 1.3838 |
lambda 0.050 | tr_acc 67.8% | val_loss 0.7431 | val_acc 71.3% *
[full_aw40_s2:UCI] epoch 5 | act_loss 0.5408 | dom_loss 1.3263 |
lambda 0.245 | tr_acc 78.3% | val_loss 0.6408 | val_acc 76.4% *
[full_aw40_s2:UCI] epoch 7 | act_loss 0.5212 | dom_loss 1.3414 |
lambda 0.336 | tr_acc 79.2% | val_loss 0.6265 | val_acc 76.7% *
[full_aw40_s2:UCI] epoch 17 | act_loss 0.4807 | dom_loss 1.3625 |
lambda 0.691 | tr_acc 81.2% | val_loss 0.5897 | val_acc 80.7% *
[full_aw40_s2:UCI] epoch 20 | act_loss 0.4734 | dom_loss 1.3668 |
lambda 0.762 | tr_acc 81.5% | val_loss 0.6305 | val_acc 77.8%
[full_aw40_s2:UCI] early stop at epoch 37
[full_aw40_s2:UCI] src-val 80.69% | TARGET-TEST 80.87% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%

```

```

[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (3, 4):100%, (4, 5):100%, (0, 1):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 60 features each)...
```

```

built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw40_s2:WISDM] trainable params: 51,531
[full_aw40_s2:WISDM] epoch 1 | act_loss 0.7825 | dom_loss 1.3348
| lambda 0.050 | tr_acc 67.5% | val_loss 0.7054 | val_acc 72.3% *
[full_aw40_s2:WISDM] epoch 2 | act_loss 0.6069 | dom_loss 1.2960
| lambda 0.100 | tr_acc 75.7% | val_loss 0.7184 | val_acc 72.8% *
[full_aw40_s2:WISDM] epoch 4 | act_loss 0.5452 | dom_loss 1.3071
| lambda 0.197 | tr_acc 78.4% | val_loss 0.7026 | val_acc 73.9% *
[full_aw40_s2:WISDM] epoch 5 | act_loss 0.5237 | dom_loss 1.3161
| lambda 0.245 | tr_acc 79.3% | val_loss 0.5910 | val_acc 78.9% *
[full_aw40_s2:WISDM] epoch 14 | act_loss 0.4624 | dom_loss 1.3313
| lambda 0.604 | tr_acc 82.4% | val_loss 0.5771 | val_acc 82.0% *
[full_aw40_s2:WISDM] epoch 20 | act_loss 0.4432 | dom_loss 1.3317
| lambda 0.762 | tr_acc 83.3% | val_loss 0.7914 | val_acc 77.2%
[full_aw40_s2:WISDM] early stop at epoch 34
[full_aw40_s2:WISDM] src-val 82.02% | TARGET-TEST 76.96% | saved ckpt + npz
[P24] train+save | arm=full_aw40_s3 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=40 | seed=3 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw40_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 4):100%, (3, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (1, 4):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 60 features each)...
```

```

built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs

```

```

done: 35396 graphs built
[full_aw40_s3:KuHar] trainable params: 51,531
[full_aw40_s3:KuHar] epoch 1 | act_loss 0.7677 | dom_loss 1.4217
| lambda 0.050 | tr_acc 68.2% | val_loss 0.6686 | val_acc 74.7% *
[full_aw40_s3:KuHar] epoch 3 | act_loss 0.5669 | dom_loss 1.3540
| lambda 0.149 | tr_acc 77.0% | val_loss 0.6871 | val_acc 75.0% *
[full_aw40_s3:KuHar] epoch 5 | act_loss 0.5325 | dom_loss 1.3698
| lambda 0.245 | tr_acc 78.5% | val_loss 0.6731 | val_acc 76.0% *
[full_aw40_s3:KuHar] epoch 8 | act_loss 0.5053 | dom_loss 1.3855
| lambda 0.380 | tr_acc 80.0% | val_loss 0.5929 | val_acc 78.9% *
[full_aw40_s3:KuHar] epoch 14 | act_loss 0.4787 | dom_loss 1.3958
| lambda 0.604 | tr_acc 81.3% | val_loss 0.6209 | val_acc 79.7% *
[full_aw40_s3:KuHar] epoch 20 | act_loss 0.4632 | dom_loss 1.4011
| lambda 0.762 | tr_acc 82.1% | val_loss 0.8442 | val_acc 70.4%
[full_aw40_s3:KuHar] epoch 22 | act_loss 0.4569 | dom_loss 1.4036
| lambda 0.800 | tr_acc 82.6% | val_loss 0.5957 | val_acc 81.7% *
[full_aw40_s3:KuHar] epoch 26 | act_loss 0.4540 | dom_loss 1.4059
| lambda 0.862 | tr_acc 82.8% | val_loss 0.5441 | val_acc 82.0% *
[full_aw40_s3:KuHar] epoch 40 | act_loss 0.4375 | dom_loss 1.4166
| lambda 0.964 | tr_acc 83.6% | val_loss 0.5781 | val_acc 79.2%
[full_aw40_s3:KuHar] epoch 41 | act_loss 0.4398 | dom_loss 1.4166
| lambda 0.967 | tr_acc 83.4% | val_loss 0.5406 | val_acc 82.1% *
[full_aw40_s3:KuHar] epoch 55 | act_loss 0.4135 | dom_loss 1.4204
| lambda 0.992 | tr_acc 84.9% | val_loss 0.5400 | val_acc 82.5% *
[full_aw40_s3:KuHar] epoch 60 | act_loss 0.4065 | dom_loss 1.4195
| lambda 0.995 | tr_acc 85.0% | val_loss 0.5354 | val_acc 82.6% *
[full_aw40_s3:KuHar] epoch 71 | act_loss 0.3824 | dom_loss 1.4196
| lambda 0.998 | tr_acc 86.3% | val_loss 0.5386 | val_acc 82.8% *
[full_aw40_s3:KuHar] epoch 75 | act_loss 0.3799 | dom_loss 1.4199
| lambda 0.999 | tr_acc 86.3% | val_loss 0.5468 | val_acc 83.1% *
[full_aw40_s3:KuHar] epoch 76 | act_loss 0.3737 | dom_loss 1.4196
| lambda 0.999 | tr_acc 86.6% | val_loss 0.5224 | val_acc 83.3% *
[full_aw40_s3:KuHar] epoch 80 | act_loss 0.3683 | dom_loss 1.4200
| lambda 0.999 | tr_acc 86.9% | val_loss 0.5644 | val_acc 81.3%
[full_aw40_s3:KuHar] epoch 84 | act_loss 0.3626 | dom_loss 1.4191
| lambda 1.000 | tr_acc 87.0% | val_loss 0.5329 | val_acc 84.0% *
[full_aw40_s3:KuHar] epoch 86 | act_loss 0.3598 | dom_loss 1.4196
| lambda 1.000 | tr_acc 87.3% | val_loss 0.5000 | val_acc 84.3% *
[full_aw40_s3:KuHar] epoch 100 | act_loss 0.3457 | dom_loss 1.4195
| lambda 1.000 | tr_acc 87.9% | val_loss 0.5894 | val_acc 79.4%
[full_aw40_s3:KuHar] src-val 84.26% | TARGET-TEST 71.53% | saved
ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 60 features eac

```

```

h)...
  built 5000/33230 graphs
  built 10000/33230 graphs
  built 15000/33230 graphs
  built 20000/33230 graphs
  built 25000/33230 graphs
  built 30000/33230 graphs
  done: 33230 graphs built
[full_aw40_s3:MotionSense] trainable params: 51,531
[full_aw40_s3:MotionSense] epoch  1 | act_loss 0.7935 | dom_loss
1.3392 | lambda 0.050 | tr_acc  66.2% | val_loss 0.8275 | val_acc 68.
4% *
[full_aw40_s3:MotionSense] epoch  4 | act_loss 0.5578 | dom_loss
1.2647 | lambda 0.197 | tr_acc  77.0% | val_loss 0.6850 | val_acc 74.
4% *
[full_aw40_s3:MotionSense] epoch  6 | act_loss 0.5329 | dom_loss
1.2900 | lambda 0.291 | tr_acc  78.2% | val_loss 0.6205 | val_acc 76.
8% *
[full_aw40_s3:MotionSense] epoch  7 | act_loss 0.5227 | dom_loss
1.2961 | lambda 0.336 | tr_acc  78.7% | val_loss 0.6172 | val_acc 77.
3% *
[full_aw40_s3:MotionSense] epoch 17 | act_loss 0.4761 | dom_loss
1.3169 | lambda 0.691 | tr_acc  81.2% | val_loss 0.6193 | val_acc 77.
8% *
[full_aw40_s3:MotionSense] epoch 20 | act_loss 0.4725 | dom_loss
1.3200 | lambda 0.762 | tr_acc  81.5% | val_loss 0.8761 | val_acc 69.
7%
[full_aw40_s3:MotionSense] epoch 21 | act_loss 0.4723 | dom_loss
1.3193 | lambda 0.782 | tr_acc  81.5% | val_loss 0.5854 | val_acc 79.
0% *
[full_aw40_s3:MotionSense] epoch 26 | act_loss 0.4651 | dom_loss
1.3217 | lambda 0.862 | tr_acc  81.9% | val_loss 0.5917 | val_acc 79.
1% *
[full_aw40_s3:MotionSense] epoch 39 | act_loss 0.4526 | dom_loss
1.3421 | lambda 0.960 | tr_acc  82.9% | val_loss 0.5701 | val_acc 79.
6% *
[full_aw40_s3:MotionSense] epoch 40 | act_loss 0.4519 | dom_loss
1.3409 | lambda 0.964 | tr_acc  82.9% | val_loss 0.6371 | val_acc 77.
2%
[full_aw40_s3:MotionSense] epoch 48 | act_loss 0.4440 | dom_loss
1.3433 | lambda 0.984 | tr_acc  83.2% | val_loss 0.5921 | val_acc 80.
4% *
[full_aw40_s3:MotionSense] epoch 50 | act_loss 0.4367 | dom_loss
1.3432 | lambda 0.987 | tr_acc  83.4% | val_loss 0.5681 | val_acc 81.
5% *
[full_aw40_s3:MotionSense] epoch 60 | act_loss 0.4198 | dom_loss
1.3420 | lambda 0.995 | tr_acc  84.4% | val_loss 0.7558 | val_acc 69.
6%
[full_aw40_s3:MotionSense] epoch 68 | act_loss 0.4053 | dom_loss
1.3402 | lambda 0.998 | tr_acc  85.2% | val_loss 0.6201 | val_acc 82.
7% *
[full_aw40_s3:MotionSense] epoch 80 | act_loss 0.3863 | dom_loss
1.3415 | lambda 0.999 | tr_acc  86.0% | val_loss 0.6048 | val_acc 79.
6%
[full_aw40_s3:MotionSense] early stop at epoch 88
[full_aw40_s3:MotionSense] src-val 82.72% | TARGET-TEST 78.15% |
saved ckpt + npz

```

```

[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (0, 1):100%, (0, 2):100%, (4, 5):100%, (3, 5):100%
    feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
    building 26450 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
[full_aw40_s3:RealWorld-Thigh] trainable params: 51,531
[full_aw40_s3:RealWorld-Thigh] epoch 1 | act_loss 0.7726 | dom_l
oss 1.2907 | lambda 0.050 | tr_acc 67.2% | val_loss 0.6483 | val_acc
73.7% *
[full_aw40_s3:RealWorld-Thigh] epoch 2 | act_loss 0.5890 | dom_l
oss 1.1977 | lambda 0.100 | tr_acc 75.3% | val_loss 0.6461 | val_acc
74.4% *
[full_aw40_s3:RealWorld-Thigh] epoch 4 | act_loss 0.5325 | dom_l
oss 1.2022 | lambda 0.197 | tr_acc 77.9% | val_loss 0.5822 | val_acc
77.4% *
[full_aw40_s3:RealWorld-Thigh] epoch 6 | act_loss 0.5065 | dom_l
oss 1.2252 | lambda 0.291 | tr_acc 78.9% | val_loss 0.5642 | val_acc
78.5% *
[full_aw40_s3:RealWorld-Thigh] epoch 9 | act_loss 0.4801 | dom_l
oss 1.2500 | lambda 0.422 | tr_acc 80.3% | val_loss 0.5489 | val_acc
81.3% *
[full_aw40_s3:RealWorld-Thigh] epoch 10 | act_loss 0.4709 | dom_l
oss 1.2507 | lambda 0.462 | tr_acc 80.5% | val_loss 0.5357 | val_acc
81.3% *
[full_aw40_s3:RealWorld-Thigh] epoch 20 | act_loss 0.4443 | dom_l
oss 1.2766 | lambda 0.762 | tr_acc 82.0% | val_loss 0.5174 | val_acc
80.9%
[full_aw40_s3:RealWorld-Thigh] epoch 24 | act_loss 0.4357 | dom_l
oss 1.2741 | lambda 0.834 | tr_acc 82.5% | val_loss 0.5096 | val_acc
83.6% *
[full_aw40_s3:RealWorld-Thigh] epoch 30 | act_loss 0.4376 | dom_l
oss 1.2906 | lambda 0.905 | tr_acc 82.4% | val_loss 0.5189 | val_acc
83.8% *
[full_aw40_s3:RealWorld-Thigh] epoch 40 | act_loss 0.4227 | dom_l
oss 1.3092 | lambda 0.964 | tr_acc 83.4% | val_loss 0.7548 | val_acc
74.9%
[full_aw40_s3:RealWorld-Thigh] early stop at epoch 50
[full_aw40_s3:RealWorld-Thigh] src-val 83.83% | TARGET-TEST 73.26%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist

```

```

[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (2, 5):100%, (0, 1):100%, (3, 5):100%, (1, 2):100%
    feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
    building 26456 sensor graphs (6 nodes, 18 edges, 60 features each)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_aw40_s3:RealWorld-Waist] trainable params: 51,531
[full_aw40_s3:RealWorld-Waist] epoch 1 | act_loss 0.7877 | dom_loss 1.3315 | lambda 0.050 | tr_acc 65.8% | val_loss 0.8478 | val_acc 65.1% *
[full_aw40_s3:RealWorld-Waist] epoch 2 | act_loss 0.6136 | dom_loss 1.2604 | lambda 0.100 | tr_acc 74.2% | val_loss 0.6312 | val_acc 75.3% *
[full_aw40_s3:RealWorld-Waist] epoch 7 | act_loss 0.5043 | dom_loss 1.2539 | lambda 0.336 | tr_acc 79.2% | val_loss 0.5862 | val_acc 77.8% *
[full_aw40_s3:RealWorld-Waist] epoch 15 | act_loss 0.4581 | dom_loss 1.2723 | lambda 0.635 | tr_acc 81.7% | val_loss 0.5381 | val_acc 79.6% *
[full_aw40_s3:RealWorld-Waist] epoch 20 | act_loss 0.4476 | dom_loss 1.2795 | lambda 0.762 | tr_acc 82.4% | val_loss 0.7256 | val_acc 73.2%
[full_aw40_s3:RealWorld-Waist] epoch 28 | act_loss 0.4407 | dom_loss 1.2860 | lambda 0.885 | tr_acc 82.5% | val_loss 0.5894 | val_acc 80.1% *
[full_aw40_s3:RealWorld-Waist] epoch 40 | act_loss 0.4252 | dom_loss 1.3054 | lambda 0.964 | tr_acc 84.0% | val_loss 0.5900 | val_acc 79.0%
[full_aw40_s3:RealWorld-Waist] early stop at epoch 48
[full_aw40_s3:RealWorld-Waist] src-val 80.10% | TARGET-TEST 66.40%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%

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```

[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw40_s3:UCI] trainable params: 51,531
[full_aw40_s3:UCI] epoch 1 | act_loss 0.7818 | dom_loss 1.3954 |
lambda 0.050 | tr_acc 67.2% | val_loss 0.7931 | val_acc 68.6% *
[full_aw40_s3:UCI] epoch 2 | act_loss 0.6179 | dom_loss 1.3272 |
lambda 0.100 | tr_acc 74.8% | val_loss 0.6895 | val_acc 73.9% *
[full_aw40_s3:UCI] epoch 6 | act_loss 0.5329 | dom_loss 1.3349 |
lambda 0.291 | tr_acc 78.5% | val_loss 0.6793 | val_acc 75.8% *
[full_aw40_s3:UCI] epoch 7 | act_loss 0.5192 | dom_loss 1.3433 |
lambda 0.336 | tr_acc 79.4% | val_loss 0.5880 | val_acc 78.5% *
[full_aw40_s3:UCI] epoch 15 | act_loss 0.4863 | dom_loss 1.3616 |
lambda 0.635 | tr_acc 80.7% | val_loss 0.5401 | val_acc 82.2% *
[full_aw40_s3:UCI] epoch 20 | act_loss 0.4754 | dom_loss 1.3661 |
lambda 0.762 | tr_acc 81.5% | val_loss 0.6268 | val_acc 78.0%
[full_aw40_s3:UCI] early stop at epoch 35
[full_aw40_s3:UCI] src-val 82.19% | TARGET-TEST 81.01% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (4, 5):100%, (0, 1):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw40_s3:WISDM] trainable params: 51,531
[full_aw40_s3:WISDM] epoch 1 | act_loss 0.7874 | dom_loss 1.3406
| lambda 0.050 | tr_acc 67.5% | val_loss 0.8570 | val_acc 66.4% *
[full_aw40_s3:WISDM] epoch 3 | act_loss 0.5638 | dom_loss 1.3099
| lambda 0.149 | tr_acc 77.6% | val_loss 0.8156 | val_acc 70.8% *
[full_aw40_s3:WISDM] epoch 4 | act_loss 0.5352 | dom_loss 1.3097
| lambda 0.197 | tr_acc 78.9% | val_loss 0.6783 | val_acc 74.9% *

```

```

[full_aw40_s3:WISDM] epoch 8 | act_loss 0.4888 | dom_loss 1.3263
| lambda 0.380 | tr_acc 81.2% | val_loss 0.6597 | val_acc 76.2% *
[full_aw40_s3:WISDM] epoch 12 | act_loss 0.4662 | dom_loss 1.3282
| lambda 0.537 | tr_acc 82.1% | val_loss 0.6771 | val_acc 76.6% *
[full_aw40_s3:WISDM] epoch 13 | act_loss 0.4593 | dom_loss 1.3276
| lambda 0.572 | tr_acc 82.3% | val_loss 0.6297 | val_acc 78.7% *
[full_aw40_s3:WISDM] epoch 15 | act_loss 0.4554 | dom_loss 1.3299
| lambda 0.635 | tr_acc 82.6% | val_loss 0.6310 | val_acc 81.3% *
[full_aw40_s3:WISDM] epoch 20 | act_loss 0.4430 | dom_loss 1.3329
| lambda 0.762 | tr_acc 83.3% | val_loss 0.9085 | val_acc 68.1%
[full_aw40_s3:WISDM] epoch 30 | act_loss 0.4215 | dom_loss 1.3367
| lambda 0.905 | tr_acc 84.5% | val_loss 0.5712 | val_acc 82.3% *
[full_aw40_s3:WISDM] epoch 40 | act_loss 0.4067 | dom_loss 1.3365
| lambda 0.964 | tr_acc 85.1% | val_loss 0.6262 | val_acc 80.5%
[full_aw40_s3:WISDM] early stop at epoch 50
[full_aw40_s3:WISDM] src-val 82.27% | TARGET-TEST 68.05% | saved
ckpt + npz
[P24] train+save | arm=full_aw40_s5 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=40 | seed=5 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw40_s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 1):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 4):100%, (3, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (1, 4):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 60 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_aw40_s5:KuHar] trainable params: 51,531
[full_aw40_s5:KuHar] epoch 1 | act_loss 0.7780 | dom_loss 1.4151
| lambda 0.050 | tr_acc 67.1% | val_loss 1.2799 | val_acc 50.9% *
[full_aw40_s5:KuHar] epoch 2 | act_loss 0.6087 | dom_loss 1.3552
| lambda 0.100 | tr_acc 75.1% | val_loss 0.7748 | val_acc 69.4% *
[full_aw40_s5:KuHar] epoch 3 | act_loss 0.5698 | dom_loss 1.3490
| lambda 0.149 | tr_acc 76.9% | val_loss 0.6667 | val_acc 73.8% *
[full_aw40_s5:KuHar] epoch 6 | act_loss 0.5269 | dom_loss 1.3790
| lambda 0.291 | tr_acc 78.7% | val_loss 0.6559 | val_acc 75.8% *
[full_aw40_s5:KuHar] epoch 8 | act_loss 0.5162 | dom_loss 1.3871
| lambda 0.380 | tr_acc 79.4% | val_loss 0.6295 | val_acc 77.2% *

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[full_aw40_s5:KuHar] epoch 11 | act_loss 0.4972 | dom_loss 1.3943
| lambda 0.501 | tr_acc 80.3% | val_loss 0.6343 | val_acc 78.3% *
[full_aw40_s5:KuHar] epoch 12 | act_loss 0.4933 | dom_loss 1.3925
| lambda 0.537 | tr_acc 80.6% | val_loss 0.5967 | val_acc 79.0% *
[full_aw40_s5:KuHar] epoch 16 | act_loss 0.4791 | dom_loss 1.3993
| lambda 0.664 | tr_acc 81.6% | val_loss 0.5983 | val_acc 79.5% *
[full_aw40_s5:KuHar] epoch 19 | act_loss 0.4748 | dom_loss 1.4024
| lambda 0.740 | tr_acc 82.0% | val_loss 0.6304 | val_acc 80.1% *
[full_aw40_s5:KuHar] epoch 20 | act_loss 0.4727 | dom_loss 1.4040
| lambda 0.762 | tr_acc 81.7% | val_loss 0.6208 | val_acc 77.0%
[full_aw40_s5:KuHar] epoch 24 | act_loss 0.4613 | dom_loss 1.4070
| lambda 0.834 | tr_acc 82.4% | val_loss 0.5957 | val_acc 80.3% *
[full_aw40_s5:KuHar] epoch 32 | act_loss 0.4554 | dom_loss 1.4126
| lambda 0.922 | tr_acc 82.8% | val_loss 0.5898 | val_acc 81.5% *
[full_aw40_s5:KuHar] epoch 40 | act_loss 0.4475 | dom_loss 1.4207
| lambda 0.964 | tr_acc 83.4% | val_loss 0.6643 | val_acc 78.6%
[full_aw40_s5:KuHar] epoch 42 | act_loss 0.4423 | dom_loss 1.4221
| lambda 0.970 | tr_acc 83.4% | val_loss 0.5522 | val_acc 82.8% *
[full_aw40_s5:KuHar] epoch 60 | act_loss 0.4074 | dom_loss 1.4236
| lambda 0.995 | tr_acc 85.2% | val_loss 0.5756 | val_acc 78.2%
[full_aw40_s5:KuHar] early stop at epoch 62
[full_aw40_s5:KuHar] src-val 82.78% | TARGET-TEST 62.50% | saved
ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_aw40_s5:MotionSense] trainable params: 51,531
[full_aw40_s5:MotionSense] epoch 1 | act_loss 0.7972 | dom_loss
1.3418 | lambda 0.050 | tr_acc 66.1% | val_loss 0.6930 | val_acc 73.
1% *
[full_aw40_s5:MotionSense] epoch 2 | act_loss 0.6263 | dom_loss
1.2473 | lambda 0.100 | tr_acc 74.1% | val_loss 0.6764 | val_acc 74.
1% *
[full_aw40_s5:MotionSense] epoch 5 | act_loss 0.5451 | dom_loss
1.2790 | lambda 0.245 | tr_acc 77.5% | val_loss 0.6486 | val_acc 75.
4% *
[full_aw40_s5:MotionSense] epoch 9 | act_loss 0.5134 | dom_loss

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1.3047 | lambda 0.422 | tr_acc 79.2% | val_loss 0.6383 | val_acc 76.
6% *
    [full_aw40_s5:MotionSense] epoch 14 | act_loss 0.4940 | dom_loss
1.3125 | lambda 0.604 | tr_acc 80.4% | val_loss 0.6197 | val_acc 77.
0% *
    [full_aw40_s5:MotionSense] epoch 15 | act_loss 0.4935 | dom_loss
1.3139 | lambda 0.635 | tr_acc 80.2% | val_loss 0.6234 | val_acc 79.
5% *
    [full_aw40_s5:MotionSense] epoch 20 | act_loss 0.4790 | dom_loss
1.3186 | lambda 0.762 | tr_acc 81.0% | val_loss 0.6306 | val_acc 78.
5%
    [full_aw40_s5:MotionSense] early stop at epoch 35
    [full_aw40_s5:MotionSense] src-val 79.46% | TARGET-TEST 79.76% |
saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (0, 1):100%, (0, 2):100%, (4, 5):100%, (3, 5):100%
    feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
    building 26450 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
    [full_aw40_s5:RealWorld-Thigh] trainable params: 51,531
    [full_aw40_s5:RealWorld-Thigh] epoch 1 | act_loss 0.7773 | dom_l
oss 1.2875 | lambda 0.050 | tr_acc 66.9% | val_loss 0.6854 | val_acc
72.1% *
    [full_aw40_s5:RealWorld-Thigh] epoch 3 | act_loss 0.5614 | dom_l
oss 1.1817 | lambda 0.149 | tr_acc 76.5% | val_loss 0.5704 | val_acc
77.6% *
    [full_aw40_s5:RealWorld-Thigh] epoch 7 | act_loss 0.4989 | dom_l
oss 1.2421 | lambda 0.336 | tr_acc 79.4% | val_loss 0.6409 | val_acc
78.1% *
    [full_aw40_s5:RealWorld-Thigh] epoch 8 | act_loss 0.4880 | dom_l
oss 1.2480 | lambda 0.380 | tr_acc 79.8% | val_loss 0.5898 | val_acc
78.9% *
    [full_aw40_s5:RealWorld-Thigh] epoch 10 | act_loss 0.4738 | dom_l
oss 1.2559 | lambda 0.462 | tr_acc 80.6% | val_loss 0.5418 | val_acc
80.0% *
    [full_aw40_s5:RealWorld-Thigh] epoch 12 | act_loss 0.4637 | dom_l
oss 1.2599 | lambda 0.537 | tr_acc 81.3% | val_loss 0.5023 | val_acc
82.5% *
    [full_aw40_s5:RealWorld-Thigh] epoch 20 | act_loss 0.4447 | dom_l
oss 1.2717 | lambda 0.762 | tr_acc 82.2% | val_loss 0.6216 | val_acc

```

79.1%

```

[full_aw40_s5:RealWorld-Thigh] early stop at epoch 32
[full_aw40_s5:RealWorld-Thigh] src-val 82.48% | TARGET-TEST 69.77%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (0, 1):100%, (3, 5):100%, (1, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_aw40_s5:RealWorld-Waist] trainable params: 51,531
[full_aw40_s5:RealWorld-Waist] epoch 1 | act_loss 0.7945 | dom_l
oss 1.3273 | lambda 0.050 | tr_acc 65.5% | val_loss 1.0023 | val_acc
56.3% *
[full_aw40_s5:RealWorld-Waist] epoch 2 | act_loss 0.6233 | dom_l
oss 1.2648 | lambda 0.100 | tr_acc 73.8% | val_loss 0.8630 | val_acc
60.7% *
[full_aw40_s5:RealWorld-Waist] epoch 3 | act_loss 0.5700 | dom_l
oss 1.2453 | lambda 0.149 | tr_acc 76.0% | val_loss 0.8080 | val_acc
71.4% *
[full_aw40_s5:RealWorld-Waist] epoch 5 | act_loss 0.5298 | dom_l
oss 1.2528 | lambda 0.245 | tr_acc 78.0% | val_loss 0.6426 | val_acc
74.8% *
[full_aw40_s5:RealWorld-Waist] epoch 6 | act_loss 0.5195 | dom_l
oss 1.2527 | lambda 0.291 | tr_acc 78.8% | val_loss 0.5603 | val_acc
78.5% *
[full_aw40_s5:RealWorld-Waist] epoch 20 | act_loss 0.4531 | dom_l
oss 1.2752 | lambda 0.762 | tr_acc 81.7% | val_loss 0.7079 | val_acc
73.9%
[full_aw40_s5:RealWorld-Waist] epoch 23 | act_loss 0.4495 | dom_l
oss 1.2775 | lambda 0.818 | tr_acc 82.5% | val_loss 0.5460 | val_acc
79.8% *
[full_aw40_s5:RealWorld-Waist] epoch 31 | act_loss 0.4427 | dom_l
oss 1.2918 | lambda 0.914 | tr_acc 82.7% | val_loss 0.5611 | val_acc
80.6% *
[full_aw40_s5:RealWorld-Waist] epoch 40 | act_loss 0.4311 | dom_l
oss 1.3093 | lambda 0.964 | tr_acc 83.6% | val_loss 0.6638 | val_acc
73.4%
[full_aw40_s5:RealWorld-Waist] epoch 42 | act_loss 0.4242 | dom_l
oss 1.3113 | lambda 0.970 | tr_acc 83.9% | val_loss 0.5483 | val_acc
80.6% *

```

```

[full_aw40_s5:RealWorld-Waist] epoch 53 | act_loss 0.4066 | dom_loss 1.3075 | lambda 0.990 | tr_acc 84.6% | val_loss 0.5475 | val_acc 82.2% *
[full_aw40_s5:RealWorld-Waist] epoch 60 | act_loss 0.3892 | dom_loss 1.3095 | lambda 0.995 | tr_acc 85.6% | val_loss 0.5661 | val_acc 79.7%
[full_aw40_s5:RealWorld-Waist] epoch 66 | act_loss 0.3793 | dom_loss 1.3129 | lambda 0.997 | tr_acc 86.1% | val_loss 0.5159 | val_acc 82.4% *
[full_aw40_s5:RealWorld-Waist] epoch 80 | act_loss 0.3512 | dom_loss 1.3120 | lambda 0.999 | tr_acc 87.5% | val_loss 0.5518 | val_acc 80.6%
[full_aw40_s5:RealWorld-Waist] early stop at epoch 86
[full_aw40_s5:RealWorld-Waist] src-val 82.38% | TARGET-TEST 68.06%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 60 features each)...)
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw40_s5:UCI] trainable params: 51,531
[full_aw40_s5:UCI] epoch 1 | act_loss 0.7874 | dom_loss 1.3906 | lambda 0.050 | tr_acc 66.6% | val_loss 0.8309 | val_acc 64.4% *
[full_aw40_s5:UCI] epoch 2 | act_loss 0.6276 | dom_loss 1.3113 | lambda 0.100 | tr_acc 74.3% | val_loss 0.7429 | val_acc 71.3% *
[full_aw40_s5:UCI] epoch 3 | act_loss 0.5808 | dom_loss 1.3121 | lambda 0.149 | tr_acc 76.6% | val_loss 0.6751 | val_acc 74.6% *
[full_aw40_s5:UCI] epoch 5 | act_loss 0.5412 | dom_loss 1.3277 | lambda 0.245 | tr_acc 78.4% | val_loss 0.6709 | val_acc 75.7% *
[full_aw40_s5:UCI] epoch 11 | act_loss 0.5075 | dom_loss 1.3577 | lambda 0.501 | tr_acc 79.7% | val_loss 0.6030 | val_acc 78.6% *
[full_aw40_s5:UCI] epoch 13 | act_loss 0.4995 | dom_loss 1.3591 | lambda 0.572 | tr_acc 80.0% | val_loss 0.6140 | val_acc 78.7% *
[full_aw40_s5:UCI] epoch 17 | act_loss 0.4918 | dom_loss 1.3666 | lambda 0.691 | tr_acc 80.5% | val_loss 0.6502 | val_acc 79.1% *
[full_aw40_s5:UCI] epoch 20 | act_loss 0.4814 | dom_loss 1.3664 | lambda 0.762 | tr_acc 81.1% | val_loss 0.6227 | val_acc 79.8% *
[full_aw40_s5:UCI] epoch 22 | act_loss 0.4768 | dom_loss 1.3689 | lambda 0.800 | tr_acc 81.6% | val_loss 0.5852 | val_acc 81.1% *

```

```

[full_aw40_s5:UCI] epoch 40 | act_loss 0.4547 | dom_loss 1.3879 |
lambda 0.964 | tr_acc 82.8% | val_loss 0.6080 | val_acc 78.8%
[full_aw40_s5:UCI] early stop at epoch 42
[full_aw40_s5:UCI] src-val 81.12% | TARGET-TEST 83.33% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 40 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 1):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (3, 5):100%, (0, 1):100%, (4, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (4, 5):100%, (0, 1):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=60, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 60 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw40_s5:WISDM] trainable params: 51,531
[full_aw40_s5:WISDM] epoch 1 | act_loss 0.8068 | dom_loss 1.3386
| lambda 0.050 | tr_acc 66.7% | val_loss 0.7052 | val_acc 73.0% *
[full_aw40_s5:WISDM] epoch 5 | act_loss 0.5371 | dom_loss 1.3159
| lambda 0.245 | tr_acc 79.0% | val_loss 0.6085 | val_acc 77.9% *
[full_aw40_s5:WISDM] epoch 6 | act_loss 0.5163 | dom_loss 1.3207
| lambda 0.291 | tr_acc 80.0% | val_loss 0.6193 | val_acc 78.9% *
[full_aw40_s5:WISDM] epoch 13 | act_loss 0.4669 | dom_loss 1.3283
| lambda 0.572 | tr_acc 82.0% | val_loss 0.6609 | val_acc 79.7% *
[full_aw40_s5:WISDM] epoch 20 | act_loss 0.4468 | dom_loss 1.3331
| lambda 0.762 | tr_acc 83.2% | val_loss 0.6236 | val_acc 79.4%
[full_aw40_s5:WISDM] early stop at epoch 33
[full_aw40_s5:WISDM] src-val 79.69% | TARGET-TEST 70.89% | saved
ckpt + npz

```

4d. aw30 (Adaptive Windowing → 30 timesteps/window)

```
In [ ]: for s in (1, 2, 3, 5):  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant full  
        --aw 30 --seed {s} --epochs 100 --log_every 20
```

```

[P24] train+save | arm=full_aw30_s1 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=30 | seed=1 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw30_s1
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 4):100%, (4, 5):100%, (1, 2):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 45 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_aw30_s1:KuHar] trainable params: 51,531
[full_aw30_s1:KuHar] epoch 1 | act_loss 0.7767 | dom_loss 1.4014
| lambda 0.050 | tr_acc 67.0% | val_loss 0.7471 | val_acc 71.1% *
[full_aw30_s1:KuHar] epoch 3 | act_loss 0.5750 | dom_loss 1.3487
| lambda 0.149 | tr_acc 76.7% | val_loss 0.7447 | val_acc 71.7% *
[full_aw30_s1:KuHar] epoch 5 | act_loss 0.5409 | dom_loss 1.3755
| lambda 0.245 | tr_acc 78.3% | val_loss 0.6408 | val_acc 77.5% *
[full_aw30_s1:KuHar] epoch 6 | act_loss 0.5338 | dom_loss 1.3864
| lambda 0.291 | tr_acc 78.9% | val_loss 0.6340 | val_acc 77.8% *
[full_aw30_s1:KuHar] epoch 11 | act_loss 0.4994 | dom_loss 1.3975
| lambda 0.501 | tr_acc 80.3% | val_loss 0.6048 | val_acc 79.4% *
[full_aw30_s1:KuHar] epoch 14 | act_loss 0.4886 | dom_loss 1.4009
| lambda 0.604 | tr_acc 80.8% | val_loss 0.5747 | val_acc 79.5% *
[full_aw30_s1:KuHar] epoch 17 | act_loss 0.4752 | dom_loss 1.4020
| lambda 0.691 | tr_acc 81.4% | val_loss 0.5724 | val_acc 81.3% *
[full_aw30_s1:KuHar] epoch 18 | act_loss 0.4765 | dom_loss 1.4033
| lambda 0.716 | tr_acc 81.7% | val_loss 0.5639 | val_acc 81.3% *
[full_aw30_s1:KuHar] epoch 20 | act_loss 0.4720 | dom_loss 1.4040
| lambda 0.762 | tr_acc 81.8% | val_loss 0.6278 | val_acc 71.4%
[full_aw30_s1:KuHar] early stop at epoch 38
[full_aw30_s1:KuHar] src-val 81.30% | TARGET-TEST 59.72% | saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)

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[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (0, 1):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_aw30_s1:MotionSense] trainable params: 51,531
[full_aw30_s1:MotionSense] epoch 1 | act_loss 0.8113 | dom_loss
1.3184 | lambda 0.050 | tr_acc 64.8% | val_loss 0.9048 | val_acc 61.
8% *
[full_aw30_s1:MotionSense] epoch 2 | act_loss 0.6329 | dom_loss
1.2483 | lambda 0.100 | tr_acc 73.7% | val_loss 0.6631 | val_acc 75.
8% *
[full_aw30_s1:MotionSense] epoch 9 | act_loss 0.5205 | dom_loss
1.3053 | lambda 0.422 | tr_acc 78.8% | val_loss 0.6588 | val_acc 76.
3% *
[full_aw30_s1:MotionSense] epoch 15 | act_loss 0.4949 | dom_loss
1.3151 | lambda 0.635 | tr_acc 80.2% | val_loss 0.5940 | val_acc 80.
0% *
[full_aw30_s1:MotionSense] epoch 17 | act_loss 0.4841 | dom_loss
1.3160 | lambda 0.691 | tr_acc 80.9% | val_loss 0.5592 | val_acc 81.
3% *
[full_aw30_s1:MotionSense] epoch 20 | act_loss 0.4805 | dom_loss
1.3193 | lambda 0.762 | tr_acc 81.1% | val_loss 0.5984 | val_acc 78.
2%
[full_aw30_s1:MotionSense] epoch 36 | act_loss 0.4663 | dom_loss
1.3403 | lambda 0.947 | tr_acc 82.1% | val_loss 0.5821 | val_acc 81.
8% *
[full_aw30_s1:MotionSense] epoch 40 | act_loss 0.4584 | dom_loss
1.3420 | lambda 0.964 | tr_acc 82.6% | val_loss 0.5436 | val_acc 81.
2%
[full_aw30_s1:MotionSense] epoch 43 | act_loss 0.4534 | dom_loss
1.3420 | lambda 0.973 | tr_acc 83.0% | val_loss 0.5439 | val_acc 82.
1% *
[full_aw30_s1:MotionSense] epoch 59 | act_loss 0.4269 | dom_loss
1.3426 | lambda 0.995 | tr_acc 84.1% | val_loss 0.5317 | val_acc 82.
7% *
[full_aw30_s1:MotionSense] epoch 60 | act_loss 0.4254 | dom_loss
1.3437 | lambda 0.995 | tr_acc 84.2% | val_loss 0.5861 | val_acc 80.
4%
[full_aw30_s1:MotionSense] epoch 66 | act_loss 0.4097 | dom_loss
1.3430 | lambda 0.997 | tr_acc 85.0% | val_loss 0.5504 | val_acc 83.
0% *
[full_aw30_s1:MotionSense] epoch 73 | act_loss 0.3957 | dom_loss
1.3433 | lambda 0.999 | tr_acc 85.5% | val_loss 0.5278 | val_acc 83.
1% *

```

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[full_aw30_s1:MotionSense] epoch 80 | act_loss 0.3856 | dom_loss
1.3427 | lambda 0.999 | tr_acc 86.2% | val_loss 0.6150 | val_acc 74.
8%
[full_aw30_s1:MotionSense] early stop at epoch 93
[full_aw30_s1:MotionSense] src-val 83.11% | TARGET-TEST 76.84% |
saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%, (0, 3):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_aw30_s1:RealWorld-Thigh] trainable params: 51,531
[full_aw30_s1:RealWorld-Thigh] epoch 1 | act_loss 0.7784 | dom_l
oss 1.2637 | lambda 0.050 | tr_acc 66.2% | val_loss 0.8281 | val_acc
61.6% *
[full_aw30_s1:RealWorld-Thigh] epoch 2 | act_loss 0.5977 | dom_l
oss 1.1751 | lambda 0.100 | tr_acc 74.7% | val_loss 0.6800 | val_acc
73.1% *
[full_aw30_s1:RealWorld-Thigh] epoch 3 | act_loss 0.5599 | dom_l
oss 1.1871 | lambda 0.149 | tr_acc 76.6% | val_loss 0.6447 | val_acc
75.4% *
[full_aw30_s1:RealWorld-Thigh] epoch 4 | act_loss 0.5411 | dom_l
oss 1.2169 | lambda 0.197 | tr_acc 77.5% | val_loss 0.6150 | val_acc
76.4% *
[full_aw30_s1:RealWorld-Thigh] epoch 10 | act_loss 0.4891 | dom_l
oss 1.2548 | lambda 0.462 | tr_acc 79.8% | val_loss 0.5425 | val_acc
79.5% *
[full_aw30_s1:RealWorld-Thigh] epoch 11 | act_loss 0.4783 | dom_l
oss 1.2601 | lambda 0.501 | tr_acc 80.4% | val_loss 0.5682 | val_acc
79.7% *
[full_aw30_s1:RealWorld-Thigh] epoch 14 | act_loss 0.4660 | dom_l
oss 1.2634 | lambda 0.604 | tr_acc 81.0% | val_loss 0.5464 | val_acc
80.0% *
[full_aw30_s1:RealWorld-Thigh] epoch 20 | act_loss 0.4447 | dom_l
oss 1.2714 | lambda 0.762 | tr_acc 82.1% | val_loss 0.6513 | val_acc
75.2%
[full_aw30_s1:RealWorld-Thigh] epoch 21 | act_loss 0.4423 | dom_l
oss 1.2748 | lambda 0.782 | tr_acc 82.1% | val_loss 0.5165 | val_acc
81.9% *
[full_aw30_s1:RealWorld-Thigh] epoch 24 | act_loss 0.4356 | dom_l

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oss 1.2755 | lambda 0.834 | tr_acc 82.5% | val_loss 0.4977 | val_acc
82.4% *
    [full_aw30_s1:RealWorld-Thigh] epoch 25 | act_loss 0.4427 | dom_l
oss 1.2811 | lambda 0.848 | tr_acc 82.1% | val_loss 0.5129 | val_acc
82.8% *
    [full_aw30_s1:RealWorld-Thigh] epoch 39 | act_loss 0.4326 | dom_l
oss 1.3085 | lambda 0.960 | tr_acc 82.9% | val_loss 0.4716 | val_acc
83.5% *
    [full_aw30_s1:RealWorld-Thigh] epoch 40 | act_loss 0.4349 | dom_l
oss 1.3121 | lambda 0.964 | tr_acc 83.0% | val_loss 0.6900 | val_acc
74.0%
    [full_aw30_s1:RealWorld-Thigh] epoch 41 | act_loss 0.4312 | dom_l
oss 1.3091 | lambda 0.967 | tr_acc 83.1% | val_loss 0.4850 | val_acc
83.7% *
    [full_aw30_s1:RealWorld-Thigh] epoch 59 | act_loss 0.3972 | dom_l
oss 1.3097 | lambda 0.995 | tr_acc 84.5% | val_loss 0.5670 | val_acc
83.7% *
    [full_aw30_s1:RealWorld-Thigh] epoch 60 | act_loss 0.3953 | dom_l
oss 1.3141 | lambda 0.995 | tr_acc 84.7% | val_loss 0.5636 | val_acc
80.2%
    [full_aw30_s1:RealWorld-Thigh] epoch 62 | act_loss 0.3866 | dom_l
oss 1.3095 | lambda 0.996 | tr_acc 85.3% | val_loss 0.5370 | val_acc
84.1% *
    [full_aw30_s1:RealWorld-Thigh] epoch 63 | act_loss 0.3911 | dom_l
oss 1.3088 | lambda 0.996 | tr_acc 85.1% | val_loss 0.4769 | val_acc
85.3% *
    [full_aw30_s1:RealWorld-Thigh] epoch 78 | act_loss 0.3569 | dom_l
oss 1.3102 | lambda 0.999 | tr_acc 86.8% | val_loss 0.4983 | val_acc
86.1% *
    [full_aw30_s1:RealWorld-Thigh] epoch 80 | act_loss 0.3581 | dom_l
oss 1.3085 | lambda 0.999 | tr_acc 86.5% | val_loss 0.5430 | val_acc
81.5%
    [full_aw30_s1:RealWorld-Thigh] early stop at epoch 98
    [full_aw30_s1:RealWorld-Thigh] src-val 86.14% | TARGET-TEST 67.53%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (0, 1):100%, (1, 2):100%, (3, 5):100%, (1, 4):100%
    feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
    building 26456 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built

```

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[full_aw30_s1:RealWorld-Waist] trainable params: 51,531
[full_aw30_s1:RealWorld-Waist] epoch 1 | act_loss 0.7948 | dom_l
oss 1.3263 | lambda 0.050 | tr_acc 65.7% | val_loss 0.8298 | val_acc
63.9% *
[full_aw30_s1:RealWorld-Waist] epoch 2 | act_loss 0.6306 | dom_l
oss 1.2555 | lambda 0.100 | tr_acc 73.4% | val_loss 0.7441 | val_acc
70.8% *
[full_aw30_s1:RealWorld-Waist] epoch 3 | act_loss 0.5898 | dom_l
oss 1.2371 | lambda 0.149 | tr_acc 75.4% | val_loss 0.7337 | val_acc
71.8% *
[full_aw30_s1:RealWorld-Waist] epoch 6 | act_loss 0.5287 | dom_l
oss 1.2579 | lambda 0.291 | tr_acc 78.3% | val_loss 0.6972 | val_acc
72.1% *
[full_aw30_s1:RealWorld-Waist] epoch 7 | act_loss 0.5192 | dom_l
oss 1.2564 | lambda 0.336 | tr_acc 78.4% | val_loss 0.6821 | val_acc
73.6% *
[full_aw30_s1:RealWorld-Waist] epoch 8 | act_loss 0.5081 | dom_l
oss 1.2630 | lambda 0.380 | tr_acc 78.9% | val_loss 0.6041 | val_acc
76.8% *
[full_aw30_s1:RealWorld-Waist] epoch 9 | act_loss 0.4954 | dom_l
oss 1.2630 | lambda 0.422 | tr_acc 79.7% | val_loss 0.6003 | val_acc
78.3% *
[full_aw30_s1:RealWorld-Waist] epoch 19 | act_loss 0.4512 | dom_l
oss 1.2767 | lambda 0.740 | tr_acc 82.2% | val_loss 0.5938 | val_acc
79.3% *
[full_aw30_s1:RealWorld-Waist] epoch 20 | act_loss 0.4491 | dom_l
oss 1.2779 | lambda 0.762 | tr_acc 82.5% | val_loss 0.7671 | val_acc
71.1%
[full_aw30_s1:RealWorld-Waist] epoch 29 | act_loss 0.4353 | dom_l
oss 1.2860 | lambda 0.896 | tr_acc 83.1% | val_loss 0.5481 | val_acc
80.8% *
[full_aw30_s1:RealWorld-Waist] epoch 38 | act_loss 0.4302 | dom_l
oss 1.3096 | lambda 0.956 | tr_acc 83.8% | val_loss 0.5653 | val_acc
81.1% *
[full_aw30_s1:RealWorld-Waist] epoch 40 | act_loss 0.4224 | dom_l
oss 1.3093 | lambda 0.964 | tr_acc 83.7% | val_loss 0.6011 | val_acc
79.3%
[full_aw30_s1:RealWorld-Waist] early stop at epoch 58
[full_aw30_s1:RealWorld-Waist] src-val 81.05% | TARGET-TEST 64.93%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (3, 4):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/34368 graphs

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built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw30_s1:UCI] trainable params: 51,531
[full_aw30_s1:UCI] epoch 1 | act_loss 0.7823 | dom_loss 1.3782 |
lambda 0.050 | tr_acc 66.8% | val_loss 0.7453 | val_acc 70.3% *
[full_aw30_s1:UCI] epoch 2 | act_loss 0.6287 | dom_loss 1.3202 |
lambda 0.100 | tr_acc 74.3% | val_loss 0.7922 | val_acc 71.6% *
[full_aw30_s1:UCI] epoch 6 | act_loss 0.5449 | dom_loss 1.3418 |
lambda 0.291 | tr_acc 78.2% | val_loss 0.7086 | val_acc 72.2% *
[full_aw30_s1:UCI] epoch 7 | act_loss 0.5313 | dom_loss 1.3443 |
lambda 0.336 | tr_acc 78.7% | val_loss 0.7451 | val_acc 72.5% *
[full_aw30_s1:UCI] epoch 8 | act_loss 0.5289 | dom_loss 1.3523 |
lambda 0.380 | tr_acc 78.9% | val_loss 0.5844 | val_acc 79.9% *
[full_aw30_s1:UCI] epoch 17 | act_loss 0.4859 | dom_loss 1.3661 |
lambda 0.691 | tr_acc 80.9% | val_loss 0.5397 | val_acc 81.6% *
[full_aw30_s1:UCI] epoch 18 | act_loss 0.4842 | dom_loss 1.3669 |
lambda 0.716 | tr_acc 81.1% | val_loss 0.5355 | val_acc 81.9% *
[full_aw30_s1:UCI] epoch 20 | act_loss 0.4811 | dom_loss 1.3691 |
lambda 0.762 | tr_acc 81.4% | val_loss 0.6623 | val_acc 76.7%
[full_aw30_s1:UCI] epoch 25 | act_loss 0.4657 | dom_loss 1.3712 |
lambda 0.848 | tr_acc 82.1% | val_loss 0.5434 | val_acc 82.3% *
[full_aw30_s1:UCI] epoch 40 | act_loss 0.4569 | dom_loss 1.3885 |
lambda 0.964 | tr_acc 82.7% | val_loss 0.6085 | val_acc 79.7%
[full_aw30_s1:UCI] early stop at epoch 45
[full_aw30_s1:UCI] src-val 82.30% | TARGET-TEST 78.55% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (3, 4):100%, (4, 5):100%, (1, 4):100%, (2, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 45 features each)...)
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw30_s1:WISDM] trainable params: 51,531
[full_aw30_s1:WISDM] epoch 1 | act_loss 0.8043 | dom_loss 1.3165
| lambda 0.050 | tr_acc 66.1% | val_loss 0.7649 | val_acc 70.1% *
[full_aw30_s1:WISDM] epoch 2 | act_loss 0.6171 | dom_loss 1.2715
| lambda 0.100 | tr_acc 75.0% | val_loss 0.6672 | val_acc 75.2% *

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[full_aw30_s1:WISDM] epoch 16 | act_loss 0.4615 | dom_loss 1.3299
| lambda 0.664 | tr_acc 82.4% | val_loss 0.7265 | val_acc 75.3% *
[full_aw30_s1:WISDM] epoch 18 | act_loss 0.4545 | dom_loss 1.3317
| lambda 0.716 | tr_acc 82.8% | val_loss 0.6971 | val_acc 78.9% *
[full_aw30_s1:WISDM] epoch 19 | act_loss 0.4488 | dom_loss 1.3314
| lambda 0.740 | tr_acc 83.1% | val_loss 0.6201 | val_acc 78.9% *
[full_aw30_s1:WISDM] epoch 20 | act_loss 0.4500 | dom_loss 1.3306
| lambda 0.762 | tr_acc 82.9% | val_loss 0.7891 | val_acc 73.1%
[full_aw30_s1:WISDM] epoch 21 | act_loss 0.4419 | dom_loss 1.3307
| lambda 0.782 | tr_acc 83.4% | val_loss 0.5638 | val_acc 82.1% *
[full_aw30_s1:WISDM] epoch 40 | act_loss 0.4139 | dom_loss 1.3364
| lambda 0.964 | tr_acc 84.8% | val_loss 0.7280 | val_acc 75.2%
[full_aw30_s1:WISDM] early stop at epoch 41
[full_aw30_s1:WISDM] src-val 82.12% | TARGET-TEST 75.42% | saved
ckpt + npz
[P24] train+save | arm=full_aw30_s2 {'cnn_embed_dim': 64, 'gnn_hidden_dim': 64, 'cnn_channels': [32, 64]} | nc=None aw=30 | seed=2 | targets=
['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI',
'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20
_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw30_s2
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(0, 2):100%, (3, 4):100%, (4, 5):100%, (1, 2):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 45 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_aw30_s2:KuHar] trainable params: 51,531
[full_aw30_s2:KuHar] epoch 1 | act_loss 0.7755 | dom_loss 1.4193
| lambda 0.050 | tr_acc 67.2% | val_loss 0.7728 | val_acc 66.1% *
[full_aw30_s2:KuHar] epoch 4 | act_loss 0.5578 | dom_loss 1.3702
| lambda 0.197 | tr_acc 77.5% | val_loss 0.6875 | val_acc 75.1% *
[full_aw30_s2:KuHar] epoch 9 | act_loss 0.5118 | dom_loss 1.3950
| lambda 0.422 | tr_acc 79.7% | val_loss 0.6408 | val_acc 77.5% *
[full_aw30_s2:KuHar] epoch 12 | act_loss 0.4921 | dom_loss 1.3963
| lambda 0.537 | tr_acc 80.7% | val_loss 0.5748 | val_acc 79.4% *
[full_aw30_s2:KuHar] epoch 16 | act_loss 0.4803 | dom_loss 1.4016
| lambda 0.664 | tr_acc 81.5% | val_loss 0.5991 | val_acc 80.0% *
[full_aw30_s2:KuHar] epoch 20 | act_loss 0.4678 | dom_loss 1.4030
| lambda 0.762 | tr_acc 82.3% | val_loss 0.5837 | val_acc 81.3% *

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[full_aw30_s2:KuHar] epoch 31 | act_loss 0.4552 | dom_loss 1.4096
| lambda 0.914 | tr_acc 82.9% | val_loss 0.5408 | val_acc 81.4% *
[full_aw30_s2:KuHar] epoch 40 | act_loss 0.4481 | dom_loss 1.4205
| lambda 0.964 | tr_acc 83.2% | val_loss 0.6275 | val_acc 79.4%
[full_aw30_s2:KuHar] epoch 42 | act_loss 0.4396 | dom_loss 1.4202
| lambda 0.970 | tr_acc 83.6% | val_loss 0.5442 | val_acc 82.6% *
[full_aw30_s2:KuHar] epoch 43 | act_loss 0.4440 | dom_loss 1.4229
| lambda 0.973 | tr_acc 83.5% | val_loss 0.5375 | val_acc 83.2% *
[full_aw30_s2:KuHar] epoch 60 | act_loss 0.4075 | dom_loss 1.4221
| lambda 0.995 | tr_acc 85.2% | val_loss 0.5610 | val_acc 80.5%
[full_aw30_s2:KuHar] early stop at epoch 63
[full_aw30_s2:KuHar] src-val 83.19% | TARGET-TEST 62.50% | saved
ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (0, 1):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_aw30_s2:MotionSense] trainable params: 51,531
[full_aw30_s2:MotionSense] epoch 1 | act_loss 0.7961 | dom_loss
1.3376 | lambda 0.050 | tr_acc 65.8% | val_loss 0.8500 | val_acc 66.
9% *
[full_aw30_s2:MotionSense] epoch 2 | act_loss 0.6306 | dom_loss
1.2610 | lambda 0.100 | tr_acc 73.7% | val_loss 0.6978 | val_acc 72.
7% *
[full_aw30_s2:MotionSense] epoch 5 | act_loss 0.5504 | dom_loss
1.2873 | lambda 0.245 | tr_acc 77.5% | val_loss 0.6316 | val_acc 74.
1% *
[full_aw30_s2:MotionSense] epoch 6 | act_loss 0.5461 | dom_loss
1.2951 | lambda 0.291 | tr_acc 77.7% | val_loss 0.6886 | val_acc 74.
9% *
[full_aw30_s2:MotionSense] epoch 8 | act_loss 0.5150 | dom_loss
1.3039 | lambda 0.380 | tr_acc 79.3% | val_loss 0.5971 | val_acc 78.
9% *
[full_aw30_s2:MotionSense] epoch 17 | act_loss 0.4816 | dom_loss
1.3180 | lambda 0.691 | tr_acc 81.2% | val_loss 0.5725 | val_acc 80.
3% *
[full_aw30_s2:MotionSense] epoch 20 | act_loss 0.4682 | dom_loss
1.3188 | lambda 0.762 | tr_acc 81.9% | val_loss 0.6839 | val_acc 74.

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3%
[full_aw30_s2:MotionSense] epoch 28 | act_loss 0.4667 | dom_loss
1.3232 | lambda 0.885 | tr_acc 82.1% | val_loss 0.5829 | val_acc 80.
9% *
[full_aw30_s2:MotionSense] epoch 40 | act_loss 0.4588 | dom_loss
1.3362 | lambda 0.964 | tr_acc 82.6% | val_loss 0.6194 | val_acc 77.
4%
[full_aw30_s2:MotionSense] epoch 48 | act_loss 0.4486 | dom_loss
1.3412 | lambda 0.984 | tr_acc 83.1% | val_loss 0.5920 | val_acc 81.
1% *
[full_aw30_s2:MotionSense] epoch 60 | act_loss 0.4254 | dom_loss
1.3436 | lambda 0.995 | tr_acc 84.4% | val_loss 0.6523 | val_acc 78.
0%
[full_aw30_s2:MotionSense] early stop at epoch 68
[full_aw30_s2:MotionSense] src-val 81.10% | TARGET-TEST 75.61% |
saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%, (0, 3):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_aw30_s2:RealWorld-Thigh] trainable params: 51,531
[full_aw30_s2:RealWorld-Thigh] epoch 1 | act_loss 0.7687 | dom_l
oss 1.2779 | lambda 0.050 | tr_acc 66.4% | val_loss 0.7840 | val_acc
65.3% *
[full_aw30_s2:RealWorld-Thigh] epoch 2 | act_loss 0.6000 | dom_l
oss 1.1940 | lambda 0.100 | tr_acc 74.6% | val_loss 0.6820 | val_acc
73.0% *
[full_aw30_s2:RealWorld-Thigh] epoch 4 | act_loss 0.5426 | dom_l
oss 1.2084 | lambda 0.197 | tr_acc 77.4% | val_loss 0.5442 | val_acc
78.9% *
[full_aw30_s2:RealWorld-Thigh] epoch 12 | act_loss 0.4702 | dom_l
oss 1.2602 | lambda 0.537 | tr_acc 80.8% | val_loss 0.5568 | val_acc
82.5% *
[full_aw30_s2:RealWorld-Thigh] epoch 20 | act_loss 0.4486 | dom_l
oss 1.2757 | lambda 0.762 | tr_acc 81.8% | val_loss 0.5306 | val_acc
81.5%
[full_aw30_s2:RealWorld-Thigh] early stop at epoch 32
[full_aw30_s2:RealWorld-Thigh] src-val 82.53% | TARGET-TEST 65.53%
| saved ckpt + npz

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[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (0, 1):100%, (1, 2):100%, (3, 5):100%, (1, 4):100%
    feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
    building 26456 sensor graphs (6 nodes, 18 edges, 45 features each)...)
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_aw30_s2:RealWorld-Waist] trainable params: 51,531
[full_aw30_s2:RealWorld-Waist] epoch 1 | act_loss 0.7830 | dom_loss 1.3264 | lambda 0.050 | tr_acc 66.0% | val_loss 0.7015 | val_acc 70.8% *
[full_aw30_s2:RealWorld-Waist] epoch 3 | act_loss 0.5855 | dom_loss 1.2490 | lambda 0.149 | tr_acc 75.8% | val_loss 0.6420 | val_acc 74.9% *
[full_aw30_s2:RealWorld-Waist] epoch 5 | act_loss 0.5402 | dom_loss 1.2635 | lambda 0.245 | tr_acc 77.8% | val_loss 0.6398 | val_acc 75.9% *
[full_aw30_s2:RealWorld-Waist] epoch 9 | act_loss 0.4993 | dom_loss 1.2649 | lambda 0.422 | tr_acc 79.5% | val_loss 0.6442 | val_acc 76.2% *
[full_aw30_s2:RealWorld-Waist] epoch 10 | act_loss 0.4917 | dom_loss 1.2641 | lambda 0.462 | tr_acc 79.9% | val_loss 0.6111 | val_acc 78.4% *
[full_aw30_s2:RealWorld-Waist] epoch 16 | act_loss 0.4586 | dom_loss 1.2782 | lambda 0.664 | tr_acc 81.6% | val_loss 0.5274 | val_acc 80.7% *
[full_aw30_s2:RealWorld-Waist] epoch 20 | act_loss 0.4481 | dom_loss 1.2813 | lambda 0.762 | tr_acc 82.2% | val_loss 0.6119 | val_acc 77.6%
[full_aw30_s2:RealWorld-Waist] early stop at epoch 36
[full_aw30_s2:RealWorld-Waist] src-val 80.73% | TARGET-TEST 70.49%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%

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irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (3, 4):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw30_s2:UCI] trainable params: 51,531
[full_aw30_s2:UCI] epoch 1 | act_loss 0.7826 | dom_loss 1.3841 |
lambda 0.050 | tr_acc 66.9% | val_loss 0.7539 | val_acc 68.8% *
[full_aw30_s2:UCI] epoch 2 | act_loss 0.6283 | dom_loss 1.3228 |
lambda 0.100 | tr_acc 74.0% | val_loss 0.7737 | val_acc 69.1% *
[full_aw30_s2:UCI] epoch 3 | act_loss 0.5903 | dom_loss 1.3253 |
lambda 0.149 | tr_acc 76.0% | val_loss 0.6563 | val_acc 70.4% *
[full_aw30_s2:UCI] epoch 5 | act_loss 0.5498 | dom_loss 1.3365 |
lambda 0.245 | tr_acc 77.8% | val_loss 0.6740 | val_acc 75.3% *
[full_aw30_s2:UCI] epoch 7 | act_loss 0.5254 | dom_loss 1.3438 |
lambda 0.336 | tr_acc 79.0% | val_loss 0.6194 | val_acc 78.0% *
[full_aw30_s2:UCI] epoch 10 | act_loss 0.5035 | dom_loss 1.3562 |
lambda 0.462 | tr_acc 80.0% | val_loss 0.5983 | val_acc 78.4% *
[full_aw30_s2:UCI] epoch 12 | act_loss 0.4961 | dom_loss 1.3601 |
lambda 0.537 | tr_acc 80.4% | val_loss 0.5989 | val_acc 78.8% *
[full_aw30_s2:UCI] epoch 17 | act_loss 0.4774 | dom_loss 1.3647 |
lambda 0.691 | tr_acc 81.5% | val_loss 0.6187 | val_acc 79.0% *
[full_aw30_s2:UCI] epoch 19 | act_loss 0.4745 | dom_loss 1.3662 |
lambda 0.740 | tr_acc 81.6% | val_loss 0.5566 | val_acc 81.1% *
[full_aw30_s2:UCI] epoch 20 | act_loss 0.4722 | dom_loss 1.3666 |
lambda 0.762 | tr_acc 81.5% | val_loss 0.5641 | val_acc 80.6%
[full_aw30_s2:UCI] epoch 34 | act_loss 0.4588 | dom_loss 1.3815 |
lambda 0.935 | tr_acc 82.6% | val_loss 0.5438 | val_acc 81.9% *
[full_aw30_s2:UCI] epoch 40 | act_loss 0.4492 | dom_loss 1.3872 |
lambda 0.964 | tr_acc 83.2% | val_loss 0.6736 | val_acc 70.9%
[full_aw30_s2:UCI] early stop at epoch 54
[full_aw30_s2:UCI] src-val 81.92% | TARGET-TEST 78.84% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (4, 5):100%, (1, 4):100%, (2, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...

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built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw30_s2:WISDM] trainable params: 51,531
[full_aw30_s2:WISDM] epoch 1 | act_loss 0.7925 | dom_loss 1.3191
| lambda 0.050 | tr_acc 66.9% | val_loss 0.8019 | val_acc 69.0% *
[full_aw30_s2:WISDM] epoch 2 | act_loss 0.6175 | dom_loss 1.2832
| lambda 0.100 | tr_acc 75.0% | val_loss 0.6545 | val_acc 75.9% *
[full_aw30_s2:WISDM] epoch 8 | act_loss 0.4983 | dom_loss 1.3215
| lambda 0.380 | tr_acc 80.7% | val_loss 0.6672 | val_acc 76.6% *
[full_aw30_s2:WISDM] epoch 11 | act_loss 0.4825 | dom_loss 1.3275
| lambda 0.501 | tr_acc 81.3% | val_loss 0.6379 | val_acc 78.4% *
[full_aw30_s2:WISDM] epoch 20 | act_loss 0.4434 | dom_loss 1.3327
| lambda 0.762 | tr_acc 83.6% | val_loss 0.7754 | val_acc 73.8%
[full_aw30_s2:WISDM] epoch 21 | act_loss 0.4462 | dom_loss 1.3322
| lambda 0.782 | tr_acc 83.2% | val_loss 0.6500 | val_acc 79.1% *
[full_aw30_s2:WISDM] epoch 23 | act_loss 0.4393 | dom_loss 1.3326
| lambda 0.818 | tr_acc 83.7% | val_loss 0.6205 | val_acc 80.3% *
[full_aw30_s2:WISDM] epoch 25 | act_loss 0.4367 | dom_loss 1.3356
| lambda 0.848 | tr_acc 83.7% | val_loss 0.6103 | val_acc 80.7% *
[full_aw30_s2:WISDM] epoch 33 | act_loss 0.4261 | dom_loss 1.3363
| lambda 0.929 | tr_acc 84.2% | val_loss 0.5855 | val_acc 81.3% *
[full_aw30_s2:WISDM] epoch 40 | act_loss 0.4112 | dom_loss 1.3364
| lambda 0.964 | tr_acc 85.0% | val_loss 0.7623 | val_acc 74.1%
[full_aw30_s2:WISDM] epoch 46 | act_loss 0.3940 | dom_loss 1.3370
| lambda 0.980 | tr_acc 85.5% | val_loss 0.6012 | val_acc 81.4% *
[full_aw30_s2:WISDM] epoch 60 | act_loss 0.3617 | dom_loss 1.3386
| lambda 0.995 | tr_acc 87.0% | val_loss 0.6177 | val_acc 79.3%
[full_aw30_s2:WISDM] epoch 62 | act_loss 0.3599 | dom_loss 1.3386
| lambda 0.996 | tr_acc 87.2% | val_loss 0.5847 | val_acc 81.7% *
[full_aw30_s2:WISDM] epoch 68 | act_loss 0.3453 | dom_loss 1.3384
| lambda 0.998 | tr_acc 87.7% | val_loss 0.5724 | val_acc 82.0% *
[full_aw30_s2:WISDM] epoch 69 | act_loss 0.3512 | dom_loss 1.3387
| lambda 0.998 | tr_acc 87.5% | val_loss 0.5749 | val_acc 82.3% *
[full_aw30_s2:WISDM] epoch 80 | act_loss 0.3239 | dom_loss 1.3382
| lambda 0.999 | tr_acc 88.6% | val_loss 0.6964 | val_acc 78.3%
[full_aw30_s2:WISDM] early stop at epoch 89
[full_aw30_s2:WISDM] src-val 82.27% | TARGET-TEST 70.81% | saved
ckpt + npz
[P24] train+save | arm=full_aw30_s3 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=30 | seed=3 | targets=
['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI',
'WISDM'] | out=/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20
_GraphRevival_DG/P24_S0_Compression/results/checkpoints/full_aw30_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa

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irs: (0, 2):100%, (3, 4):100%, (4, 5):100%, (1, 2):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_aw30_s3:KuHar] trainable params: 51,531
[full_aw30_s3:KuHar] epoch 1 | act_loss 0.7712 | dom_loss 1.4165
| lambda 0.050 | tr_acc 67.6% | val_loss 0.6681 | val_acc 74.3% *
[full_aw30_s3:KuHar] epoch 2 | act_loss 0.6074 | dom_loss 1.3548
| lambda 0.100 | tr_acc 75.1% | val_loss 0.6488 | val_acc 74.9% *
[full_aw30_s3:KuHar] epoch 6 | act_loss 0.5274 | dom_loss 1.3846
| lambda 0.291 | tr_acc 78.9% | val_loss 0.5861 | val_acc 78.5% *
[full_aw30_s3:KuHar] epoch 7 | act_loss 0.5187 | dom_loss 1.3902
| lambda 0.336 | tr_acc 79.3% | val_loss 0.5736 | val_acc 79.0% *
[full_aw30_s3:KuHar] epoch 8 | act_loss 0.5139 | dom_loss 1.3916
| lambda 0.380 | tr_acc 79.5% | val_loss 0.5651 | val_acc 80.0% *
[full_aw30_s3:KuHar] epoch 17 | act_loss 0.4727 | dom_loss 1.4011
| lambda 0.691 | tr_acc 81.8% | val_loss 0.5585 | val_acc 81.5% *
[full_aw30_s3:KuHar] epoch 20 | act_loss 0.4702 | dom_loss 1.4039
| lambda 0.762 | tr_acc 82.2% | val_loss 0.9403 | val_acc 67.4%
[full_aw30_s3:KuHar] early stop at epoch 37
[full_aw30_s3:KuHar] src-val 81.54% | TARGET-TEST 56.94% | saved
ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (0, 1):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_aw30_s3:MotionSense] trainable params: 51,531
[full_aw30_s3:MotionSense] epoch 1 | act_loss 0.8017 | dom_loss

```

```

1.3284 | lambda 0.050 | tr_acc 65.4% | val_loss 0.9250 | val_acc 64.9% *
[full_aw30_s3:MotionSense] epoch 2 | act_loss 0.6312 | dom_loss
1.2542 | lambda 0.100 | tr_acc 73.8% | val_loss 0.7410 | val_acc 67.7% *
[full_aw30_s3:MotionSense] epoch 3 | act_loss 0.5882 | dom_loss
1.2682 | lambda 0.149 | tr_acc 75.9% | val_loss 0.6274 | val_acc 75.7% *
[full_aw30_s3:MotionSense] epoch 5 | act_loss 0.5536 | dom_loss
1.2838 | lambda 0.245 | tr_acc 77.2% | val_loss 0.6434 | val_acc 76.3% *
[full_aw30_s3:MotionSense] epoch 8 | act_loss 0.5205 | dom_loss
1.3026 | lambda 0.380 | tr_acc 78.9% | val_loss 0.6133 | val_acc 78.0% *
[full_aw30_s3:MotionSense] epoch 10 | act_loss 0.5075 | dom_loss
1.3075 | lambda 0.462 | tr_acc 79.6% | val_loss 0.6038 | val_acc 78.5% *
[full_aw30_s3:MotionSense] epoch 20 | act_loss 0.4766 | dom_loss
1.3186 | lambda 0.762 | tr_acc 81.4% | val_loss 0.7077 | val_acc 73.7%
[full_aw30_s3:MotionSense] epoch 23 | act_loss 0.4738 | dom_loss
1.3196 | lambda 0.818 | tr_acc 81.5% | val_loss 0.5435 | val_acc 81.1% *
[full_aw30_s3:MotionSense] epoch 40 | act_loss 0.4569 | dom_loss
1.3398 | lambda 0.964 | tr_acc 82.7% | val_loss 0.5894 | val_acc 79.6%
[full_aw30_s3:MotionSense] epoch 41 | act_loss 0.4576 | dom_loss
1.3385 | lambda 0.967 | tr_acc 82.5% | val_loss 0.5427 | val_acc 82.0% *
[full_aw30_s3:MotionSense] epoch 46 | act_loss 0.4487 | dom_loss
1.3434 | lambda 0.980 | tr_acc 83.1% | val_loss 0.5876 | val_acc 82.4% *
[full_aw30_s3:MotionSense] epoch 60 | act_loss 0.4206 | dom_loss
1.3407 | lambda 0.995 | tr_acc 84.5% | val_loss 0.6517 | val_acc 72.9%
[full_aw30_s3:MotionSense] early stop at epoch 66
[full_aw30_s3:MotionSense] src-val 82.36% | TARGET-TEST 75.71% |
saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(1, 2):100%, (0, 1):100%, (0, 2):100%, (0, 3):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 45 features each)
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs

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```

    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
[full_aw30_s3:RealWorld-Thigh] trainable params: 51,531
[full_aw30_s3:RealWorld-Thigh] epoch 1 | act_loss 0.7746 | dom_l
oss 1.2791 | lambda 0.050 | tr_acc 67.1% | val_loss 0.6621 | val_acc
72.9% *
[full_aw30_s3:RealWorld-Thigh] epoch 4 | act_loss 0.5371 | dom_l
oss 1.2162 | lambda 0.197 | tr_acc 77.7% | val_loss 0.5895 | val_acc
77.0% *
[full_aw30_s3:RealWorld-Thigh] epoch 5 | act_loss 0.5179 | dom_l
oss 1.2143 | lambda 0.245 | tr_acc 78.3% | val_loss 0.5691 | val_acc
79.0% *
[full_aw30_s3:RealWorld-Thigh] epoch 9 | act_loss 0.4872 | dom_l
oss 1.2544 | lambda 0.422 | tr_acc 80.0% | val_loss 0.5373 | val_acc
79.2% *
[full_aw30_s3:RealWorld-Thigh] epoch 10 | act_loss 0.4791 | dom_l
oss 1.2560 | lambda 0.462 | tr_acc 80.2% | val_loss 0.5585 | val_acc
79.6% *
[full_aw30_s3:RealWorld-Thigh] epoch 13 | act_loss 0.4654 | dom_l
oss 1.2653 | lambda 0.572 | tr_acc 81.1% | val_loss 0.5012 | val_acc
81.5% *
[full_aw30_s3:RealWorld-Thigh] epoch 17 | act_loss 0.4521 | dom_l
oss 1.2707 | lambda 0.691 | tr_acc 81.8% | val_loss 0.4942 | val_acc
83.7% *
[full_aw30_s3:RealWorld-Thigh] epoch 20 | act_loss 0.4429 | dom_l
oss 1.2747 | lambda 0.762 | tr_acc 82.1% | val_loss 0.5674 | val_acc
74.3%
[full_aw30_s3:RealWorld-Thigh] early stop at epoch 37
[full_aw30_s3:RealWorld-Thigh] src-val 83.71% | TARGET-TEST 69.57%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (0, 1):100%, (1, 2):100%, (3, 5):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
[full_aw30_s3:RealWorld-Waist] trainable params: 51,531
[full_aw30_s3:RealWorld-Waist] epoch 1 | act_loss 0.7982 | dom_l
oss 1.3304 | lambda 0.050 | tr_acc 65.4% | val_loss 0.8931 | val_acc
62.8% *

```

```

[full_aw30_s3:RealWorld-Waist] epoch 2 | act_loss 0.6205 | dom_loss 1.2661 | lambda 0.100 | tr_acc 73.8% | val_loss 0.6281 | val_acc 76.0% *
[full_aw30_s3:RealWorld-Waist] epoch 7 | act_loss 0.5127 | dom_loss 1.2637 | lambda 0.336 | tr_acc 78.8% | val_loss 0.6096 | val_acc 77.3% *
[full_aw30_s3:RealWorld-Waist] epoch 8 | act_loss 0.5014 | dom_loss 1.2660 | lambda 0.380 | tr_acc 79.3% | val_loss 0.5665 | val_acc 79.1% *
[full_aw30_s3:RealWorld-Waist] epoch 16 | act_loss 0.4566 | dom_loss 1.2764 | lambda 0.664 | tr_acc 81.8% | val_loss 0.5583 | val_acc 80.1% *
[full_aw30_s3:RealWorld-Waist] epoch 20 | act_loss 0.4442 | dom_loss 1.2789 | lambda 0.762 | tr_acc 82.7% | val_loss 0.7013 | val_acc 73.9%
[full_aw30_s3:RealWorld-Waist] epoch 34 | act_loss 0.4302 | dom_loss 1.3056 | lambda 0.935 | tr_acc 83.6% | val_loss 0.5289 | val_acc 81.2% *
[full_aw30_s3:RealWorld-Waist] epoch 40 | act_loss 0.4205 | dom_loss 1.3100 | lambda 0.964 | tr_acc 84.0% | val_loss 0.5728 | val_acc 80.5%
[full_aw30_s3:RealWorld-Waist] early stop at epoch 54
[full_aw30_s3:RealWorld-Waist] src-val 81.15% | TARGET-TEST 62.81%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (4, 5):100%, (1, 2):100%, (3, 4):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 45 features each)...)
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_aw30_s3:UCI] trainable params: 51,531
[full_aw30_s3:UCI] epoch 1 | act_loss 0.7875 | dom_loss 1.3892 | lambda 0.050 | tr_acc 66.9% | val_loss 0.7486 | val_acc 70.8% *
[full_aw30_s3:UCI] epoch 4 | act_loss 0.5690 | dom_loss 1.3290 | lambda 0.197 | tr_acc 77.1% | val_loss 0.6887 | val_acc 74.2% *
[full_aw30_s3:UCI] epoch 6 | act_loss 0.5428 | dom_loss 1.3465 | lambda 0.291 | tr_acc 78.2% | val_loss 0.6218 | val_acc 77.3% *
[full_aw30_s3:UCI] epoch 9 | act_loss 0.5142 | dom_loss 1.3567 | lambda 0.422 | tr_acc 79.7% | val_loss 0.5841 | val_acc 79.7% *
[full_aw30_s3:UCI] epoch 11 | act_loss 0.5037 | dom_loss 1.3610 |

```

```

lambda 0.501 | tr_acc 80.1% | val_loss 0.5574 | val_acc 80.8% *
[full_aw30_s3:UCI] epoch 20 | act_loss 0.4759 | dom_loss 1.3678 |
lambda 0.762 | tr_acc 81.6% | val_loss 0.6121 | val_acc 76.9%
[full_aw30_s3:UCI] epoch 28 | act_loss 0.4670 | dom_loss 1.3725 |
lambda 0.885 | tr_acc 82.3% | val_loss 0.5636 | val_acc 82.0% *
[full_aw30_s3:UCI] epoch 40 | act_loss 0.4567 | dom_loss 1.3863 |
lambda 0.964 | tr_acc 82.9% | val_loss 0.6304 | val_acc 79.2%
[full_aw30_s3:UCI] early stop at epoch 48
[full_aw30_s3:UCI] src-val 82.01% | TARGET-TEST 80.00% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (3, 4):100%, (4, 5):100%, (1, 4):100%, (2, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 45 features each)...)
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_aw30_s3:WISDM] trainable params: 51,531
[full_aw30_s3:WISDM] epoch 1 | act_loss 0.7961 | dom_loss 1.3289
| lambda 0.050 | tr_acc 66.7% | val_loss 0.7540 | val_acc 70.5% *
[full_aw30_s3:WISDM] epoch 3 | act_loss 0.5677 | dom_loss 1.3027
| lambda 0.149 | tr_acc 77.4% | val_loss 0.7262 | val_acc 72.7% *
[full_aw30_s3:WISDM] epoch 4 | act_loss 0.5409 | dom_loss 1.3033
| lambda 0.197 | tr_acc 78.9% | val_loss 0.6864 | val_acc 74.4% *
[full_aw30_s3:WISDM] epoch 5 | act_loss 0.5321 | dom_loss 1.3149
| lambda 0.245 | tr_acc 79.0% | val_loss 0.6561 | val_acc 76.9% *
[full_aw30_s3:WISDM] epoch 8 | act_loss 0.4958 | dom_loss 1.3236
| lambda 0.380 | tr_acc 80.9% | val_loss 0.6708 | val_acc 78.4% *
[full_aw30_s3:WISDM] epoch 15 | act_loss 0.4592 | dom_loss 1.3288
| lambda 0.635 | tr_acc 82.5% | val_loss 0.6155 | val_acc 79.7% *
[full_aw30_s3:WISDM] epoch 16 | act_loss 0.4557 | dom_loss 1.3297
| lambda 0.664 | tr_acc 82.8% | val_loss 0.6074 | val_acc 80.8% *
[full_aw30_s3:WISDM] epoch 20 | act_loss 0.4472 | dom_loss 1.3330
| lambda 0.762 | tr_acc 83.4% | val_loss 0.7009 | val_acc 74.8%
[full_aw30_s3:WISDM] epoch 28 | act_loss 0.4279 | dom_loss 1.3356
| lambda 0.885 | tr_acc 84.1% | val_loss 0.6807 | val_acc 80.8% *
[full_aw30_s3:WISDM] epoch 34 | act_loss 0.4207 | dom_loss 1.3351
| lambda 0.935 | tr_acc 84.4% | val_loss 0.5597 | val_acc 81.4% *
[full_aw30_s3:WISDM] epoch 40 | act_loss 0.4130 | dom_loss 1.3364
| lambda 0.964 | tr_acc 84.7% | val_loss 0.6081 | val_acc 77.4%
[full_aw30_s3:WISDM] early stop at epoch 54
[full_aw30_s3:WISDM] src-val 81.37% | TARGET-TEST 71.58% | saved

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```

ckpt + npz
[P24] train+save | arm=full_aw30_s5 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=None aw=30 | seed=5 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_aw30_s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 4):100%, (4, 5):100%, (1, 2):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (1, 2):100%, (3, 4):100%, (2, 5):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 45 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_aw30_s5:KuHar] trainable params: 51,531
[full_aw30_s5:KuHar] epoch 1 | act_loss 0.7814 | dom_loss 1.4123
| lambda 0.050 | tr_acc 66.8% | val_loss 1.2387 | val_acc 49.2% *
[full_aw30_s5:KuHar] epoch 2 | act_loss 0.6151 | dom_loss 1.3442
| lambda 0.100 | tr_acc 74.8% | val_loss 0.8039 | val_acc 67.1% *
[full_aw30_s5:KuHar] epoch 3 | act_loss 0.5806 | dom_loss 1.3611
| lambda 0.149 | tr_acc 76.5% | val_loss 0.6821 | val_acc 69.4% *
[full_aw30_s5:KuHar] epoch 5 | act_loss 0.5422 | dom_loss 1.3761
| lambda 0.245 | tr_acc 77.9% | val_loss 0.7531 | val_acc 72.5% *
[full_aw30_s5:KuHar] epoch 6 | act_loss 0.5379 | dom_loss 1.3830
| lambda 0.291 | tr_acc 78.3% | val_loss 0.6193 | val_acc 76.6% *
[full_aw30_s5:KuHar] epoch 7 | act_loss 0.5273 | dom_loss 1.3911
| lambda 0.336 | tr_acc 78.8% | val_loss 0.5796 | val_acc 79.2% *
[full_aw30_s5:KuHar] epoch 11 | act_loss 0.5051 | dom_loss 1.3993
| lambda 0.501 | tr_acc 80.0% | val_loss 0.6129 | val_acc 80.2% *
[full_aw30_s5:KuHar] epoch 16 | act_loss 0.4851 | dom_loss 1.4033
| lambda 0.664 | tr_acc 81.1% | val_loss 0.5609 | val_acc 80.2% *
[full_aw30_s5:KuHar] epoch 18 | act_loss 0.4811 | dom_loss 1.4033
| lambda 0.716 | tr_acc 81.2% | val_loss 0.5565 | val_acc 81.7% *
[full_aw30_s5:KuHar] epoch 20 | act_loss 0.4747 | dom_loss 1.4044
| lambda 0.762 | tr_acc 81.7% | val_loss 0.6062 | val_acc 78.0% *
[full_aw30_s5:KuHar] epoch 33 | act_loss 0.4565 | dom_loss 1.4103
| lambda 0.929 | tr_acc 82.6% | val_loss 0.5271 | val_acc 82.5% *
[full_aw30_s5:KuHar] epoch 39 | act_loss 0.4504 | dom_loss 1.4164
| lambda 0.960 | tr_acc 83.0% | val_loss 0.5540 | val_acc 82.6% *
[full_aw30_s5:KuHar] epoch 40 | act_loss 0.4495 | dom_loss 1.4198
| lambda 0.964 | tr_acc 83.1% | val_loss 0.5632 | val_acc 80.5%

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[full_aw30_s5:KuHar] epoch 54 | act_loss 0.4269 | dom_loss 1.4253
| lambda 0.991 | tr_acc 84.5% | val_loss 0.5211 | val_acc 83.7% *
[full_aw30_s5:KuHar] epoch 60 | act_loss 0.4147 | dom_loss 1.4239
| lambda 0.995 | tr_acc 84.8% | val_loss 0.5276 | val_acc 82.6%
[full_aw30_s5:KuHar] early stop at epoch 74
[full_aw30_s5:KuHar] src-val 83.67% | TARGET-TEST 65.97% | saved
ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (0, 1):100%, (4, 5):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_aw30_s5:MotionSense] trainable params: 51,531
[full_aw30_s5:MotionSense] epoch 1 | act_loss 0.8088 | dom_loss
1.3321 | lambda 0.050 | tr_acc 65.2% | val_loss 0.7278 | val_acc 67.
3% *
[full_aw30_s5:MotionSense] epoch 2 | act_loss 0.6315 | dom_loss
1.2501 | lambda 0.100 | tr_acc 73.4% | val_loss 0.7444 | val_acc 72.
7% *
[full_aw30_s5:MotionSense] epoch 3 | act_loss 0.5951 | dom_loss
1.2656 | lambda 0.149 | tr_acc 75.3% | val_loss 0.6000 | val_acc 77.
1% *
[full_aw30_s5:MotionSense] epoch 10 | act_loss 0.5180 | dom_loss
1.3111 | lambda 0.462 | tr_acc 79.1% | val_loss 0.5750 | val_acc 79.
9% *
[full_aw30_s5:MotionSense] epoch 20 | act_loss 0.4789 | dom_loss
1.3193 | lambda 0.762 | tr_acc 81.0% | val_loss 0.6302 | val_acc 79.
1%
[full_aw30_s5:MotionSense] epoch 25 | act_loss 0.4753 | dom_loss
1.3238 | lambda 0.848 | tr_acc 81.5% | val_loss 0.5656 | val_acc 81.
0% *
[full_aw30_s5:MotionSense] epoch 35 | act_loss 0.4755 | dom_loss
1.3390 | lambda 0.941 | tr_acc 81.7% | val_loss 0.5642 | val_acc 81.
3% *
[full_aw30_s5:MotionSense] epoch 40 | act_loss 0.4644 | dom_loss
1.3402 | lambda 0.964 | tr_acc 82.4% | val_loss 0.7048 | val_acc 71.
1%
[full_aw30_s5:MotionSense] early stop at epoch 55
[full_aw30_s5:MotionSense] src-val 81.34% | TARGET-TEST 73.35% |

```

```

saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 2):100%, (0, 3):100%, (3, 5):100%
    feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
    building 26450 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
[full_aw30_s5:RealWorld-Thigh] trainable params: 51,531
[full_aw30_s5:RealWorld-Thigh] epoch 1 | act_loss 0.7753 | dom_l
oss 1.2816 | lambda 0.050 | tr_acc 66.8% | val_loss 0.6994 | val_acc
70.7% *
[full_aw30_s5:RealWorld-Thigh] epoch 3 | act_loss 0.5663 | dom_l
oss 1.1969 | lambda 0.149 | tr_acc 76.2% | val_loss 0.6307 | val_acc
74.0% *
[full_aw30_s5:RealWorld-Thigh] epoch 5 | act_loss 0.5259 | dom_l
oss 1.2205 | lambda 0.245 | tr_acc 78.2% | val_loss 0.6142 | val_acc
77.3% *
[full_aw30_s5:RealWorld-Thigh] epoch 8 | act_loss 0.4945 | dom_l
oss 1.2506 | lambda 0.380 | tr_acc 79.5% | val_loss 0.5389 | val_acc
79.9% *
[full_aw30_s5:RealWorld-Thigh] epoch 10 | act_loss 0.4808 | dom_l
oss 1.2599 | lambda 0.462 | tr_acc 80.4% | val_loss 0.5205 | val_acc
81.4% *
[full_aw30_s5:RealWorld-Thigh] epoch 17 | act_loss 0.4576 | dom_l
oss 1.2715 | lambda 0.691 | tr_acc 81.5% | val_loss 0.4819 | val_acc
82.5% *
[full_aw30_s5:RealWorld-Thigh] epoch 20 | act_loss 0.4487 | dom_l
oss 1.2710 | lambda 0.762 | tr_acc 82.0% | val_loss 0.5562 | val_acc
82.1%
[full_aw30_s5:RealWorld-Thigh] epoch 22 | act_loss 0.4478 | dom_l
oss 1.2751 | lambda 0.800 | tr_acc 81.9% | val_loss 0.5029 | val_acc
83.4% *
[full_aw30_s5:RealWorld-Thigh] epoch 34 | act_loss 0.4407 | dom_l
oss 1.3068 | lambda 0.935 | tr_acc 82.6% | val_loss 0.5022 | val_acc
83.8% *
[full_aw30_s5:RealWorld-Thigh] epoch 40 | act_loss 0.4334 | dom_l
oss 1.3120 | lambda 0.964 | tr_acc 83.2% | val_loss 0.5359 | val_acc
82.2%
[full_aw30_s5:RealWorld-Thigh] epoch 49 | act_loss 0.4151 | dom_l
oss 1.3120 | lambda 0.985 | tr_acc 83.8% | val_loss 0.4868 | val_acc
84.1% *

```

```

[full_aw30_s5:RealWorld-Thigh] epoch 60 | act_loss 0.3921 | dom_l
oss 1.3112 | lambda 0.995 | tr_acc 85.1% | val_loss 0.5427 | val_acc
81.3%
[full_aw30_s5:RealWorld-Thigh] early stop at epoch 69
[full_aw30_s5:RealWorld-Thigh] src-val 84.14% | TARGET-TEST 66.60%
| saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (0, 1):100%, (1, 2):100%, (3, 5):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_aw30_s5:RealWorld-Waist] trainable params: 51,531
[full_aw30_s5:RealWorld-Waist] epoch 1 | act_loss 0.8012 | dom_l
oss 1.3267 | lambda 0.050 | tr_acc 65.1% | val_loss 0.9518 | val_acc
59.1% *
[full_aw30_s5:RealWorld-Waist] epoch 2 | act_loss 0.6224 | dom_l
oss 1.2626 | lambda 0.100 | tr_acc 73.9% | val_loss 0.6689 | val_acc
71.2% *
[full_aw30_s5:RealWorld-Waist] epoch 6 | act_loss 0.5264 | dom_l
oss 1.2601 | lambda 0.291 | tr_acc 78.3% | val_loss 0.6228 | val_acc
76.8% *
[full_aw30_s5:RealWorld-Waist] epoch 14 | act_loss 0.4659 | dom_l
oss 1.2721 | lambda 0.604 | tr_acc 81.3% | val_loss 0.5743 | val_acc
79.3% *
[full_aw30_s5:RealWorld-Waist] epoch 20 | act_loss 0.4492 | dom_l
oss 1.2827 | lambda 0.762 | tr_acc 82.1% | val_loss 0.6192 | val_acc
76.5%
[full_aw30_s5:RealWorld-Waist] epoch 28 | act_loss 0.4432 | dom_l
oss 1.3001 | lambda 0.885 | tr_acc 82.9% | val_loss 0.5536 | val_acc
80.2% *
[full_aw30_s5:RealWorld-Waist] epoch 40 | act_loss 0.4259 | dom_l
oss 1.3066 | lambda 0.964 | tr_acc 84.0% | val_loss 0.7644 | val_acc
70.9%
[full_aw30_s5:RealWorld-Waist] early stop at epoch 48
[full_aw30_s5:RealWorld-Waist] src-val 80.18% | TARGET-TEST 66.05%
| saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel

```

```

[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (3, 4):100%, (0, 3):100%, (0, 2):100%
    feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
    building 34368 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs
    built 30000/34368 graphs
    done: 34368 graphs built
[full_aw30_s5:UCI] trainable params: 51,531
[full_aw30_s5:UCI] epoch 1 | act_loss 0.7957 | dom_loss 1.3910 |
lambda 0.050 | tr_acc 66.2% | val_loss 1.0515 | val_acc 61.3% *
[full_aw30_s5:UCI] epoch 2 | act_loss 0.6373 | dom_loss 1.3215 |
lambda 0.100 | tr_acc 73.6% | val_loss 0.6938 | val_acc 72.4% *
[full_aw30_s5:UCI] epoch 3 | act_loss 0.5921 | dom_loss 1.3142 |
lambda 0.149 | tr_acc 76.0% | val_loss 0.7421 | val_acc 73.5% *
[full_aw30_s5:UCI] epoch 8 | act_loss 0.5295 | dom_loss 1.3511 |
lambda 0.380 | tr_acc 78.8% | val_loss 0.6155 | val_acc 78.0% *
[full_aw30_s5:UCI] epoch 13 | act_loss 0.5031 | dom_loss 1.3651 |
lambda 0.572 | tr_acc 80.2% | val_loss 0.6161 | val_acc 78.2% *
[full_aw30_s5:UCI] epoch 15 | act_loss 0.4957 | dom_loss 1.3663 |
lambda 0.635 | tr_acc 80.4% | val_loss 0.5409 | val_acc 82.1% *
[full_aw30_s5:UCI] epoch 20 | act_loss 0.4806 | dom_loss 1.3680 |
lambda 0.762 | tr_acc 81.2% | val_loss 0.6102 | val_acc 76.9%
[full_aw30_s5:UCI] early stop at epoch 35
[full_aw30_s5:UCI] src-val 82.10% | TARGET-TEST 82.90% | saved c
kpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 AW: 60 -> 30 timesteps/window (strided)
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (0, 3):100%, (1, 4):100%, (4, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 3):100%, (3, 5):100%, (0, 2):100%, (4, 5):100%, (1, 2):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (4, 5):100%, (1, 4):100%, (2, 5):100%
    feature_mode=hybrid, feat_dim=45, normalize_mode=per_channel
    building 28040 sensor graphs (6 nodes, 18 edges, 45 features eac
h)...
    built 5000/28040 graphs
    built 10000/28040 graphs
    built 15000/28040 graphs
    built 20000/28040 graphs

```

```

built 25000/28040 graphs
done: 28040 graphs built
[full_aw30_s5:WISDM] trainable params: 51,531
[full_aw30_s5:WISDM] epoch 1 | act_loss 0.8128 | dom_loss 1.3357
| lambda 0.050 | tr_acc 66.0% | val_loss 0.7044 | val_acc 72.6% *
[full_aw30_s5:WISDM] epoch 4 | act_loss 0.5554 | dom_loss 1.2998
| lambda 0.197 | tr_acc 78.1% | val_loss 0.6739 | val_acc 76.0% *
[full_aw30_s5:WISDM] epoch 5 | act_loss 0.5407 | dom_loss 1.3116
| lambda 0.245 | tr_acc 78.5% | val_loss 0.6244 | val_acc 77.6% *
[full_aw30_s5:WISDM] epoch 7 | act_loss 0.5077 | dom_loss 1.3250
| lambda 0.336 | tr_acc 80.1% | val_loss 0.6303 | val_acc 78.2% *
[full_aw30_s5:WISDM] epoch 13 | act_loss 0.4685 | dom_loss 1.3287
| lambda 0.572 | tr_acc 81.9% | val_loss 0.6270 | val_acc 80.4% *
[full_aw30_s5:WISDM] epoch 19 | act_loss 0.4530 | dom_loss 1.3323
| lambda 0.740 | tr_acc 82.9% | val_loss 0.6052 | val_acc 82.4% *
[full_aw30_s5:WISDM] epoch 20 | act_loss 0.4503 | dom_loss 1.3321
| lambda 0.762 | tr_acc 82.9% | val_loss 0.7014 | val_acc 75.6%
[full_aw30_s5:WISDM] early stop at epoch 39
[full_aw30_s5:WISDM] src-val 82.37% | TARGET-TEST 68.20% | saved
ckpt + npz

```

5. full at the 4 missing seeds — for the 5-seed RED/GS table

The edge-budget study is *inference-only*, but it needs a `full` checkpoint per seed to swap edge policies on. Seed 42 (`results/checkpoints/full`) already exists and is skipped.

```
In [ ]: for s in (1, 2, 3, 5):  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant full  
        --seed {s} --epochs 100 --log_every 20
```

```

[P24] train+save | arm=full_s1 {'cnn_embed_dim': 64, 'gnn_hidden': 64,
'cnn_channels': [32, 64]} | nc=None aw=- | seed=1 | targets=['KuHar',
'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] |
out=/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphReviv
al_DG/P24_S0_Compression/results/checkpoints/full_s1
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_s1:KuHar] trainable params: 51,531
[full_s1:KuHar] epoch 1 | act_loss 0.7617 | dom_loss 1.4183 | la
mbda 0.050 | tr_acc 68.5% | val_loss 0.7096 | val_acc 72.7% *
[full_s1:KuHar] epoch 4 | act_loss 0.5282 | dom_loss 1.3482 | la
mbda 0.197 | tr_acc 78.5% | val_loss 0.5764 | val_acc 79.6% *
[full_s1:KuHar] epoch 13 | act_loss 0.4664 | dom_loss 1.3925 | la
mbda 0.572 | tr_acc 81.6% | val_loss 0.5877 | val_acc 80.0% *
[full_s1:KuHar] epoch 17 | act_loss 0.4503 | dom_loss 1.3973 | la
mbda 0.691 | tr_acc 82.5% | val_loss 0.6126 | val_acc 80.3% *
[full_s1:KuHar] epoch 20 | act_loss 0.4459 | dom_loss 1.4000 | la
mbda 0.762 | tr_acc 82.9% | val_loss 0.6451 | val_acc 79.9%
[full_s1:KuHar] epoch 22 | act_loss 0.4392 | dom_loss 1.3996 | la
mbda 0.800 | tr_acc 83.1% | val_loss 0.6226 | val_acc 81.7% *
[full_s1:KuHar] epoch 26 | act_loss 0.4386 | dom_loss 1.4042 | la
mbda 0.862 | tr_acc 83.5% | val_loss 0.5254 | val_acc 82.9% *
[full_s1:KuHar] epoch 40 | act_loss 0.4273 | dom_loss 1.4231 | la
mbda 0.964 | tr_acc 84.2% | val_loss 0.6602 | val_acc 76.0%
[full_s1:KuHar] early stop at epoch 46
[full_s1:KuHar] src-val 82.91% | TARGET-TEST 70.83% | saved ckpt
+ npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa

```

```

irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_s1:MotionSense] trainable params: 51,531
[full_s1:MotionSense] epoch 1 | act_loss 0.7919 | dom_loss 1.339
0 | lambda 0.050 | tr_acc 66.6% | val_loss 0.7124 | val_acc 72.8% *
[full_s1:MotionSense] epoch 4 | act_loss 0.5386 | dom_loss 1.258
5 | lambda 0.197 | tr_acc 77.6% | val_loss 0.6627 | val_acc 75.4% *
[full_s1:MotionSense] epoch 6 | act_loss 0.5074 | dom_loss 1.278
8 | lambda 0.291 | tr_acc 79.1% | val_loss 0.6760 | val_acc 75.9% *
[full_s1:MotionSense] epoch 9 | act_loss 0.4881 | dom_loss 1.298
6 | lambda 0.422 | tr_acc 80.3% | val_loss 0.6163 | val_acc 77.9% *
[full_s1:MotionSense] epoch 20 | act_loss 0.4537 | dom_loss 1.317
4 | lambda 0.762 | tr_acc 82.2% | val_loss 0.7158 | val_acc 74.4%
[full_s1:MotionSense] epoch 25 | act_loss 0.4483 | dom_loss 1.324
8 | lambda 0.848 | tr_acc 82.6% | val_loss 0.6366 | val_acc 78.2% *
[full_s1:MotionSense] epoch 27 | act_loss 0.4531 | dom_loss 1.326
3 | lambda 0.874 | tr_acc 82.5% | val_loss 0.5951 | val_acc 79.4% *
[full_s1:MotionSense] epoch 31 | act_loss 0.4472 | dom_loss 1.329
9 | lambda 0.914 | tr_acc 82.8% | val_loss 0.5805 | val_acc 79.9% *
[full_s1:MotionSense] epoch 32 | act_loss 0.4415 | dom_loss 1.330
4 | lambda 0.922 | tr_acc 83.0% | val_loss 0.5948 | val_acc 80.3% *
[full_s1:MotionSense] epoch 40 | act_loss 0.4337 | dom_loss 1.339
8 | lambda 0.964 | tr_acc 83.6% | val_loss 0.6783 | val_acc 79.3%
[full_s1:MotionSense] early stop at epoch 52
[full_s1:MotionSense] src-val 80.31% | TARGET-TEST 81.73% | save
d ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs

```

```

done: 26450 graphs built
[full_s1:RealWorld-Thigh] trainable params: 51,531
[full_s1:RealWorld-Thigh] epoch 1 | act_loss 0.7682 | dom_loss
1.2897 | lambda 0.050 | tr_acc 66.7% | val_loss 0.6724 | val_acc 72.
9% *
[full_s1:RealWorld-Thigh] epoch 2 | act_loss 0.5875 | dom_loss
1.2050 | lambda 0.100 | tr_acc 75.1% | val_loss 0.5998 | val_acc 75.
6% *
[full_s1:RealWorld-Thigh] epoch 3 | act_loss 0.5421 | dom_loss
1.2047 | lambda 0.149 | tr_acc 77.2% | val_loss 0.5660 | val_acc 78.
5% *
[full_s1:RealWorld-Thigh] epoch 7 | act_loss 0.4846 | dom_loss
1.2313 | lambda 0.336 | tr_acc 79.8% | val_loss 0.5657 | val_acc 78.
8% *
[full_s1:RealWorld-Thigh] epoch 8 | act_loss 0.4765 | dom_loss
1.2398 | lambda 0.380 | tr_acc 80.3% | val_loss 0.5277 | val_acc 81.
0% *
[full_s1:RealWorld-Thigh] epoch 13 | act_loss 0.4403 | dom_loss
1.2583 | lambda 0.572 | tr_acc 81.8% | val_loss 0.5336 | val_acc 81.
0% *
[full_s1:RealWorld-Thigh] epoch 20 | act_loss 0.4191 | dom_loss
1.2705 | lambda 0.762 | tr_acc 83.2% | val_loss 0.7083 | val_acc 74.
5%
[full_s1:RealWorld-Thigh] epoch 22 | act_loss 0.4178 | dom_loss
1.2743 | lambda 0.800 | tr_acc 83.1% | val_loss 0.6005 | val_acc 81.
9% *
[full_s1:RealWorld-Thigh] epoch 37 | act_loss 0.4126 | dom_loss
1.3099 | lambda 0.952 | tr_acc 84.0% | val_loss 0.5003 | val_acc 83.
0% *
[full_s1:RealWorld-Thigh] epoch 40 | act_loss 0.4078 | dom_loss
1.3092 | lambda 0.964 | tr_acc 84.0% | val_loss 0.5067 | val_acc 83.
7% *
[full_s1:RealWorld-Thigh] epoch 42 | act_loss 0.4028 | dom_loss
1.3088 | lambda 0.970 | tr_acc 84.5% | val_loss 0.4612 | val_acc 85.
4% *
[full_s1:RealWorld-Thigh] epoch 60 | act_loss 0.3673 | dom_loss
1.3046 | lambda 0.995 | tr_acc 86.1% | val_loss 0.5905 | val_acc 78.
1%
[full_s1:RealWorld-Thigh] early stop at epoch 62
[full_s1:RealWorld-Thigh] src-val 85.44% | TARGET-TEST 72.39% |
saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs

```

```

built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_s1:RealWorld-Waist] trainable params: 51,531
[full_s1:RealWorld-Waist] epoch 1 | act_loss 0.7785 | dom_loss
1.3310 | lambda 0.050 | tr_acc 66.7% | val_loss 0.8499 | val_acc 65.
6% *
[full_s1:RealWorld-Waist] epoch 2 | act_loss 0.6056 | dom_loss
1.2596 | lambda 0.100 | tr_acc 74.5% | val_loss 0.8193 | val_acc 68.
0% *
[full_s1:RealWorld-Waist] epoch 3 | act_loss 0.5522 | dom_loss
1.2404 | lambda 0.149 | tr_acc 77.3% | val_loss 0.6073 | val_acc 75.
5% *
[full_s1:RealWorld-Waist] epoch 4 | act_loss 0.5205 | dom_loss
1.2504 | lambda 0.197 | tr_acc 78.3% | val_loss 0.5813 | val_acc 77.
8% *
[full_s1:RealWorld-Waist] epoch 6 | act_loss 0.4858 | dom_loss
1.2419 | lambda 0.291 | tr_acc 80.1% | val_loss 0.5514 | val_acc 79.
4% *
[full_s1:RealWorld-Waist] epoch 7 | act_loss 0.4775 | dom_loss
1.2462 | lambda 0.336 | tr_acc 80.3% | val_loss 0.5349 | val_acc 80.
0% *
[full_s1:RealWorld-Waist] epoch 10 | act_loss 0.4542 | dom_loss
1.2577 | lambda 0.462 | tr_acc 81.6% | val_loss 0.5385 | val_acc 80.
6% *
[full_s1:RealWorld-Waist] epoch 19 | act_loss 0.4266 | dom_loss
1.2703 | lambda 0.740 | tr_acc 82.9% | val_loss 0.5786 | val_acc 82.
3% *
[full_s1:RealWorld-Waist] epoch 20 | act_loss 0.4261 | dom_loss
1.2717 | lambda 0.762 | tr_acc 83.0% | val_loss 0.7941 | val_acc 68.
1%
[full_s1:RealWorld-Waist] early stop at epoch 39
[full_s1:RealWorld-Waist] src-val 82.26% | TARGET-TEST 67.40% |
saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features each)...
```

```

built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs

```

```

done: 34368 graphs built
[full_s1:UCI] trainable params: 51,531
[full_s1:UCI] epoch 1 | act_loss 0.7650 | dom_loss 1.3880 | lamb
da 0.050 | tr_acc 68.3% | val_loss 0.6922 | val_acc 71.6% *
[full_s1:UCI] epoch 2 | act_loss 0.6119 | dom_loss 1.3241 | lamb
da 0.100 | tr_acc 75.1% | val_loss 0.6764 | val_acc 75.4% *
[full_s1:UCI] epoch 5 | act_loss 0.5172 | dom_loss 1.3267 | lamb
da 0.245 | tr_acc 79.0% | val_loss 0.6220 | val_acc 76.4% *
[full_s1:UCI] epoch 7 | act_loss 0.4947 | dom_loss 1.3351 | lamb
da 0.336 | tr_acc 80.0% | val_loss 0.6438 | val_acc 77.4% *
[full_s1:UCI] epoch 9 | act_loss 0.4844 | dom_loss 1.3445 | lamb
da 0.422 | tr_acc 80.6% | val_loss 0.5866 | val_acc 78.5% *
[full_s1:UCI] epoch 14 | act_loss 0.4655 | dom_loss 1.3584 | lamb
da 0.604 | tr_acc 81.8% | val_loss 0.6336 | val_acc 79.5% *
[full_s1:UCI] epoch 20 | act_loss 0.4591 | dom_loss 1.3659 | lamb
da 0.762 | tr_acc 82.1% | val_loss 0.7570 | val_acc 76.9%
[full_s1:UCI] epoch 21 | act_loss 0.4518 | dom_loss 1.3679 | lamb
da 0.782 | tr_acc 82.4% | val_loss 0.5878 | val_acc 80.4% *
[full_s1:UCI] epoch 23 | act_loss 0.4498 | dom_loss 1.3662 | lamb
da 0.818 | tr_acc 82.4% | val_loss 0.5880 | val_acc 81.5% *
[full_s1:UCI] epoch 40 | act_loss 0.4325 | dom_loss 1.3883 | lamb
da 0.964 | tr_acc 83.6% | val_loss 0.7383 | val_acc 73.6%
[full_s1:UCI] early stop at epoch 43
[full_s1:UCI] src-val 81.50% | TARGET-TEST 85.51% | saved ckpt +
npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_s1:WISDM] trainable params: 51,531
[full_s1:WISDM] epoch 1 | act_loss 0.7820 | dom_loss 1.3262 | la
mbda 0.050 | tr_acc 67.9% | val_loss 1.0824 | val_acc 57.7% *
[full_s1:WISDM] epoch 2 | act_loss 0.5862 | dom_loss 1.2948 | la
mbda 0.100 | tr_acc 76.5% | val_loss 0.8681 | val_acc 64.2% *
[full_s1:WISDM] epoch 3 | act_loss 0.5449 | dom_loss 1.3076 | la
mbda 0.149 | tr_acc 78.4% | val_loss 0.6431 | val_acc 77.1% *
[full_s1:WISDM] epoch 11 | act_loss 0.4520 | dom_loss 1.3265 | la
mbda 0.501 | tr_acc 82.6% | val_loss 0.7030 | val_acc 77.2% *
[full_s1:WISDM] epoch 18 | act_loss 0.4297 | dom_loss 1.3324 | la
mbda 0.716 | tr_acc 83.9% | val_loss 0.7099 | val_acc 77.6% *

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[full_s1:WISDM] epoch 19 | act_loss 0.4236 | dom_loss 1.3321 | la
mbda 0.740 | tr_acc 84.1% | val_loss 0.6459 | val_acc 80.3% *
[full_s1:WISDM] epoch 20 | act_loss 0.4200 | dom_loss 1.3315 | la
mbda 0.762 | tr_acc 84.3% | val_loss 0.9373 | val_acc 68.3%
[full_s1:WISDM] epoch 28 | act_loss 0.4058 | dom_loss 1.3384 | la
mbda 0.885 | tr_acc 85.1% | val_loss 0.6033 | val_acc 80.8% *
[full_s1:WISDM] epoch 40 | act_loss 0.3802 | dom_loss 1.3387 | la
mbda 0.964 | tr_acc 86.1% | val_loss 0.7874 | val_acc 71.8%
[full_s1:WISDM] epoch 46 | act_loss 0.3744 | dom_loss 1.3396 | la
mbda 0.980 | tr_acc 86.5% | val_loss 0.6472 | val_acc 81.1% *
[full_s1:WISDM] epoch 49 | act_loss 0.3673 | dom_loss 1.3407 | la
mbda 0.985 | tr_acc 86.8% | val_loss 0.5624 | val_acc 81.7% *
[full_s1:WISDM] epoch 60 | act_loss 0.3464 | dom_loss 1.3399 | la
mbda 0.995 | tr_acc 87.8% | val_loss 0.7015 | val_acc 79.2%
[full_s1:WISDM] early stop at epoch 69
[full_s1:WISDM] src-val 81.73% | TARGET-TEST 76.00% | saved ckpt
+ npz
[P24] train+save | arm=full_s2 {'cnn_embed_dim': 64, 'gnn_hidden': 64,
'cnn_channels': [32, 64]} | nc=None aw=- | seed=2 | targets=['KuHar',
'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] |
out=/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphReviv
al_DG/P24_S0_Compression/results/checkpoints/full_s2
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_s2:KuHar] trainable params: 51,531
[full_s2:KuHar] epoch 1 | act_loss 0.7466 | dom_loss 1.4256 | la
mbda 0.050 | tr_acc 69.3% | val_loss 0.7279 | val_acc 72.9% *
[full_s2:KuHar] epoch 5 | act_loss 0.5081 | dom_loss 1.3676 | la
mbda 0.245 | tr_acc 79.7% | val_loss 0.6132 | val_acc 77.7% *
[full_s2:KuHar] epoch 15 | act_loss 0.4575 | dom_loss 1.3972 | la
mbda 0.635 | tr_acc 82.3% | val_loss 0.6044 | val_acc 80.2% *
[full_s2:KuHar] epoch 20 | act_loss 0.4440 | dom_loss 1.4015 | la
mbda 0.762 | tr_acc 83.0% | val_loss 0.7791 | val_acc 68.8%
[full_s2:KuHar] epoch 30 | act_loss 0.4318 | dom_loss 1.4129 | la
mbda 0.905 | tr_acc 83.7% | val_loss 0.6179 | val_acc 81.1% *
[full_s2:KuHar] epoch 40 | act_loss 0.4250 | dom_loss 1.4219 | la

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mbda 0.964 | tr_acc 84.4% | val_loss 0.6022 | val_acc 78.8%
[full_s2:KuHar] epoch 45 | act_loss 0.4152 | dom_loss 1.4221 | la
mbda 0.978 | tr_acc 84.8% | val_loss 0.5905 | val_acc 82.1% *
[full_s2:KuHar] epoch 53 | act_loss 0.3970 | dom_loss 1.4202 | la
mbda 0.990 | tr_acc 85.5% | val_loss 0.5753 | val_acc 82.4% *
[full_s2:KuHar] epoch 60 | act_loss 0.3864 | dom_loss 1.4189 | la
mbda 0.995 | tr_acc 86.1% | val_loss 0.6047 | val_acc 79.9%
[full_s2:KuHar] epoch 72 | act_loss 0.3677 | dom_loss 1.4199 | la
mbda 0.999 | tr_acc 86.9% | val_loss 0.5449 | val_acc 82.5% *
[full_s2:KuHar] epoch 74 | act_loss 0.3628 | dom_loss 1.4212 | la
mbda 0.999 | tr_acc 87.1% | val_loss 0.5262 | val_acc 83.3% *
[full_s2:KuHar] epoch 80 | act_loss 0.3518 | dom_loss 1.4200 | la
mbda 0.999 | tr_acc 87.6% | val_loss 0.5532 | val_acc 79.6%
[full_s2:KuHar] epoch 81 | act_loss 0.3483 | dom_loss 1.4192 | la
mbda 0.999 | tr_acc 87.9% | val_loss 0.5528 | val_acc 83.9% *
[full_s2:KuHar] epoch 83 | act_loss 0.3443 | dom_loss 1.4192 | la
mbda 1.000 | tr_acc 87.9% | val_loss 0.5155 | val_acc 84.7% *
[full_s2:KuHar] epoch 99 | act_loss 0.3355 | dom_loss 1.4195 | la
mbda 1.000 | tr_acc 88.4% | val_loss 0.5174 | val_acc 84.9% *
[full_s2:KuHar] epoch 100 | act_loss 0.3342 | dom_loss 1.4194 | la
mbda 1.000 | tr_acc 88.4% | val_loss 0.9916 | val_acc 70.1%
[full_s2:KuHar] src-val 84.94% | TARGET-TEST 71.53% | saved ckpt
+ npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_s2:MotionSense] trainable params: 51,531
[full_s2:MotionSense] epoch 1 | act_loss 0.7716 | dom_loss 1.340
5 | lambda 0.050 | tr_acc 67.7% | val_loss 0.7971 | val_acc 66.1% *
[full_s2:MotionSense] epoch 2 | act_loss 0.6031 | dom_loss 1.270
6 | lambda 0.100 | tr_acc 75.0% | val_loss 0.6370 | val_acc 75.3% *
[full_s2:MotionSense] epoch 6 | act_loss 0.5094 | dom_loss 1.285
9 | lambda 0.291 | tr_acc 79.2% | val_loss 0.6245 | val_acc 75.3% *
[full_s2:MotionSense] epoch 8 | act_loss 0.4854 | dom_loss 1.296
9 | lambda 0.380 | tr_acc 80.3% | val_loss 0.6466 | val_acc 77.6% *
[full_s2:MotionSense] epoch 12 | act_loss 0.4700 | dom_loss 1.307
8 | lambda 0.537 | tr_acc 81.0% | val_loss 0.6746 | val_acc 78.2% *
[full_s2:MotionSense] epoch 16 | act_loss 0.4601 | dom_loss 1.312

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```

1 | lambda 0.664 | tr_acc 81.6% | val_loss 0.5813 | val_acc 79.4% *
  [full_s2:MotionSense] epoch 20 | act_loss 0.4577 | dom_loss 1.318
3 | lambda 0.762 | tr_acc 82.0% | val_loss 0.6806 | val_acc 74.8%
  [full_s2:MotionSense] early stop at epoch 36
  [full_s2:MotionSense] src-val 79.41% | TARGET-TEST 85.78% | save
d ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
  [full_s2:RealWorld-Thigh] trainable params: 51,531
  [full_s2:RealWorld-Thigh] epoch 1 | act_loss 0.7425 | dom_loss
1.2906 | lambda 0.050 | tr_acc 68.3% | val_loss 0.7689 | val_acc 69.
0% *
  [full_s2:RealWorld-Thigh] epoch 3 | act_loss 0.5442 | dom_loss
1.2091 | lambda 0.149 | tr_acc 77.0% | val_loss 0.6262 | val_acc 71.
9% *
  [full_s2:RealWorld-Thigh] epoch 4 | act_loss 0.5279 | dom_loss
1.2136 | lambda 0.197 | tr_acc 77.9% | val_loss 0.6391 | val_acc 75.
7% *
  [full_s2:RealWorld-Thigh] epoch 5 | act_loss 0.5077 | dom_loss
1.2190 | lambda 0.245 | tr_acc 78.6% | val_loss 0.6796 | val_acc 75.
9% *
  [full_s2:RealWorld-Thigh] epoch 7 | act_loss 0.4821 | dom_loss
1.2421 | lambda 0.336 | tr_acc 79.7% | val_loss 0.5880 | val_acc 77.
6% *
  [full_s2:RealWorld-Thigh] epoch 12 | act_loss 0.4422 | dom_loss
1.2589 | lambda 0.537 | tr_acc 81.8% | val_loss 0.5999 | val_acc 80.
1% *
  [full_s2:RealWorld-Thigh] epoch 14 | act_loss 0.4376 | dom_loss
1.2685 | lambda 0.604 | tr_acc 82.3% | val_loss 0.5683 | val_acc 81.
0% *
  [full_s2:RealWorld-Thigh] epoch 20 | act_loss 0.4211 | dom_loss
1.2723 | lambda 0.762 | tr_acc 82.9% | val_loss 0.7357 | val_acc 72.
0%
  [full_s2:RealWorld-Thigh] epoch 27 | act_loss 0.4224 | dom_loss
1.2996 | lambda 0.874 | tr_acc 83.2% | val_loss 0.5771 | val_acc 81.
8% *
  [full_s2:RealWorld-Thigh] epoch 29 | act_loss 0.4196 | dom_loss
1.3051 | lambda 0.896 | tr_acc 83.3% | val_loss 0.5017 | val_acc 83.

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9% *
[full_s2:RealWorld-Thigh] epoch 40 | act_loss 0.4041 | dom_loss
1.3129 | lambda 0.964 | tr_acc 84.2% | val_loss 0.5699 | val_acc 80.
9%
[full_s2:RealWorld-Thigh] early stop at epoch 49
[full_s2:RealWorld-Thigh] src-val 83.91% | TARGET-TEST 64.67% |
saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_s2:RealWorld-Waist] trainable params: 51,531
[full_s2:RealWorld-Waist] epoch 1 | act_loss 0.7582 | dom_loss
1.3282 | lambda 0.050 | tr_acc 68.0% | val_loss 0.6859 | val_acc 73.
9% *
[full_s2:RealWorld-Waist] epoch 7 | act_loss 0.4751 | dom_loss
1.2541 | lambda 0.336 | tr_acc 80.4% | val_loss 0.6569 | val_acc 76.
6% *
[full_s2:RealWorld-Waist] epoch 12 | act_loss 0.4429 | dom_loss
1.2622 | lambda 0.537 | tr_acc 82.3% | val_loss 0.6214 | val_acc 77.
4% *
[full_s2:RealWorld-Waist] epoch 16 | act_loss 0.4312 | dom_loss
1.2697 | lambda 0.664 | tr_acc 82.7% | val_loss 0.5785 | val_acc 78.
5% *
[full_s2:RealWorld-Waist] epoch 20 | act_loss 0.4207 | dom_loss
1.2743 | lambda 0.762 | tr_acc 83.7% | val_loss 0.7385 | val_acc 70.
4%
[full_s2:RealWorld-Waist] epoch 24 | act_loss 0.4166 | dom_loss
1.2827 | lambda 0.834 | tr_acc 83.9% | val_loss 0.5623 | val_acc 79.
9% *
[full_s2:RealWorld-Waist] epoch 31 | act_loss 0.4145 | dom_loss
1.3008 | lambda 0.914 | tr_acc 84.3% | val_loss 0.5486 | val_acc 82.
0% *
[full_s2:RealWorld-Waist] epoch 40 | act_loss 0.3985 | dom_loss
1.3020 | lambda 0.964 | tr_acc 85.1% | val_loss 0.7229 | val_acc 70.
6%
[full_s2:RealWorld-Waist] epoch 41 | act_loss 0.4014 | dom_loss
1.3063 | lambda 0.967 | tr_acc 84.8% | val_loss 0.5465 | val_acc 82.
2% *
[full_s2:RealWorld-Waist] epoch 52 | act_loss 0.3818 | dom_loss

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1.3083 | lambda 0.989 | tr_acc 86.1% | val_loss 0.5272 | val_acc 82.3% *
[full_s2:RealWorld-Waist] epoch 60 | act_loss 0.3696 | dom_loss
1.3072 | lambda 0.995 | tr_acc 86.2% | val_loss 0.6078 | val_acc 75.3%
[full_s2:RealWorld-Waist] epoch 68 | act_loss 0.3494 | dom_loss
1.3039 | lambda 0.998 | tr_acc 87.4% | val_loss 0.5297 | val_acc 82.4% *
[full_s2:RealWorld-Waist] epoch 79 | act_loss 0.3312 | dom_loss
1.3068 | lambda 0.999 | tr_acc 88.3% | val_loss 0.5062 | val_acc 83.1% *
[full_s2:RealWorld-Waist] epoch 80 | act_loss 0.3304 | dom_loss
1.3072 | lambda 0.999 | tr_acc 88.4% | val_loss 0.5399 | val_acc 80.8%
[full_s2:RealWorld-Waist] epoch 89 | act_loss 0.3223 | dom_loss
1.3078 | lambda 1.000 | tr_acc 88.7% | val_loss 0.5249 | val_acc 83.3% *
[full_s2:RealWorld-Waist] epoch 100 | act_loss 0.3129 | dom_loss
1.3070 | lambda 1.000 | tr_acc 89.3% | val_loss 0.5348 | val_acc 82.1%
[full_s2:RealWorld-Waist] src-val 83.31% | TARGET-TEST 74.07% |
saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features each)...)
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_s2:UCI] trainable params: 51,531
[full_s2:UCI] epoch 1 | act_loss 0.7488 | dom_loss 1.3853 | lambda 0.050 | tr_acc 69.1% | val_loss 0.7777 | val_acc 65.0% *
[full_s2:UCI] epoch 2 | act_loss 0.5976 | dom_loss 1.3256 | lambda 0.100 | tr_acc 75.8% | val_loss 0.7947 | val_acc 67.8% *
[full_s2:UCI] epoch 3 | act_loss 0.5550 | dom_loss 1.3290 | lambda 0.149 | tr_acc 77.6% | val_loss 0.7034 | val_acc 73.8% *
[full_s2:UCI] epoch 7 | act_loss 0.4943 | dom_loss 1.3374 | lambda 0.336 | tr_acc 80.0% | val_loss 0.6603 | val_acc 75.8% *
[full_s2:UCI] epoch 14 | act_loss 0.4656 | dom_loss 1.3588 | lambda 0.604 | tr_acc 81.4% | val_loss 0.6555 | val_acc 79.2% *
[full_s2:UCI] epoch 20 | act_loss 0.4532 | dom_loss 1.3656 | lambda 0.762 | tr_acc 82.1% | val_loss 0.6762 | val_acc 79.3% *

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[full_s2:UCI] epoch 24 | act_loss 0.4438 | dom_loss 1.3700 | lamb
da 0.834 | tr_acc 82.8% | val_loss 0.5858 | val_acc 79.6% *
[full_s2:UCI] epoch 35 | act_loss 0.4369 | dom_loss 1.3865 | lamb
da 0.941 | tr_acc 83.4% | val_loss 0.6222 | val_acc 80.0% *
[full_s2:UCI] epoch 36 | act_loss 0.4393 | dom_loss 1.3869 | lamb
da 0.947 | tr_acc 83.3% | val_loss 0.6601 | val_acc 80.2% *
[full_s2:UCI] epoch 40 | act_loss 0.4253 | dom_loss 1.3896 | lamb
da 0.964 | tr_acc 84.2% | val_loss 0.9050 | val_acc 69.8%
[full_s2:UCI] epoch 43 | act_loss 0.4218 | dom_loss 1.3907 | lamb
da 0.973 | tr_acc 84.2% | val_loss 0.5876 | val_acc 80.8% *
[full_s2:UCI] epoch 44 | act_loss 0.4254 | dom_loss 1.3905 | lamb
da 0.976 | tr_acc 84.1% | val_loss 0.6031 | val_acc 81.2% *
[full_s2:UCI] epoch 51 | act_loss 0.4128 | dom_loss 1.3885 | lamb
da 0.988 | tr_acc 84.8% | val_loss 0.5805 | val_acc 81.3% *
[full_s2:UCI] epoch 53 | act_loss 0.4074 | dom_loss 1.3889 | lamb
da 0.990 | tr_acc 85.0% | val_loss 0.5736 | val_acc 81.8% *
[full_s2:UCI] epoch 55 | act_loss 0.4046 | dom_loss 1.3895 | lamb
da 0.992 | tr_acc 85.0% | val_loss 0.5911 | val_acc 81.8% *
[full_s2:UCI] epoch 60 | act_loss 0.3929 | dom_loss 1.3879 | lamb
da 0.995 | tr_acc 85.3% | val_loss 0.6547 | val_acc 75.8%
[full_s2:UCI] epoch 69 | act_loss 0.3793 | dom_loss 1.3862 | lamb
da 0.998 | tr_acc 86.2% | val_loss 0.5706 | val_acc 82.1% *
[full_s2:UCI] epoch 76 | act_loss 0.3644 | dom_loss 1.3843 | lamb
da 0.999 | tr_acc 87.0% | val_loss 0.5911 | val_acc 82.5% *
[full_s2:UCI] epoch 80 | act_loss 0.3566 | dom_loss 1.3839 | lamb
da 0.999 | tr_acc 87.2% | val_loss 0.6273 | val_acc 79.5%
[full_s2:UCI] epoch 88 | act_loss 0.3452 | dom_loss 1.3853 | lamb
da 1.000 | tr_acc 87.8% | val_loss 0.5507 | val_acc 82.6% *
[full_s2:UCI] epoch 96 | act_loss 0.3415 | dom_loss 1.3850 | lamb
da 1.000 | tr_acc 87.8% | val_loss 0.5503 | val_acc 83.0% *
[full_s2:UCI] epoch 97 | act_loss 0.3419 | dom_loss 1.3850 | lamb
da 1.000 | tr_acc 87.8% | val_loss 0.5572 | val_acc 83.4% *
[full_s2:UCI] epoch 100 | act_loss 0.3403 | dom_loss 1.3854 | lamb
da 1.000 | tr_acc 87.9% | val_loss 0.5887 | val_acc 81.3%
[full_s2:UCI] src-val 83.45% | TARGET-TEST 84.78% | saved ckpt +
npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built

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[full_s2:WISDM] trainable params: 51,531
[full_s2:WISDM] epoch 1 | act_loss 0.7527 | dom_loss 1.3288 | la
mbda 0.050 | tr_acc 69.7% | val_loss 0.6844 | val_acc 72.2% *
[full_s2:WISDM] epoch 2 | act_loss 0.5833 | dom_loss 1.3129 | la
mbda 0.100 | tr_acc 76.9% | val_loss 0.6475 | val_acc 75.5% *
[full_s2:WISDM] epoch 3 | act_loss 0.5427 | dom_loss 1.3059 | la
mbda 0.149 | tr_acc 78.4% | val_loss 0.6484 | val_acc 76.1% *
[full_s2:WISDM] epoch 4 | act_loss 0.5191 | dom_loss 1.3123 | la
mbda 0.197 | tr_acc 79.5% | val_loss 0.6379 | val_acc 76.2% *
[full_s2:WISDM] epoch 6 | act_loss 0.4880 | dom_loss 1.3173 | la
mbda 0.291 | tr_acc 80.6% | val_loss 0.6466 | val_acc 78.3% *
[full_s2:WISDM] epoch 20 | act_loss 0.4210 | dom_loss 1.3334 | la
mbda 0.762 | tr_acc 84.1% | val_loss 0.6804 | val_acc 78.9% *
[full_s2:WISDM] epoch 28 | act_loss 0.4003 | dom_loss 1.3393 | la
mbda 0.885 | tr_acc 85.6% | val_loss 0.6095 | val_acc 79.1% *
[full_s2:WISDM] epoch 30 | act_loss 0.4006 | dom_loss 1.3385 | la
mbda 0.905 | tr_acc 85.4% | val_loss 0.6405 | val_acc 80.4% *
[full_s2:WISDM] epoch 31 | act_loss 0.3990 | dom_loss 1.3387 | la
mbda 0.914 | tr_acc 85.2% | val_loss 0.6183 | val_acc 81.3% *
[full_s2:WISDM] epoch 40 | act_loss 0.3884 | dom_loss 1.3380 | la
mbda 0.964 | tr_acc 85.9% | val_loss 0.7704 | val_acc 73.5%
[full_s2:WISDM] early stop at epoch 51
[full_s2:WISDM] src-val 81.35% | TARGET-TEST 69.97% | saved ckpt
+ npz
[P24] train+save | arm=full_s3 {'cnn_embed_dim': 64, 'gnn_hidden': 64,
'cnn_channels': [32, 64]} | nc=None aw=- | seed=3 | targets=['KuHar',
'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] |
out=/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphReviv
al_DG/P24_S0_Compression/results/checkpoints/full_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_s3:KuHar] trainable params: 51,531
[full_s3:KuHar] epoch 1 | act_loss 0.7510 | dom_loss 1.4286 | la
mbda 0.050 | tr_acc 69.1% | val_loss 0.6381 | val_acc 75.7% *
[full_s3:KuHar] epoch 3 | act_loss 0.5454 | dom_loss 1.3688 | la
mbda 0.149 | tr_acc 77.8% | val_loss 0.6645 | val_acc 75.8% *

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[full_s3:KuHar] epoch 5 | act_loss 0.5101 | dom_loss 1.3706 | la
mbda 0.245 | tr_acc 79.3% | val_loss 0.6219 | val_acc 76.3% *
[full_s3:KuHar] epoch 6 | act_loss 0.4978 | dom_loss 1.3740 | la
mbda 0.291 | tr_acc 79.8% | val_loss 0.6223 | val_acc 77.2% *
[full_s3:KuHar] epoch 14 | act_loss 0.4631 | dom_loss 1.3943 | la
mbda 0.604 | tr_acc 81.6% | val_loss 0.6414 | val_acc 77.9% *
[full_s3:KuHar] epoch 19 | act_loss 0.4522 | dom_loss 1.4005 | la
mbda 0.740 | tr_acc 82.4% | val_loss 0.5954 | val_acc 80.6% *
[full_s3:KuHar] epoch 20 | act_loss 0.4479 | dom_loss 1.4010 | la
mbda 0.762 | tr_acc 82.5% | val_loss 0.9690 | val_acc 67.1%
[full_s3:KuHar] epoch 26 | act_loss 0.4426 | dom_loss 1.4038 | la
mbda 0.862 | tr_acc 83.0% | val_loss 0.5615 | val_acc 81.6% *
[full_s3:KuHar] epoch 40 | act_loss 0.4244 | dom_loss 1.4125 | la
mbda 0.964 | tr_acc 84.2% | val_loss 0.6487 | val_acc 78.4%
[full_s3:KuHar] early stop at epoch 46
[full_s3:KuHar] src-val 81.58% | TARGET-TEST 63.89% | saved ckpt
+ npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_s3:MotionSense] trainable params: 51,531
[full_s3:MotionSense] epoch 1 | act_loss 0.7765 | dom_loss 1.341
2 | lambda 0.050 | tr_acc 67.6% | val_loss 0.9257 | val_acc 63.4% *
[full_s3:MotionSense] epoch 2 | act_loss 0.6088 | dom_loss 1.279
3 | lambda 0.100 | tr_acc 75.0% | val_loss 0.7844 | val_acc 71.2% *
[full_s3:MotionSense] epoch 3 | act_loss 0.5560 | dom_loss 1.267
1 | lambda 0.149 | tr_acc 77.2% | val_loss 0.7779 | val_acc 71.8% *
[full_s3:MotionSense] epoch 4 | act_loss 0.5375 | dom_loss 1.271
6 | lambda 0.197 | tr_acc 77.7% | val_loss 0.6359 | val_acc 75.8% *
[full_s3:MotionSense] epoch 5 | act_loss 0.5220 | dom_loss 1.274
2 | lambda 0.245 | tr_acc 78.5% | val_loss 0.6435 | val_acc 77.1% *
[full_s3:MotionSense] epoch 6 | act_loss 0.5087 | dom_loss 1.284
3 | lambda 0.291 | tr_acc 79.0% | val_loss 0.6286 | val_acc 77.6% *
[full_s3:MotionSense] epoch 11 | act_loss 0.4766 | dom_loss 1.302
2 | lambda 0.501 | tr_acc 80.5% | val_loss 0.6451 | val_acc 77.8% *
[full_s3:MotionSense] epoch 17 | act_loss 0.4621 | dom_loss 1.314
9 | lambda 0.691 | tr_acc 81.6% | val_loss 0.6171 | val_acc 77.9% *
[full_s3:MotionSense] epoch 20 | act_loss 0.4597 | dom_loss 1.315

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7 | lambda 0.762 | tr_acc 81.8% | val_loss 0.6167 | val_acc 80.1% *
  [full_s3:MotionSense] epoch 40 | act_loss 0.4404 | dom_loss 1.339
8 | lambda 0.964 | tr_acc 83.3% | val_loss 0.6220 | val_acc 80.5% *
  [full_s3:MotionSense] epoch 41 | act_loss 0.4358 | dom_loss 1.338
2 | lambda 0.967 | tr_acc 83.3% | val_loss 0.5794 | val_acc 80.7% *
  [full_s3:MotionSense] epoch 60 | act_loss 0.4054 | dom_loss 1.337
4 | lambda 0.995 | tr_acc 85.1% | val_loss 0.7182 | val_acc 69.8%
  [full_s3:MotionSense] early stop at epoch 61
  [full_s3:MotionSense] src-val 80.70% | TARGET-TEST 83.33% | save
d ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
  [full_s3:RealWorld-Thigh] trainable params: 51,531
  [full_s3:RealWorld-Thigh] epoch 1 | act_loss 0.7589 | dom_loss
1.3001 | lambda 0.050 | tr_acc 68.1% | val_loss 0.7469 | val_acc 69.
5% *
  [full_s3:RealWorld-Thigh] epoch 2 | act_loss 0.5782 | dom_loss
1.2096 | lambda 0.100 | tr_acc 75.2% | val_loss 0.7521 | val_acc 69.
8% *
  [full_s3:RealWorld-Thigh] epoch 3 | act_loss 0.5384 | dom_loss
1.2118 | lambda 0.149 | tr_acc 77.1% | val_loss 0.6370 | val_acc 75.
0% *
  [full_s3:RealWorld-Thigh] epoch 4 | act_loss 0.5190 | dom_loss
1.2108 | lambda 0.197 | tr_acc 78.2% | val_loss 0.5939 | val_acc 76.
4% *
  [full_s3:RealWorld-Thigh] epoch 5 | act_loss 0.4976 | dom_loss
1.2187 | lambda 0.245 | tr_acc 79.0% | val_loss 0.6069 | val_acc 77.
0% *
  [full_s3:RealWorld-Thigh] epoch 10 | act_loss 0.4532 | dom_loss
1.2463 | lambda 0.462 | tr_acc 80.9% | val_loss 0.5910 | val_acc 77.
9% *
  [full_s3:RealWorld-Thigh] epoch 12 | act_loss 0.4484 | dom_loss
1.2571 | lambda 0.537 | tr_acc 81.4% | val_loss 0.5481 | val_acc 80.
8% *
  [full_s3:RealWorld-Thigh] epoch 13 | act_loss 0.4419 | dom_loss
1.2563 | lambda 0.572 | tr_acc 81.6% | val_loss 0.5322 | val_acc 81.
6% *
  [full_s3:RealWorld-Thigh] epoch 20 | act_loss 0.4200 | dom_loss

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1.2699 | lambda 0.762 | tr_acc 82.9% | val_loss 0.5766 | val_acc 78.2%
    [full_s3:RealWorld-Thigh] epoch 29 | act_loss 0.4179 | dom_loss
1.2934 | lambda 0.896 | tr_acc 83.4% | val_loss 0.5259 | val_acc 82.8% *
    [full_s3:RealWorld-Thigh] epoch 40 | act_loss 0.4073 | dom_loss
1.3116 | lambda 0.964 | tr_acc 84.2% | val_loss 0.5644 | val_acc 79.6%
    [full_s3:RealWorld-Thigh] epoch 44 | act_loss 0.4023 | dom_loss
1.3112 | lambda 0.976 | tr_acc 84.3% | val_loss 0.5247 | val_acc 83.6% *
    [full_s3:RealWorld-Thigh] epoch 52 | act_loss 0.3872 | dom_loss
1.3088 | lambda 0.989 | tr_acc 84.9% | val_loss 0.4780 | val_acc 84.9% *
    [full_s3:RealWorld-Thigh] epoch 53 | act_loss 0.3849 | dom_loss
1.3103 | lambda 0.990 | tr_acc 84.9% | val_loss 0.4690 | val_acc 84.9% *
    [full_s3:RealWorld-Thigh] epoch 59 | act_loss 0.3772 | dom_loss
1.3104 | lambda 0.995 | tr_acc 85.6% | val_loss 0.4886 | val_acc 85.6% *
    [full_s3:RealWorld-Thigh] epoch 60 | act_loss 0.3749 | dom_loss
1.3116 | lambda 0.995 | tr_acc 85.5% | val_loss 0.5102 | val_acc 83.8%
    [full_s3:RealWorld-Thigh] epoch 64 | act_loss 0.3676 | dom_loss
1.3099 | lambda 0.997 | tr_acc 85.9% | val_loss 0.4638 | val_acc 85.9% *
    [full_s3:RealWorld-Thigh] epoch 72 | act_loss 0.3502 | dom_loss
1.3099 | lambda 0.999 | tr_acc 86.6% | val_loss 0.4626 | val_acc 86.3% *
    [full_s3:RealWorld-Thigh] epoch 80 | act_loss 0.3365 | dom_loss
1.3101 | lambda 0.999 | tr_acc 87.3% | val_loss 0.4473 | val_acc 85.4%
    [full_s3:RealWorld-Thigh] epoch 85 | act_loss 0.3313 | dom_loss
1.3089 | lambda 1.000 | tr_acc 87.6% | val_loss 0.4322 | val_acc 87.1% *
    [full_s3:RealWorld-Thigh] epoch 100 | act_loss 0.3172 | dom_loss
1.3095 | lambda 1.000 | tr_acc 88.3% | val_loss 0.4509 | val_acc 85.6%
    [full_s3:RealWorld-Thigh] src-val 87.12% | TARGET-TEST 69.25% |
saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (6 nodes, 18 edges, 90 features each)
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs

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built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_s3:RealWorld-Waist] trainable params: 51,531
[full_s3:RealWorld-Waist] epoch 1 | act_loss 0.7705 | dom_loss
1.3360 | lambda 0.050 | tr_acc 67.1% | val_loss 0.7501 | val_acc 70.
7% *
[full_s3:RealWorld-Waist] epoch 2 | act_loss 0.5968 | dom_loss
1.2592 | lambda 0.100 | tr_acc 75.1% | val_loss 0.6332 | val_acc 75.
2% *
[full_s3:RealWorld-Waist] epoch 4 | act_loss 0.5212 | dom_loss
1.2488 | lambda 0.197 | tr_acc 78.5% | val_loss 0.6274 | val_acc 75.
8% *
[full_s3:RealWorld-Waist] epoch 6 | act_loss 0.4900 | dom_loss
1.2546 | lambda 0.291 | tr_acc 80.0% | val_loss 0.6010 | val_acc 76.
3% *
[full_s3:RealWorld-Waist] epoch 7 | act_loss 0.4822 | dom_loss
1.2560 | lambda 0.336 | tr_acc 80.1% | val_loss 0.6139 | val_acc 77.
6% *
[full_s3:RealWorld-Waist] epoch 15 | act_loss 0.4382 | dom_loss
1.2696 | lambda 0.635 | tr_acc 82.3% | val_loss 0.5672 | val_acc 79.
0% *
[full_s3:RealWorld-Waist] epoch 20 | act_loss 0.4243 | dom_loss
1.2726 | lambda 0.762 | tr_acc 83.1% | val_loss 0.8504 | val_acc 65.
5%
[full_s3:RealWorld-Waist] epoch 27 | act_loss 0.4196 | dom_loss
1.2888 | lambda 0.874 | tr_acc 83.8% | val_loss 0.5592 | val_acc 80.
8% *
[full_s3:RealWorld-Waist] epoch 40 | act_loss 0.4022 | dom_loss
1.3040 | lambda 0.964 | tr_acc 85.1% | val_loss 0.9072 | val_acc 70.
1%
[full_s3:RealWorld-Waist] epoch 45 | act_loss 0.3929 | dom_loss
1.3035 | lambda 0.978 | tr_acc 85.3% | val_loss 0.5819 | val_acc 80.
9% *
[full_s3:RealWorld-Waist] epoch 49 | act_loss 0.3896 | dom_loss
1.3055 | lambda 0.985 | tr_acc 85.4% | val_loss 0.5549 | val_acc 81.
7% *
[full_s3:RealWorld-Waist] epoch 57 | act_loss 0.3698 | dom_loss
1.3049 | lambda 0.993 | tr_acc 86.5% | val_loss 0.5308 | val_acc 81.
8% *
[full_s3:RealWorld-Waist] epoch 60 | act_loss 0.3670 | dom_loss
1.3054 | lambda 0.995 | tr_acc 86.5% | val_loss 0.5342 | val_acc 80.
8%
[full_s3:RealWorld-Waist] epoch 70 | act_loss 0.3454 | dom_loss
1.3047 | lambda 0.998 | tr_acc 87.6% | val_loss 0.5644 | val_acc 82.
0% *
[full_s3:RealWorld-Waist] epoch 75 | act_loss 0.3365 | dom_loss
1.3047 | lambda 0.999 | tr_acc 88.1% | val_loss 0.5575 | val_acc 82.
3% *
[full_s3:RealWorld-Waist] epoch 78 | act_loss 0.3314 | dom_loss
1.3056 | lambda 0.999 | tr_acc 88.2% | val_loss 0.5571 | val_acc 82.
8% *
[full_s3:RealWorld-Waist] epoch 80 | act_loss 0.3275 | dom_loss
1.3038 | lambda 0.999 | tr_acc 88.6% | val_loss 0.6256 | val_acc 77.
3%
[full_s3:RealWorld-Waist] epoch 90 | act_loss 0.3151 | dom_loss

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1.3062 | lambda 1.000 | tr_acc 89.1% | val_loss 0.5430 | val_acc 82.
8% *
[full_s3:RealWorld-Waist] epoch 92 | act_loss 0.3108 | dom_loss
1.3041 | lambda 1.000 | tr_acc 89.1% | val_loss 0.5328 | val_acc 83.
2% *
[full_s3:RealWorld-Waist] epoch 100 | act_loss 0.3102 | dom_loss
1.3053 | lambda 1.000 | tr_acc 89.3% | val_loss 0.5414 | val_acc 82.
3%
[full_s3:RealWorld-Waist] src-val 83.16% | TARGET-TEST 71.03% |
saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_s3:UCI] trainable params: 51,531
[full_s3:UCI] epoch 1 | act_loss 0.7631 | dom_loss 1.3935 | lamb
da 0.050 | tr_acc 68.5% | val_loss 0.7157 | val_acc 71.3% *
[full_s3:UCI] epoch 3 | act_loss 0.5592 | dom_loss 1.3347 | lamb
da 0.149 | tr_acc 77.2% | val_loss 0.7464 | val_acc 72.7% *
[full_s3:UCI] epoch 4 | act_loss 0.5300 | dom_loss 1.3252 | lamb
da 0.197 | tr_acc 78.5% | val_loss 0.6805 | val_acc 74.9% *
[full_s3:UCI] epoch 5 | act_loss 0.5162 | dom_loss 1.3265 | lamb
da 0.245 | tr_acc 79.0% | val_loss 0.6333 | val_acc 77.5% *
[full_s3:UCI] epoch 17 | act_loss 0.4593 | dom_loss 1.3611 | lamb
da 0.691 | tr_acc 81.7% | val_loss 0.6349 | val_acc 79.1% *
[full_s3:UCI] epoch 20 | act_loss 0.4546 | dom_loss 1.3631 | lamb
da 0.762 | tr_acc 82.4% | val_loss 0.6258 | val_acc 79.3% *
[full_s3:UCI] epoch 24 | act_loss 0.4473 | dom_loss 1.3677 | lamb
da 0.834 | tr_acc 82.5% | val_loss 0.6378 | val_acc 80.1% *
[full_s3:UCI] epoch 30 | act_loss 0.4415 | dom_loss 1.3725 | lamb
da 0.905 | tr_acc 83.1% | val_loss 0.5800 | val_acc 80.7% *
[full_s3:UCI] epoch 40 | act_loss 0.4304 | dom_loss 1.3907 | lamb
da 0.964 | tr_acc 84.0% | val_loss 0.6633 | val_acc 77.4%
[full_s3:UCI] epoch 43 | act_loss 0.4283 | dom_loss 1.3894 | lamb
da 0.973 | tr_acc 83.9% | val_loss 0.5902 | val_acc 80.8% *
[full_s3:UCI] epoch 45 | act_loss 0.4227 | dom_loss 1.3888 | lamb
da 0.978 | tr_acc 84.3% | val_loss 0.5592 | val_acc 81.2% *
[full_s3:UCI] epoch 48 | act_loss 0.4182 | dom_loss 1.3889 | lamb
da 0.984 | tr_acc 84.5% | val_loss 0.5599 | val_acc 82.5% *

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[full_s3:UCI] epoch 52 | act_loss 0.4123 | dom_loss 1.3880 | lamb
da 0.989 | tr_acc 84.6% | val_loss 0.5461 | val_acc 82.8% *
[full_s3:UCI] epoch 60 | act_loss 0.3958 | dom_loss 1.3860 | lamb
da 0.995 | tr_acc 85.7% | val_loss 0.5766 | val_acc 80.4%
[full_s3:UCI] early stop at epoch 72
[full_s3:UCI] src-val 82.79% | TARGET-TEST 84.06% | saved ckpt +
npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_s3:WISDM] trainable params: 51,531
[full_s3:WISDM] epoch 1 | act_loss 0.7629 | dom_loss 1.3410 | la
mbda 0.050 | tr_acc 69.1% | val_loss 0.7861 | val_acc 68.3% *
[full_s3:WISDM] epoch 3 | act_loss 0.5414 | dom_loss 1.3098 | la
mbda 0.149 | tr_acc 78.5% | val_loss 0.6172 | val_acc 77.2% *
[full_s3:WISDM] epoch 10 | act_loss 0.4577 | dom_loss 1.3259 | la
mbda 0.462 | tr_acc 82.2% | val_loss 0.7358 | val_acc 77.7% *
[full_s3:WISDM] epoch 12 | act_loss 0.4439 | dom_loss 1.3268 | la
mbda 0.537 | tr_acc 82.6% | val_loss 0.6378 | val_acc 78.8% *
[full_s3:WISDM] epoch 20 | act_loss 0.4198 | dom_loss 1.3333 | la
mbda 0.762 | tr_acc 84.2% | val_loss 0.6768 | val_acc 78.1%
[full_s3:WISDM] epoch 26 | act_loss 0.4084 | dom_loss 1.3378 | la
mbda 0.862 | tr_acc 84.9% | val_loss 0.5743 | val_acc 81.3% *
[full_s3:WISDM] epoch 40 | act_loss 0.3838 | dom_loss 1.3386 | la
mbda 0.964 | tr_acc 86.2% | val_loss 0.5896 | val_acc 80.9%
[full_s3:WISDM] early stop at epoch 46
[full_s3:WISDM] src-val 81.27% | TARGET-TEST 74.16% | saved ckpt
+ npz
[P24] train+save | arm=full_s5 {'cnn_embed_dim': 64, 'gnn_hidden': 64,
'cnn_channels': [32, 64]} | nc=None aw=- | seed=5 | targets=['KuHar',
'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] |
out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphReviv
al_DG/P24_SO_Compression/results/checkpoints/full_s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5

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[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_s5:KuHar] trainable params: 51,531
[full_s5:KuHar] epoch 1 | act_loss 0.7598 | dom_loss 1.4281 | la
mbda 0.050 | tr_acc 68.3% | val_loss 1.1015 | val_acc 55.2% *
[full_s5:KuHar] epoch 2 | act_loss 0.5945 | dom_loss 1.3573 | la
mbda 0.100 | tr_acc 75.8% | val_loss 1.0264 | val_acc 62.6% *
[full_s5:KuHar] epoch 4 | act_loss 0.5272 | dom_loss 1.3526 | la
mbda 0.197 | tr_acc 78.6% | val_loss 0.8077 | val_acc 66.0% *
[full_s5:KuHar] epoch 5 | act_loss 0.5071 | dom_loss 1.3605 | la
mbda 0.245 | tr_acc 79.1% | val_loss 0.6697 | val_acc 76.5% *
[full_s5:KuHar] epoch 7 | act_loss 0.4950 | dom_loss 1.3771 | la
mbda 0.336 | tr_acc 79.9% | val_loss 0.5911 | val_acc 78.8% *
[full_s5:KuHar] epoch 11 | act_loss 0.4728 | dom_loss 1.3893 | la
mbda 0.501 | tr_acc 81.2% | val_loss 0.6306 | val_acc 80.0% *
[full_s5:KuHar] epoch 18 | act_loss 0.4531 | dom_loss 1.3989 | la
mbda 0.716 | tr_acc 82.6% | val_loss 0.5684 | val_acc 80.6% *
[full_s5:KuHar] epoch 20 | act_loss 0.4477 | dom_loss 1.3999 | la
mbda 0.762 | tr_acc 82.8% | val_loss 0.5725 | val_acc 80.9% *
[full_s5:KuHar] epoch 28 | act_loss 0.4373 | dom_loss 1.4053 | la
mbda 0.885 | tr_acc 83.6% | val_loss 0.5310 | val_acc 82.0% *
[full_s5:KuHar] epoch 32 | act_loss 0.4387 | dom_loss 1.4134 | la
mbda 0.922 | tr_acc 83.5% | val_loss 0.5654 | val_acc 82.5% *
[full_s5:KuHar] epoch 40 | act_loss 0.4295 | dom_loss 1.4230 | la
mbda 0.964 | tr_acc 84.1% | val_loss 0.7558 | val_acc 71.9%
[full_s5:KuHar] early stop at epoch 52
[full_s5:KuHar] src-val 82.48% | TARGET-TEST 65.28% | saved ckpt
+ npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac

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h)...
  built 5000/33230 graphs
  built 10000/33230 graphs
  built 15000/33230 graphs
  built 20000/33230 graphs
  built 25000/33230 graphs
  built 30000/33230 graphs
  done: 33230 graphs built
[full_s5:MotionSense] trainable params: 51,531
[full_s5:MotionSense] epoch 1 | act_loss 0.7804 | dom_loss 1.344
8 | lambda 0.050 | tr_acc 67.2% | val_loss 0.8234 | val_acc 65.4% *
[full_s5:MotionSense] epoch 2 | act_loss 0.6107 | dom_loss 1.265
6 | lambda 0.100 | tr_acc 74.8% | val_loss 0.7243 | val_acc 72.2% *
[full_s5:MotionSense] epoch 3 | act_loss 0.5654 | dom_loss 1.269
6 | lambda 0.149 | tr_acc 76.8% | val_loss 0.6365 | val_acc 75.4% *
[full_s5:MotionSense] epoch 5 | act_loss 0.5239 | dom_loss 1.272
8 | lambda 0.245 | tr_acc 78.5% | val_loss 0.6607 | val_acc 76.1% *
[full_s5:MotionSense] epoch 9 | act_loss 0.4919 | dom_loss 1.296
2 | lambda 0.422 | tr_acc 79.8% | val_loss 0.6368 | val_acc 77.3% *
[full_s5:MotionSense] epoch 12 | act_loss 0.4781 | dom_loss 1.305
3 | lambda 0.537 | tr_acc 80.5% | val_loss 0.6249 | val_acc 77.7% *
[full_s5:MotionSense] epoch 15 | act_loss 0.4683 | dom_loss 1.309
6 | lambda 0.635 | tr_acc 81.1% | val_loss 0.6365 | val_acc 78.8% *
[full_s5:MotionSense] epoch 20 | act_loss 0.4609 | dom_loss 1.317
3 | lambda 0.762 | tr_acc 81.7% | val_loss 0.7620 | val_acc 72.6%
[full_s5:MotionSense] epoch 22 | act_loss 0.4581 | dom_loss 1.318
1 | lambda 0.800 | tr_acc 81.9% | val_loss 0.5976 | val_acc 80.0% *
[full_s5:MotionSense] epoch 24 | act_loss 0.4563 | dom_loss 1.321
8 | lambda 0.834 | tr_acc 82.3% | val_loss 0.5740 | val_acc 81.5% *
[full_s5:MotionSense] epoch 40 | act_loss 0.4401 | dom_loss 1.344
7 | lambda 0.964 | tr_acc 83.3% | val_loss 0.6466 | val_acc 78.2%
[full_s5:MotionSense] early stop at epoch 44
[full_s5:MotionSense] src-val 81.53% | TARGET-TEST 83.24% | save
d ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/26450 graphs
  built 10000/26450 graphs
  built 15000/26450 graphs
  built 20000/26450 graphs
  built 25000/26450 graphs
  done: 26450 graphs built
[full_s5:RealWorld-Thigh] trainable params: 51,531
[full_s5:RealWorld-Thigh] epoch 1 | act_loss 0.7654 | dom_loss

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1.3042 | lambda 0.050 | tr_acc 67.5% | val_loss 0.7234 | val_acc 69.
4% *
[full_s5:RealWorld-Thigh] epoch 3 | act_loss 0.5454 | dom_loss
1.1940 | lambda 0.149 | tr_acc 76.9% | val_loss 0.5950 | val_acc 72.
3% *
[full_s5:RealWorld-Thigh] epoch 4 | act_loss 0.5146 | dom_loss
1.2062 | lambda 0.197 | tr_acc 78.4% | val_loss 0.5983 | val_acc 78.
1% *
[full_s5:RealWorld-Thigh] epoch 5 | act_loss 0.5001 | dom_loss
1.2164 | lambda 0.245 | tr_acc 78.9% | val_loss 0.5630 | val_acc 78.
3% *
[full_s5:RealWorld-Thigh] epoch 6 | act_loss 0.4846 | dom_loss
1.2354 | lambda 0.291 | tr_acc 79.7% | val_loss 0.5820 | val_acc 79.
1% *
[full_s5:RealWorld-Thigh] epoch 12 | act_loss 0.4423 | dom_loss
1.2584 | lambda 0.537 | tr_acc 82.0% | val_loss 0.5089 | val_acc 82.
8% *
[full_s5:RealWorld-Thigh] epoch 20 | act_loss 0.4216 | dom_loss
1.2682 | lambda 0.762 | tr_acc 83.1% | val_loss 0.4856 | val_acc 82.
8%
[full_s5:RealWorld-Thigh] early stop at epoch 32
[full_s5:RealWorld-Thigh] src-val 82.83% | TARGET-TEST 73.95% |
saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[full_s5:RealWorld-Waist] trainable params: 51,531
[full_s5:RealWorld-Waist] epoch 1 | act_loss 0.7778 | dom_loss
1.3269 | lambda 0.050 | tr_acc 66.9% | val_loss 0.7597 | val_acc 66.
6% *
[full_s5:RealWorld-Waist] epoch 2 | act_loss 0.6094 | dom_loss
1.2601 | lambda 0.100 | tr_acc 74.6% | val_loss 0.6397 | val_acc 75.
8% *
[full_s5:RealWorld-Waist] epoch 5 | act_loss 0.5039 | dom_loss
1.2424 | lambda 0.245 | tr_acc 79.1% | val_loss 0.5835 | val_acc 77.
5% *
[full_s5:RealWorld-Waist] epoch 7 | act_loss 0.4826 | dom_loss
1.2513 | lambda 0.336 | tr_acc 80.0% | val_loss 0.5673 | val_acc 78.
4% *

```

```

[full_s5:RealWorld-Waist] epoch 8 | act_loss 0.4759 | dom_loss
1.2525 | lambda 0.380 | tr_acc 80.5% | val_loss 0.5701 | val_acc 79.
2% *
[full_s5:RealWorld-Waist] epoch 20 | act_loss 0.4341 | dom_loss
1.2740 | lambda 0.762 | tr_acc 82.8% | val_loss 0.6385 | val_acc 75.
7%
[full_s5:RealWorld-Waist] epoch 22 | act_loss 0.4307 | dom_loss
1.2759 | lambda 0.800 | tr_acc 82.8% | val_loss 0.5750 | val_acc 79.
3% *
[full_s5:RealWorld-Waist] epoch 31 | act_loss 0.4285 | dom_loss
1.2998 | lambda 0.914 | tr_acc 83.7% | val_loss 0.6047 | val_acc 80.
3% *
[full_s5:RealWorld-Waist] epoch 39 | act_loss 0.4107 | dom_loss
1.3019 | lambda 0.960 | tr_acc 84.3% | val_loss 0.5976 | val_acc 80.
4% *
[full_s5:RealWorld-Waist] epoch 40 | act_loss 0.4126 | dom_loss
1.3013 | lambda 0.964 | tr_acc 84.5% | val_loss 0.6943 | val_acc 75.
0%
[full_s5:RealWorld-Waist] epoch 44 | act_loss 0.4025 | dom_loss
1.3037 | lambda 0.976 | tr_acc 84.9% | val_loss 0.6338 | val_acc 80.
7% *
[full_s5:RealWorld-Waist] epoch 53 | act_loss 0.3832 | dom_loss
1.3020 | lambda 0.990 | tr_acc 85.7% | val_loss 0.6334 | val_acc 81.
0% *
[full_s5:RealWorld-Waist] epoch 60 | act_loss 0.3702 | dom_loss
1.3043 | lambda 0.995 | tr_acc 86.5% | val_loss 0.6808 | val_acc 79.
3%
[full_s5:RealWorld-Waist] epoch 64 | act_loss 0.3645 | dom_loss
1.3061 | lambda 0.997 | tr_acc 86.8% | val_loss 0.5959 | val_acc 81.
0% *
[full_s5:RealWorld-Waist] epoch 73 | act_loss 0.3439 | dom_loss
1.3067 | lambda 0.999 | tr_acc 87.6% | val_loss 0.5793 | val_acc 82.
9% *
[full_s5:RealWorld-Waist] epoch 80 | act_loss 0.3319 | dom_loss
1.3057 | lambda 0.999 | tr_acc 88.4% | val_loss 0.5906 | val_acc 81.
3%
[full_s5:RealWorld-Waist] early stop at epoch 93
[full_s5:RealWorld-Waist] src-val 82.86% | TARGET-TEST 71.84% |
saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs

```

```

built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_s5:UCI] trainable params: 51,531
[full_s5:UCI] epoch 1 | act_loss 0.7681 | dom_loss 1.3975 | lamb
da 0.050 | tr_acc 68.1% | val_loss 0.9891 | val_acc 55.7% *
[full_s5:UCI] epoch 2 | act_loss 0.6108 | dom_loss 1.3176 | lamb
da 0.100 | tr_acc 75.4% | val_loss 0.7014 | val_acc 74.3% *
[full_s5:UCI] epoch 5 | act_loss 0.5195 | dom_loss 1.3232 | lamb
da 0.245 | tr_acc 79.0% | val_loss 0.6245 | val_acc 77.2% *
[full_s5:UCI] epoch 18 | act_loss 0.4621 | dom_loss 1.3633 | lamb
da 0.716 | tr_acc 81.6% | val_loss 0.7402 | val_acc 77.2% *
[full_s5:UCI] epoch 20 | act_loss 0.4562 | dom_loss 1.3642 | lamb
da 0.762 | tr_acc 82.1% | val_loss 0.6316 | val_acc 77.7% *
[full_s5:UCI] epoch 21 | act_loss 0.4551 | dom_loss 1.3653 | lamb
da 0.782 | tr_acc 82.1% | val_loss 0.6874 | val_acc 77.8% *
[full_s5:UCI] epoch 23 | act_loss 0.4542 | dom_loss 1.3669 | lamb
da 0.818 | tr_acc 82.4% | val_loss 0.6226 | val_acc 79.3% *
[full_s5:UCI] epoch 28 | act_loss 0.4438 | dom_loss 1.3704 | lamb
da 0.885 | tr_acc 83.0% | val_loss 0.5910 | val_acc 80.9% *
[full_s5:UCI] epoch 31 | act_loss 0.4439 | dom_loss 1.3808 | lamb
da 0.914 | tr_acc 82.9% | val_loss 0.6247 | val_acc 81.0% *
[full_s5:UCI] epoch 40 | act_loss 0.4335 | dom_loss 1.3902 | lamb
da 0.964 | tr_acc 83.7% | val_loss 0.8668 | val_acc 66.5%
[full_s5:UCI] epoch 45 | act_loss 0.4254 | dom_loss 1.3916 | lamb
da 0.978 | tr_acc 84.1% | val_loss 0.5952 | val_acc 81.4% *
[full_s5:UCI] epoch 60 | act_loss 0.3996 | dom_loss 1.3887 | lamb
da 0.995 | tr_acc 85.2% | val_loss 0.6145 | val_acc 79.1%
[full_s5:UCI] early stop at epoch 65
[full_s5:UCI] src-val 81.43% | TARGET-TEST 79.71% | saved ckpt +
npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_s5:WISDM] trainable params: 51,531
[full_s5:WISDM] epoch 1 | act_loss 0.7793 | dom_loss 1.3323 | la
mbda 0.050 | tr_acc 68.2% | val_loss 0.6766 | val_acc 74.0% *
[full_s5:WISDM] epoch 4 | act_loss 0.5209 | dom_loss 1.3010 | la

```

```

mbda 0.197 | tr_acc 79.3% | val_loss 0.6537 | val_acc 76.2% *
[full_s5:WISDM] epoch 6 | act_loss 0.4987 | dom_loss 1.3130 | la
mbda 0.291 | tr_acc 80.4% | val_loss 0.6372 | val_acc 77.4% *
[full_s5:WISDM] epoch 9 | act_loss 0.4724 | dom_loss 1.3230 | la
mbda 0.422 | tr_acc 81.7% | val_loss 0.6187 | val_acc 78.8% *
[full_s5:WISDM] epoch 12 | act_loss 0.4539 | dom_loss 1.3279 | la
mbda 0.537 | tr_acc 82.4% | val_loss 0.6499 | val_acc 79.3% *
[full_s5:WISDM] epoch 13 | act_loss 0.4450 | dom_loss 1.3289 | la
mbda 0.572 | tr_acc 82.7% | val_loss 0.6408 | val_acc 80.0% *
[full_s5:WISDM] epoch 16 | act_loss 0.4340 | dom_loss 1.3321 | la
mbda 0.664 | tr_acc 83.4% | val_loss 0.6158 | val_acc 80.0% *
[full_s5:WISDM] epoch 20 | act_loss 0.4243 | dom_loss 1.3339 | la
mbda 0.762 | tr_acc 84.2% | val_loss 0.6621 | val_acc 77.4%
[full_s5:WISDM] epoch 27 | act_loss 0.4170 | dom_loss 1.3388 | la
mbda 0.874 | tr_acc 84.4% | val_loss 0.6311 | val_acc 81.0% *
[full_s5:WISDM] epoch 31 | act_loss 0.4020 | dom_loss 1.3389 | la
mbda 0.914 | tr_acc 85.4% | val_loss 0.6443 | val_acc 81.9% *
[full_s5:WISDM] epoch 40 | act_loss 0.3889 | dom_loss 1.3389 | la
mbda 0.964 | tr_acc 85.7% | val_loss 0.6815 | val_acc 77.6%
[full_s5:WISDM] early stop at epoch 51
[full_s5:WISDM] src-val 81.90% | TARGET-TEST 71.51% | saved ckpt
+ npz

```

6. Top up the headline arms no_gz & tiny to 5 seeds

They already have seeds {42,1,2}; add {3,5} so every headline P24 number uses the same canonical 5-seed protocol (cheap — no_gz is 5-node, tiny is quarter-width).

```
In [9]: for s in (3, 5):  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant full  
        --nc no_gz --seed {s} --epochs 100 --log_every 20  
        !python -m p24.train_save --data_path "$DATA_PATH" --variant tiny  
        --seed {s} --epochs 100 --log_every 20
```

```

[P24] train+save | arm=full_nc-no_gz_s3 {'cnn_embed_dim': 64, 'gnn_hidden': 64, 'cnn_channels': [32, 64]} | nc=no_gz aw=- | seed=3 | targets=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_nc-no_gz_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (1, 4):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (0, 1):100%, (1, 4):100%, (3, 4):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (5 nodes, 18 edges, 90 features each)...)
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-no_gz_s3:KuHar] trainable params: 51,339
[full_nc-no_gz_s3:KuHar] epoch 1 | act_loss 0.7457 | dom_loss 1.4214 | lambda 0.050 | tr_acc 69.6% | val_loss 0.7712 | val_acc 72.5%
*
[full_nc-no_gz_s3:KuHar] epoch 2 | act_loss 0.6077 | dom_loss 1.3694 | lambda 0.100 | tr_acc 75.3% | val_loss 0.6876 | val_acc 75.1%
*
[full_nc-no_gz_s3:KuHar] epoch 5 | act_loss 0.5258 | dom_loss 1.3688 | lambda 0.245 | tr_acc 79.1% | val_loss 0.7129 | val_acc 75.3%
*
[full_nc-no_gz_s3:KuHar] epoch 6 | act_loss 0.5155 | dom_loss 1.3772 | lambda 0.291 | tr_acc 79.8% | val_loss 0.6438 | val_acc 78.3%
*
[full_nc-no_gz_s3:KuHar] epoch 11 | act_loss 0.4871 | dom_loss 1.3885 | lambda 0.501 | tr_acc 81.0% | val_loss 0.6102 | val_acc 79.0%
*
[full_nc-no_gz_s3:KuHar] epoch 20 | act_loss 0.4660 | dom_loss 1.4059 | lambda 0.762 | tr_acc 82.0% | val_loss 0.7937 | val_acc 74.2%
[full_nc-no_gz_s3:KuHar] early stop at epoch 31
[full_nc-no_gz_s3:KuHar] src-val 79.01% | TARGET-TEST 61.11% | saved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph

```

```

[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (1, 4):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 33230 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
  [full_nc-no_gz_s3:MotionSense] trainable params: 51,339
  [full_nc-no_gz_s3:MotionSense] epoch 1 | act_loss 0.7751 | dom_l
oss 1.3425 | lambda 0.050 | tr_acc 68.0% | val_loss 0.6692 | val_acc
74.1% *
  [full_nc-no_gz_s3:MotionSense] epoch 9 | act_loss 0.5150 | dom_l
oss 1.2947 | lambda 0.422 | tr_acc 79.3% | val_loss 0.6346 | val_acc
76.4% *
  [full_nc-no_gz_s3:MotionSense] epoch 20 | act_loss 0.4784 | dom_l
oss 1.3172 | lambda 0.762 | tr_acc 81.2% | val_loss 0.7205 | val_acc
70.4%
  [full_nc-no_gz_s3:MotionSense] early stop at epoch 29
  [full_nc-no_gz_s3:MotionSense] src-val 76.36% | TARGET-TEST 83.90%
| saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
  [full_nc-no_gz_s3:RealWorld-Thigh] trainable params: 51,339
  [full_nc-no_gz_s3:RealWorld-Thigh] epoch 1 | act_loss 0.7505 | d
om_loss 1.2888 | lambda 0.050 | tr_acc 68.5% | val_loss 0.6602 | val_
acc 72.2% *
  [full_nc-no_gz_s3:RealWorld-Thigh] epoch 3 | act_loss 0.5615 | d

```

```

om_loss 1.2024 | lambda 0.149 | tr_acc 76.0% | val_loss 0.6745 | val_
acc 73.8% *
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 5 | act_loss 0.5225 | d
om_loss 1.2161 | lambda 0.245 | tr_acc 77.8% | val_loss 0.6008 | val_
acc 75.8% *
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 14 | act_loss 0.4673 | d
om_loss 1.2629 | lambda 0.604 | tr_acc 79.9% | val_loss 0.5976 | val_
acc 76.7% *
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 16 | act_loss 0.4615 | d
om_loss 1.2684 | lambda 0.664 | tr_acc 80.4% | val_loss 0.6085 | val_
acc 76.8% *
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 20 | act_loss 0.4549 | d
om_loss 1.2737 | lambda 0.762 | tr_acc 80.6% | val_loss 0.8138 | val_
acc 71.2%
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 29 | act_loss 0.4512 | d
om_loss 1.2855 | lambda 0.896 | tr_acc 81.0% | val_loss 0.5903 | val_
acc 77.7% *
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 35 | act_loss 0.4427 | d
om_loss 1.2963 | lambda 0.941 | tr_acc 81.4% | val_loss 0.6038 | val_
acc 77.9% *
    [full_nc-no_gz_s3:RealWorld-Thigh] epoch 40 | act_loss 0.4423 | d
om_loss 1.3045 | lambda 0.964 | tr_acc 82.0% | val_loss 0.8753 | val_
acc 66.3%
    [full_nc-no_gz_s3:RealWorld-Thigh] early stop at epoch 55
    [full_nc-no_gz_s3:RealWorld-Thigh] src-val 77.94% | TARGET-TEST 6
6.29% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (1, 4):100%, (0, 3):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
    [full_nc-no_gz_s3:RealWorld-Waist] trainable params: 51,339
    [full_nc-no_gz_s3:RealWorld-Waist] epoch 1 | act_loss 0.7722 | d
om_loss 1.3281 | lambda 0.050 | tr_acc 67.5% | val_loss 0.7010 | val_
acc 72.3% *
    [full_nc-no_gz_s3:RealWorld-Waist] epoch 2 | act_loss 0.6213 | d
om_loss 1.2477 | lambda 0.100 | tr_acc 74.4% | val_loss 0.6968 | val_
acc 74.2% *
    [full_nc-no_gz_s3:RealWorld-Waist] epoch 4 | act_loss 0.5494 | d
om_loss 1.2572 | lambda 0.197 | tr_acc 77.4% | val_loss 0.6573 | val_

```

```

acc 76.1% *
[full_nc-no_gz_s3:RealWorld-Waist] epoch 16 | act_loss 0.4518 | d
om_loss 1.2697 | lambda 0.664 | tr_acc 82.3% | val_loss 0.6577 | val_
acc 77.1% *
[full_nc-no_gz_s3:RealWorld-Waist] epoch 20 | act_loss 0.4463 | d
om_loss 1.2779 | lambda 0.762 | tr_acc 82.4% | val_loss 0.7496 | val_
acc 73.9%
[full_nc-no_gz_s3:RealWorld-Waist] epoch 31 | act_loss 0.4377 | d
om_loss 1.2969 | lambda 0.914 | tr_acc 83.1% | val_loss 0.6176 | val_
acc 79.3% *
[full_nc-no_gz_s3:RealWorld-Waist] epoch 40 | act_loss 0.4261 | d
om_loss 1.3031 | lambda 0.964 | tr_acc 84.0% | val_loss 0.8503 | val_
acc 66.9%
[full_nc-no_gz_s3:RealWorld-Waist] early stop at epoch 51
[full_nc-no_gz_s3:RealWorld-Waist] src-val 79.27% | TARGET-TEST 7
2.72% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%, (3, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[full_nc-no_gz_s3:UCI] trainable params: 51,339
[full_nc-no_gz_s3:UCI] epoch 1 | act_loss 0.7587 | dom_loss 1.39
45 | lambda 0.050 | tr_acc 68.6% | val_loss 0.6917 | val_acc 74.6%
*
[full_nc-no_gz_s3:UCI] epoch 3 | act_loss 0.5784 | dom_loss 1.32
64 | lambda 0.149 | tr_acc 76.5% | val_loss 0.6468 | val_acc 76.1%
*
[full_nc-no_gz_s3:UCI] epoch 5 | act_loss 0.5367 | dom_loss 1.32
43 | lambda 0.245 | tr_acc 78.6% | val_loss 0.6610 | val_acc 76.4%
*
[full_nc-no_gz_s3:UCI] epoch 9 | act_loss 0.5017 | dom_loss 1.34
41 | lambda 0.422 | tr_acc 80.1% | val_loss 0.6670 | val_acc 76.6%
*
[full_nc-no_gz_s3:UCI] epoch 20 | act_loss 0.4711 | dom_loss 1.36
67 | lambda 0.762 | tr_acc 81.8% | val_loss 0.7191 | val_acc 74.8%
[full_nc-no_gz_s3:UCI] epoch 27 | act_loss 0.4656 | dom_loss 1.37
26 | lambda 0.874 | tr_acc 82.3% | val_loss 0.6704 | val_acc 77.0%
*

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[full_nc-no_gz_s3:UCI] epoch 29 | act_loss 0.4660 | dom_loss 1.37
43 | lambda 0.896 | tr_acc 82.1% | val_loss 0.6373 | val_acc 78.9%
*
[full_nc-no_gz_s3:UCI] epoch 40 | act_loss 0.4545 | dom_loss 1.38
80 | lambda 0.964 | tr_acc 82.6% | val_loss 0.7021 | val_acc 73.9%
[full_nc-no_gz_s3:UCI] epoch 47 | act_loss 0.4451 | dom_loss 1.39
17 | lambda 0.982 | tr_acc 83.1% | val_loss 0.5696 | val_acc 80.9%
*
[full_nc-no_gz_s3:UCI] epoch 60 | act_loss 0.4237 | dom_loss 1.38
65 | lambda 0.995 | tr_acc 84.0% | val_loss 0.5776 | val_acc 80.3%
[full_nc-no_gz_s3:UCI] early stop at epoch 67
[full_nc-no_gz_s3:UCI] src-val 80.89% | TARGET-TEST 81.45% | sav
ed ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (0, 3):100%, (0, 1):100%, (0, 2):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-no_gz_s3:WISDM] trainable params: 51,339
[full_nc-no_gz_s3:WISDM] epoch 1 | act_loss 0.7654 | dom_loss 1.
3325 | lambda 0.050 | tr_acc 69.3% | val_loss 0.9362 | val_acc 61.4%
*
[full_nc-no_gz_s3:WISDM] epoch 2 | act_loss 0.6110 | dom_loss 1.
2931 | lambda 0.100 | tr_acc 75.5% | val_loss 0.8931 | val_acc 66.5%
*
[full_nc-no_gz_s3:WISDM] epoch 4 | act_loss 0.5398 | dom_loss 1.
2976 | lambda 0.197 | tr_acc 78.7% | val_loss 0.6775 | val_acc 77.1%
*
[full_nc-no_gz_s3:WISDM] epoch 20 | act_loss 0.4402 | dom_loss 1.
3315 | lambda 0.762 | tr_acc 83.3% | val_loss 0.7625 | val_acc 70.3%
[full_nc-no_gz_s3:WISDM] early stop at epoch 24
[full_nc-no_gz_s3:WISDM] src-val 77.08% | TARGET-TEST 76.00% | s
aved ckpt + npz
[P24] train+save | arm=tiny_s3 {'cnn_embed_dim': 16, 'gnn_hidden': 16,
'cnn_channels': [8, 16]} | nc=None aw=- | seed=3 | targets=['KuHar',
'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] |
out=/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphReviv
al_DG/P24_S0_Compression/results/checkpoints/tiny_s3
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',

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'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/35396 graphs
    built 10000/35396 graphs
    built 15000/35396 graphs
    built 20000/35396 graphs
    built 25000/35396 graphs
    built 30000/35396 graphs
    built 35000/35396 graphs
    done: 35396 graphs built
[tiny_s3:KuHar] trainable params: 4,731
    [tiny_s3:KuHar] epoch 1 | act_loss 0.9950 | dom_loss 1.4448 | la
mbda 0.050 | tr_acc 57.1% | val_loss 0.7195 | val_acc 71.6% *
    [tiny_s3:KuHar] epoch 2 | act_loss 0.6736 | dom_loss 1.4031 | la
mbda 0.100 | tr_acc 71.9% | val_loss 0.7546 | val_acc 72.1% *
    [tiny_s3:KuHar] epoch 9 | act_loss 0.5492 | dom_loss 1.3964 | la
mbda 0.422 | tr_acc 78.1% | val_loss 0.6453 | val_acc 76.8% *
    [tiny_s3:KuHar] epoch 11 | act_loss 0.5442 | dom_loss 1.3986 | la
mbda 0.501 | tr_acc 78.3% | val_loss 0.5968 | val_acc 77.6% *
    [tiny_s3:KuHar] epoch 20 | act_loss 0.5224 | dom_loss 1.4064 | la
mbda 0.762 | tr_acc 79.3% | val_loss 0.7003 | val_acc 70.0%
    [tiny_s3:KuHar] epoch 21 | act_loss 0.5248 | dom_loss 1.4046 | la
mbda 0.782 | tr_acc 79.1% | val_loss 0.5846 | val_acc 79.9% *
    [tiny_s3:KuHar] epoch 40 | act_loss 0.5022 | dom_loss 1.4112 | la
mbda 0.964 | tr_acc 80.5% | val_loss 1.2461 | val_acc 60.7%
    [tiny_s3:KuHar] early stop at epoch 41
[tiny_s3:KuHar] src-val 79.86% | TARGET-TEST 64.58% | saved ckpt
+ npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs

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built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[tiny_s3:MotionSense] trainable params: 4,731
[tiny_s3:MotionSense] epoch 1 | act_loss 1.0103 | dom_loss 1.372
5 | lambda 0.050 | tr_acc 57.0% | val_loss 0.7364 | val_acc 71.7% *
[tiny_s3:MotionSense] epoch 5 | act_loss 0.5972 | dom_loss 1.303
6 | lambda 0.245 | tr_acc 75.4% | val_loss 0.6970 | val_acc 73.2% *
[tiny_s3:MotionSense] epoch 7 | act_loss 0.5816 | dom_loss 1.305
5 | lambda 0.336 | tr_acc 76.4% | val_loss 0.6588 | val_acc 74.5% *
[tiny_s3:MotionSense] epoch 18 | act_loss 0.5374 | dom_loss 1.322
4 | lambda 0.716 | tr_acc 78.4% | val_loss 0.6666 | val_acc 76.1% *
[tiny_s3:MotionSense] epoch 20 | act_loss 0.5394 | dom_loss 1.324
6 | lambda 0.762 | tr_acc 78.2% | val_loss 0.7969 | val_acc 71.2%
[tiny_s3:MotionSense] epoch 32 | act_loss 0.5249 | dom_loss 1.328
1 | lambda 0.922 | tr_acc 78.7% | val_loss 0.6660 | val_acc 76.2% *
[tiny_s3:MotionSense] epoch 33 | act_loss 0.5243 | dom_loss 1.328
4 | lambda 0.929 | tr_acc 78.8% | val_loss 0.6442 | val_acc 76.3% *
[tiny_s3:MotionSense] epoch 38 | act_loss 0.5245 | dom_loss 1.331
4 | lambda 0.956 | tr_acc 79.0% | val_loss 0.5640 | val_acc 79.6% *
[tiny_s3:MotionSense] epoch 40 | act_loss 0.5204 | dom_loss 1.330
2 | lambda 0.964 | tr_acc 79.3% | val_loss 1.1828 | val_acc 70.7%
[tiny_s3:MotionSense] epoch 58 | act_loss 0.4981 | dom_loss 1.329
6 | lambda 0.994 | tr_acc 80.5% | val_loss 0.5683 | val_acc 81.0% *
[tiny_s3:MotionSense] epoch 60 | act_loss 0.4946 | dom_loss 1.332
0 | lambda 0.995 | tr_acc 80.7% | val_loss 0.8552 | val_acc 70.0%
[tiny_s3:MotionSense] early stop at epoch 78
[tiny_s3:MotionSense] src-val 81.05% | TARGET-TEST 81.17% | save
d ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[tiny_s3:RealWorld-Thigh] trainable params: 4,731
[tiny_s3:RealWorld-Thigh] epoch 1 | act_loss 1.0475 | dom_loss
1.3411 | lambda 0.050 | tr_acc 54.8% | val_loss 0.7494 | val_acc 66.
8% *
[tiny_s3:RealWorld-Thigh] epoch 2 | act_loss 0.6738 | dom_loss
1.2806 | lambda 0.100 | tr_acc 70.5% | val_loss 0.6751 | val_acc 70.

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9% *
[tiny_s3:RealWorld-Thigh] epoch 3 | act_loss 0.6155 | dom_loss
1.2579 | lambda 0.149 | tr_acc 73.6% | val_loss 0.6421 | val_acc 74.
5% *
[tiny_s3:RealWorld-Thigh] epoch 10 | act_loss 0.5376 | dom_loss
1.2657 | lambda 0.462 | tr_acc 76.9% | val_loss 0.5650 | val_acc 77.
4% *
[tiny_s3:RealWorld-Thigh] epoch 20 | act_loss 0.5118 | dom_loss
1.2771 | lambda 0.762 | tr_acc 78.3% | val_loss 0.7745 | val_acc 67.
4%
[tiny_s3:RealWorld-Thigh] early stop at epoch 30
[tiny_s3:RealWorld-Thigh] src-val 77.44% | TARGET-TEST 70.36% |
saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[tiny_s3:RealWorld-Waist] trainable params: 4,731
[tiny_s3:RealWorld-Waist] epoch 1 | act_loss 1.0543 | dom_loss
1.3555 | lambda 0.050 | tr_acc 55.3% | val_loss 0.7260 | val_acc 68.
6% *
[tiny_s3:RealWorld-Waist] epoch 3 | act_loss 0.6427 | dom_loss
1.2985 | lambda 0.149 | tr_acc 73.0% | val_loss 0.6732 | val_acc 69.
4% *
[tiny_s3:RealWorld-Waist] epoch 5 | act_loss 0.5927 | dom_loss
1.2842 | lambda 0.245 | tr_acc 75.7% | val_loss 0.7331 | val_acc 71.
0% *
[tiny_s3:RealWorld-Waist] epoch 6 | act_loss 0.5804 | dom_loss
1.2766 | lambda 0.291 | tr_acc 75.9% | val_loss 0.7267 | val_acc 73.
2% *
[tiny_s3:RealWorld-Waist] epoch 7 | act_loss 0.5681 | dom_loss
1.2740 | lambda 0.336 | tr_acc 76.6% | val_loss 0.6376 | val_acc 74.
2% *
[tiny_s3:RealWorld-Waist] epoch 8 | act_loss 0.5548 | dom_loss
1.2750 | lambda 0.380 | tr_acc 77.5% | val_loss 0.6615 | val_acc 75.
6% *
[tiny_s3:RealWorld-Waist] epoch 12 | act_loss 0.5254 | dom_loss
1.2783 | lambda 0.537 | tr_acc 78.7% | val_loss 0.5659 | val_acc 79.
1% *
[tiny_s3:RealWorld-Waist] epoch 20 | act_loss 0.5037 | dom_loss

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1.2847 | lambda 0.762 | tr_acc 79.8% | val_loss 0.8808 | val_acc 64.7%
[tiny_s3:RealWorld-Waist] epoch 25 | act_loss 0.4992 | dom_loss
1.2893 | lambda 0.848 | tr_acc 80.1% | val_loss 0.5853 | val_acc 79.5% *
[tiny_s3:RealWorld-Waist] epoch 40 | act_loss 0.4824 | dom_loss
1.2921 | lambda 0.964 | tr_acc 81.2% | val_loss 0.9953 | val_acc 65.7%
[tiny_s3:RealWorld-Waist] early stop at epoch 45
[tiny_s3:RealWorld-Waist] src-val 79.47% | TARGET-TEST 73.57% |
saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features each)...)
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[tiny_s3:UCI] trainable params: 4,731
[tiny_s3:UCI] epoch 1 | act_loss 0.9906 | dom_loss 1.4279 | lambda 0.050 | tr_acc 57.9% | val_loss 0.7364 | val_acc 69.8% *
[tiny_s3:UCI] epoch 5 | act_loss 0.5940 | dom_loss 1.3442 | lambda 0.245 | tr_acc 76.4% | val_loss 0.7343 | val_acc 72.7% *
[tiny_s3:UCI] epoch 11 | act_loss 0.5604 | dom_loss 1.3605 | lambda 0.501 | tr_acc 77.5% | val_loss 0.7015 | val_acc 73.7% *
[tiny_s3:UCI] epoch 14 | act_loss 0.5442 | dom_loss 1.3669 | lambda 0.604 | tr_acc 78.6% | val_loss 0.6524 | val_acc 75.1% *
[tiny_s3:UCI] epoch 15 | act_loss 0.5443 | dom_loss 1.3676 | lambda 0.635 | tr_acc 78.5% | val_loss 0.6649 | val_acc 76.6% *
[tiny_s3:UCI] epoch 17 | act_loss 0.5421 | dom_loss 1.3700 | lambda 0.691 | tr_acc 78.6% | val_loss 0.6286 | val_acc 76.8% *
[tiny_s3:UCI] epoch 20 | act_loss 0.5355 | dom_loss 1.3723 | lambda 0.762 | tr_acc 78.8% | val_loss 0.7318 | val_acc 71.8%
[tiny_s3:UCI] epoch 29 | act_loss 0.5271 | dom_loss 1.3747 | lambda 0.896 | tr_acc 79.1% | val_loss 0.6687 | val_acc 77.9% *
[tiny_s3:UCI] epoch 40 | act_loss 0.5182 | dom_loss 1.3789 | lambda 0.964 | tr_acc 79.6% | val_loss 0.7160 | val_acc 69.4%
[tiny_s3:UCI] epoch 45 | act_loss 0.5141 | dom_loss 1.3784 | lambda 0.978 | tr_acc 79.9% | val_loss 0.5923 | val_acc 79.8% *
[tiny_s3:UCI] epoch 48 | act_loss 0.5117 | dom_loss 1.3791 | lambda 0.984 | tr_acc 80.0% | val_loss 0.6077 | val_acc 80.1% *
[tiny_s3:UCI] epoch 60 | act_loss 0.4968 | dom_loss 1.3824 | lambda

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da 0.995 | tr_acc 80.9% | val_loss 0.8102 | val_acc 71.0%
[tiny_s3:UCI] early stop at epoch 68
[tiny_s3:UCI] src-val 80.14% | TARGET-TEST 78.41% | saved ckpt +
npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features each)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[tiny_s3:WISDM] trainable params: 4,731
[tiny_s3:WISDM] epoch 1 | act_loss 1.0369 | dom_loss 1.3631 | lambda
mbda 0.050 | tr_acc 57.7% | val_loss 0.8352 | val_acc 64.1% *
[tiny_s3:WISDM] epoch 2 | act_loss 0.6745 | dom_loss 1.3236 | lambda
mbda 0.100 | tr_acc 72.4% | val_loss 0.6794 | val_acc 70.7% *
[tiny_s3:WISDM] epoch 3 | act_loss 0.6183 | dom_loss 1.3171 | lambda
mbda 0.149 | tr_acc 75.4% | val_loss 0.7381 | val_acc 73.3% *
[tiny_s3:WISDM] epoch 4 | act_loss 0.5910 | dom_loss 1.3190 | lambda
mbda 0.197 | tr_acc 76.4% | val_loss 0.6689 | val_acc 75.0% *
[tiny_s3:WISDM] epoch 6 | act_loss 0.5577 | dom_loss 1.3163 | lambda
mbda 0.291 | tr_acc 78.1% | val_loss 0.6209 | val_acc 78.1% *
[tiny_s3:WISDM] epoch 18 | act_loss 0.5126 | dom_loss 1.3297 | lambda
mbda 0.716 | tr_acc 80.0% | val_loss 0.6367 | val_acc 78.3% *
[tiny_s3:WISDM] epoch 20 | act_loss 0.5092 | dom_loss 1.3299 | lambda
mbda 0.762 | tr_acc 80.0% | val_loss 1.3235 | val_acc 59.5%
[tiny_s3:WISDM] early stop at epoch 38
[tiny_s3:WISDM] src-val 78.27% | TARGET-TEST 75.65% | saved ckpt +
npz
[P24] train+save | arm=full_nc-no_gz_s5 {'cnn_embed_dim': 64, 'gnn_hidden':
64, 'cnn_channels': [32, 64]} | nc=no_gz aw=- | seed=5 | targets
=['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI',
'WISDM'] | out=/content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20
_GraphRevival_DG/P24_SO_Compression/results/checkpoints/full_nc-no_gz_
s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept

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pairs: (3, 4):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (1, 4):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (1, 4):100%, (3, 4):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[full_nc-no_gz_s5:KuHar] trainable params: 51,339
[full_nc-no_gz_s5:KuHar] epoch 1 | act_loss 0.7501 | dom_loss 1.
4204 | lambda 0.050 | tr_acc 69.4% | val_loss 0.8064 | val_acc 64.7%
*
[full_nc-no_gz_s5:KuHar] epoch 2 | act_loss 0.6110 | dom_loss 1.
3711 | lambda 0.100 | tr_acc 75.4% | val_loss 0.6411 | val_acc 76.3%
*
[full_nc-no_gz_s5:KuHar] epoch 7 | act_loss 0.5065 | dom_loss 1.
3785 | lambda 0.336 | tr_acc 80.0% | val_loss 0.6725 | val_acc 77.5%
*
[full_nc-no_gz_s5:KuHar] epoch 17 | act_loss 0.4751 | dom_loss 1.
4026 | lambda 0.691 | tr_acc 81.5% | val_loss 0.6397 | val_acc 78.0%
*
[full_nc-no_gz_s5:KuHar] epoch 20 | act_loss 0.4680 | dom_loss 1.
4008 | lambda 0.762 | tr_acc 81.9% | val_loss 0.7573 | val_acc 71.8%
[full_nc-no_gz_s5:KuHar] epoch 22 | act_loss 0.4631 | dom_loss 1.
4039 | lambda 0.800 | tr_acc 82.0% | val_loss 0.6085 | val_acc 80.0%
*
[full_nc-no_gz_s5:KuHar] epoch 40 | act_loss 0.4551 | dom_loss 1.
4185 | lambda 0.964 | tr_acc 82.6% | val_loss 0.6028 | val_acc 79.8%
[full_nc-no_gz_s5:KuHar] early stop at epoch 42
[full_nc-no_gz_s5:KuHar] src-val 80.03% | TARGET-TEST 65.97% | s
aved ckpt + npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs

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built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[full_nc-no_gz_s5:MotionSense] trainable params: 51,339
[full_nc-no_gz_s5:MotionSense] epoch 1 | act_loss 0.7826 | dom_loss 1.3320 | lambda 0.050 | tr_acc 67.3% | val_loss 0.7266 | val_acc 72.2% *
[full_nc-no_gz_s5:MotionSense] epoch 6 | act_loss 0.5309 | dom_loss 1.2761 | lambda 0.291 | tr_acc 78.5% | val_loss 0.6253 | val_acc 77.5% *
[full_nc-no_gz_s5:MotionSense] epoch 17 | act_loss 0.4827 | dom_loss 1.3186 | lambda 0.691 | tr_acc 80.7% | val_loss 0.6184 | val_acc 78.9% *
[full_nc-no_gz_s5:MotionSense] epoch 20 | act_loss 0.4792 | dom_loss 1.3199 | lambda 0.762 | tr_acc 81.2% | val_loss 0.7815 | val_acc 67.1%
[full_nc-no_gz_s5:MotionSense] early stop at epoch 37
[full_nc-no_gz_s5:MotionSense] src-val 78.89% | TARGET-TEST 83.24%
| saved ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (5 nodes, 18 edges, 90 features each)...)
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[full_nc-no_gz_s5:RealWorld-Thigh] trainable params: 51,339
[full_nc-no_gz_s5:RealWorld-Thigh] epoch 1 | act_loss 0.7540 | dom_loss 1.2867 | lambda 0.050 | tr_acc 68.1% | val_loss 0.7173 | val_acc 71.1% *
[full_nc-no_gz_s5:RealWorld-Thigh] epoch 7 | act_loss 0.5070 | dom_loss 1.2363 | lambda 0.336 | tr_acc 78.4% | val_loss 0.7773 | val_acc 71.5% *
[full_nc-no_gz_s5:RealWorld-Thigh] epoch 9 | act_loss 0.4971 | dom_loss 1.2521 | lambda 0.422 | tr_acc 79.0% | val_loss 0.6206 | val_acc 75.6% *
[full_nc-no_gz_s5:RealWorld-Thigh] epoch 12 | act_loss 0.4785 | dom_loss 1.2648 | lambda 0.537 | tr_acc 79.5% | val_loss 0.5465 | val_acc 76.6% *
[full_nc-no_gz_s5:RealWorld-Thigh] epoch 13 | act_loss 0.4738 | d

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om_loss 1.2629 | lambda 0.572 | tr_acc 79.9% | val_loss 0.5785 | val_
acc 77.8% *
    [full_nc-no_gz_s5:RealWorld-Thigh] epoch 20 | act_loss 0.4593 | d
om_loss 1.2709 | lambda 0.762 | tr_acc 80.6% | val_loss 0.6665 | val_
acc 70.7%
    [full_nc-no_gz_s5:RealWorld-Thigh] epoch 30 | act_loss 0.4561 | d
om_loss 1.3020 | lambda 0.905 | tr_acc 81.2% | val_loss 0.5744 | val_
acc 79.2% *
    [full_nc-no_gz_s5:RealWorld-Thigh] epoch 40 | act_loss 0.4386 | d
om_loss 1.3104 | lambda 0.964 | tr_acc 82.0% | val_loss 0.6357 | val_
acc 76.7%
    [full_nc-no_gz_s5:RealWorld-Thigh] early stop at epoch 50
    [full_nc-no_gz_s5:RealWorld-Thigh] src-val 79.22% | TARGET-TEST 7
1.36% | saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (1, 4):100%, (0, 3):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
    [full_nc-no_gz_s5:RealWorld-Waist] trainable params: 51,339
    [full_nc-no_gz_s5:RealWorld-Waist] epoch 1 | act_loss 0.7723 | d
om_loss 1.3240 | lambda 0.050 | tr_acc 67.6% | val_loss 0.8131 | val_
acc 67.6% *
    [full_nc-no_gz_s5:RealWorld-Waist] epoch 2 | act_loss 0.6357 | d
om_loss 1.2610 | lambda 0.100 | tr_acc 73.6% | val_loss 0.6465 | val_
acc 75.3% *
    [full_nc-no_gz_s5:RealWorld-Waist] epoch 6 | act_loss 0.5163 | d
om_loss 1.2468 | lambda 0.291 | tr_acc 78.9% | val_loss 0.6858 | val_
acc 76.1% *
    [full_nc-no_gz_s5:RealWorld-Waist] epoch 7 | act_loss 0.4990 | d
om_loss 1.2485 | lambda 0.336 | tr_acc 79.7% | val_loss 0.6103 | val_
acc 77.6% *
    [full_nc-no_gz_s5:RealWorld-Waist] epoch 20 | act_loss 0.4446 | d
om_loss 1.2789 | lambda 0.762 | tr_acc 82.6% | val_loss 0.8017 | val_
acc 67.4%
    [full_nc-no_gz_s5:RealWorld-Waist] early stop at epoch 27
    [full_nc-no_gz_s5:RealWorld-Waist] src-val 77.59% | TARGET-TEST 6
8.87% | saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl

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d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%, (3, 4):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 34368 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs
    built 30000/34368 graphs
    done: 34368 graphs built
[full_nc-no_gz_s5:UCI] trainable params: 51,339
[full_nc-no_gz_s5:UCI] epoch 1 | act_loss 0.7598 | dom_loss 1.38
68 | lambda 0.050 | tr_acc 69.0% | val_loss 0.7252 | val_acc 73.7%
*
[full_nc-no_gz_s5:UCI] epoch 2 | act_loss 0.6213 | dom_loss 1.31
10 | lambda 0.100 | tr_acc 74.9% | val_loss 0.6700 | val_acc 75.1%
*
[full_nc-no_gz_s5:UCI] epoch 6 | act_loss 0.5266 | dom_loss 1.32
40 | lambda 0.291 | tr_acc 79.1% | val_loss 0.6774 | val_acc 77.1%
*
[full_nc-no_gz_s5:UCI] epoch 20 | act_loss 0.4734 | dom_loss 1.36
86 | lambda 0.762 | tr_acc 81.6% | val_loss 0.8429 | val_acc 68.2%
[full_nc-no_gz_s5:UCI] epoch 25 | act_loss 0.4671 | dom_loss 1.37
10 | lambda 0.848 | tr_acc 82.0% | val_loss 0.6396 | val_acc 78.5%
*
[full_nc-no_gz_s5:UCI] epoch 40 | act_loss 0.4561 | dom_loss 1.38
76 | lambda 0.964 | tr_acc 82.8% | val_loss 0.8650 | val_acc 68.7%
[full_nc-no_gz_s5:UCI] early stop at epoch 45
[full_nc-no_gz_s5:UCI] src-val 78.52% | TARGET-TEST 74.35% | sav
ed ckpt + npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P24 NC: keeping channels [0, 1, 2, 3, 4] -> 5 nodes/graph
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (1, 4):100%, (1, 2):100%, (0, 3):100%, (0, 1):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (0, 3):100%, (0, 1):100%, (3, 4):100%, (1, 4):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 4):100%, (0, 3):100%, (0, 1):100%, (0, 2):100%, (1, 4):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 28040 sensor graphs (5 nodes, 18 edges, 90 features eac
h)...

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```

built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[full_nc-no_gz_s5:WISDM] trainable params: 51,339
[full_nc-no_gz_s5:WISDM] epoch 1 | act_loss 0.7700 | dom_loss 1.
3201 | lambda 0.050 | tr_acc 68.6% | val_loss 0.7021 | val_acc 74.1%
*
[full_nc-no_gz_s5:WISDM] epoch 2 | act_loss 0.6183 | dom_loss 1.
2866 | lambda 0.100 | tr_acc 75.1% | val_loss 0.6437 | val_acc 76.2%
*
[full_nc-no_gz_s5:WISDM] epoch 18 | act_loss 0.4515 | dom_loss 1.
3286 | lambda 0.716 | tr_acc 83.1% | val_loss 0.7277 | val_acc 77.3%
*
[full_nc-no_gz_s5:WISDM] epoch 20 | act_loss 0.4451 | dom_loss 1.
3309 | lambda 0.762 | tr_acc 83.1% | val_loss 0.8959 | val_acc 67.9%
[full_nc-no_gz_s5:WISDM] early stop at epoch 38
[full_nc-no_gz_s5:WISDM] src-val 77.33% | TARGET-TEST 68.09% | s
aved ckpt + npz
[P24] train+save | arm=tiny_s5 {'cnn_embed_dim': 16, 'gnn_hidden': 16,
'cnn_channels': [8, 16]} | nc=None aw=- | seed=5 | targets=['KuHar',
'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM'] |
out=/content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphReviv
al_DG/P24_S0_Compression/results/checkpoints/tiny_s5
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[tiny_s5:KuHar] trainable params: 4,731
[tiny_s5:KuHar] epoch 1 | act_loss 1.0298 | dom_loss 1.4557 | la
mbda 0.050 | tr_acc 56.1% | val_loss 0.7635 | val_acc 72.5% *
[tiny_s5:KuHar] epoch 2 | act_loss 0.7049 | dom_loss 1.4114 | la
mbda 0.100 | tr_acc 71.7% | val_loss 0.6928 | val_acc 74.2% *
[tiny_s5:KuHar] epoch 9 | act_loss 0.5659 | dom_loss 1.3889 | la
mbda 0.422 | tr_acc 77.5% | val_loss 0.6515 | val_acc 76.6% *
[tiny_s5:KuHar] epoch 13 | act_loss 0.5531 | dom_loss 1.4035 | la

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mbda 0.572 | tr_acc 77.8% | val_loss 0.6324 | val_acc 77.1% *
[tiny_s5:KuHar] epoch 15 | act_loss 0.5442 | dom_loss 1.4050 | la
mbda 0.635 | tr_acc 78.3% | val_loss 0.6222 | val_acc 77.2% *
[tiny_s5:KuHar] epoch 20 | act_loss 0.5393 | dom_loss 1.4067 | la
mbda 0.762 | tr_acc 78.7% | val_loss 0.7587 | val_acc 72.4%
[tiny_s5:KuHar] early stop at epoch 35
[tiny_s5:KuHar] src-val 77.24% | TARGET-TEST 53.47% | saved ckpt
+ npz
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[tiny_s5:MotionSense] trainable params: 4,731
[tiny_s5:MotionSense] epoch 1 | act_loss 1.0634 | dom_loss 1.383
2 | lambda 0.050 | tr_acc 54.3% | val_loss 0.8487 | val_acc 68.7% *
[tiny_s5:MotionSense] epoch 2 | act_loss 0.7532 | dom_loss 1.334
3 | lambda 0.100 | tr_acc 69.0% | val_loss 0.7602 | val_acc 71.3% *
[tiny_s5:MotionSense] epoch 3 | act_loss 0.6842 | dom_loss 1.313
1 | lambda 0.149 | tr_acc 71.9% | val_loss 0.7406 | val_acc 72.5% *
[tiny_s5:MotionSense] epoch 5 | act_loss 0.6253 | dom_loss 1.295
6 | lambda 0.245 | tr_acc 74.4% | val_loss 0.6721 | val_acc 73.7% *
[tiny_s5:MotionSense] epoch 7 | act_loss 0.5969 | dom_loss 1.299
2 | lambda 0.336 | tr_acc 75.7% | val_loss 0.6858 | val_acc 74.1% *
[tiny_s5:MotionSense] epoch 11 | act_loss 0.5665 | dom_loss 1.308
8 | lambda 0.501 | tr_acc 77.1% | val_loss 0.6668 | val_acc 74.4% *
[tiny_s5:MotionSense] epoch 13 | act_loss 0.5606 | dom_loss 1.315
0 | lambda 0.572 | tr_acc 76.9% | val_loss 0.6576 | val_acc 75.1% *
[tiny_s5:MotionSense] epoch 14 | act_loss 0.5619 | dom_loss 1.314
5 | lambda 0.604 | tr_acc 77.3% | val_loss 0.6469 | val_acc 75.1% *
[tiny_s5:MotionSense] epoch 20 | act_loss 0.5455 | dom_loss 1.324
6 | lambda 0.762 | tr_acc 77.8% | val_loss 0.8518 | val_acc 63.7%
[tiny_s5:MotionSense] epoch 21 | act_loss 0.5448 | dom_loss 1.324
2 | lambda 0.782 | tr_acc 77.8% | val_loss 0.6066 | val_acc 78.2% *
[tiny_s5:MotionSense] epoch 38 | act_loss 0.5262 | dom_loss 1.332
7 | lambda 0.956 | tr_acc 78.9% | val_loss 0.6214 | val_acc 78.4% *
[tiny_s5:MotionSense] epoch 40 | act_loss 0.5262 | dom_loss 1.333
8 | lambda 0.964 | tr_acc 78.7% | val_loss 0.7001 | val_acc 71.5%
[tiny_s5:MotionSense] epoch 44 | act_loss 0.5200 | dom_loss 1.335
8 | lambda 0.976 | tr_acc 79.2% | val_loss 0.6016 | val_acc 78.7% *

```

```

[tiny_s5:MotionSense] epoch 47 | act_loss 0.5161 | dom_loss 1.337
2 | lambda 0.982 | tr_acc 79.5% | val_loss 0.5796 | val_acc 79.9% *
[tiny_s5:MotionSense] epoch 60 | act_loss 0.5068 | dom_loss 1.339
4 | lambda 0.995 | tr_acc 79.8% | val_loss 0.6326 | val_acc 77.5%
[tiny_s5:MotionSense] early stop at epoch 67
[tiny_s5:MotionSense] src-val 79.87% | TARGET-TEST 76.93% | save
d ckpt + npz
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[tiny_s5:RealWorld-Thigh] trainable params: 4,731
[tiny_s5:RealWorld-Thigh] epoch 1 | act_loss 1.0700 | dom_loss
1.3457 | lambda 0.050 | tr_acc 53.8% | val_loss 0.8972 | val_acc 59.
8% *
[tiny_s5:RealWorld-Thigh] epoch 2 | act_loss 0.7247 | dom_loss
1.2886 | lambda 0.100 | tr_acc 69.6% | val_loss 0.6977 | val_acc 72.
1% *
[tiny_s5:RealWorld-Thigh] epoch 3 | act_loss 0.6477 | dom_loss
1.2706 | lambda 0.149 | tr_acc 72.4% | val_loss 0.6585 | val_acc 74.
2% *
[tiny_s5:RealWorld-Thigh] epoch 12 | act_loss 0.5323 | dom_loss
1.2707 | lambda 0.537 | tr_acc 77.7% | val_loss 0.5747 | val_acc 78.
7% *
[tiny_s5:RealWorld-Thigh] epoch 14 | act_loss 0.5346 | dom_loss
1.2732 | lambda 0.604 | tr_acc 77.9% | val_loss 0.5636 | val_acc 78.
8% *
[tiny_s5:RealWorld-Thigh] epoch 19 | act_loss 0.5212 | dom_loss
1.2794 | lambda 0.740 | tr_acc 77.9% | val_loss 0.5881 | val_acc 78.
9% *
[tiny_s5:RealWorld-Thigh] epoch 20 | act_loss 0.5245 | dom_loss
1.2806 | lambda 0.762 | tr_acc 78.1% | val_loss 0.5625 | val_acc 77.
5%
[tiny_s5:RealWorld-Thigh] epoch 21 | act_loss 0.5192 | dom_loss
1.2832 | lambda 0.782 | tr_acc 78.7% | val_loss 0.5913 | val_acc 79.
0% *
[tiny_s5:RealWorld-Thigh] epoch 33 | act_loss 0.5154 | dom_loss
1.2932 | lambda 0.929 | tr_acc 78.3% | val_loss 0.5843 | val_acc 79.
1% *
[tiny_s5:RealWorld-Thigh] epoch 40 | act_loss 0.5044 | dom_loss

```

```

1.2942 | lambda 0.964 | tr_acc 78.8% | val_loss 0.5905 | val_acc 76.2%
    [tiny_s5:RealWorld-Thigh] epoch 46 | act_loss 0.5057 | dom_loss
1.3011 | lambda 0.980 | tr_acc 79.2% | val_loss 0.5618 | val_acc 80.3% *
    [tiny_s5:RealWorld-Thigh] epoch 59 | act_loss 0.4899 | dom_loss
1.3041 | lambda 0.995 | tr_acc 80.1% | val_loss 0.6001 | val_acc 80.3% *
    [tiny_s5:RealWorld-Thigh] epoch 60 | act_loss 0.4854 | dom_loss
1.3031 | lambda 0.995 | tr_acc 80.4% | val_loss 0.5842 | val_acc 79.9%
    [tiny_s5:RealWorld-Thigh] epoch 63 | act_loss 0.4834 | dom_loss
1.3056 | lambda 0.996 | tr_acc 80.3% | val_loss 0.5938 | val_acc 81.4% *
    [tiny_s5:RealWorld-Thigh] epoch 69 | act_loss 0.4772 | dom_loss
1.3063 | lambda 0.998 | tr_acc 81.1% | val_loss 0.5621 | val_acc 82.0% *
    [tiny_s5:RealWorld-Thigh] epoch 80 | act_loss 0.4654 | dom_loss
1.3075 | lambda 0.999 | tr_acc 81.5% | val_loss 0.5595 | val_acc 80.9%
    [tiny_s5:RealWorld-Thigh] epoch 84 | act_loss 0.4622 | dom_loss
1.3070 | lambda 1.000 | tr_acc 81.7% | val_loss 0.5384 | val_acc 82.6% *
    [tiny_s5:RealWorld-Thigh] epoch 100 | act_loss 0.4580 | dom_loss
1.3066 | lambda 1.000 | tr_acc 82.1% | val_loss 0.5328 | val_acc 81.9%
    [tiny_s5:RealWorld-Thigh] src-val 82.63% | TARGET-TEST 70.26% |
saved ckpt + npz
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26456 sensor graphs (6 nodes, 18 edges, 90 features each)...)
    built 5000/26456 graphs
    built 10000/26456 graphs
    built 15000/26456 graphs
    built 20000/26456 graphs
    built 25000/26456 graphs
    done: 26456 graphs built
    [tiny_s5:RealWorld-Waist] trainable params: 4,731
    [tiny_s5:RealWorld-Waist] epoch 1 | act_loss 1.0870 | dom_loss
1.3584 | lambda 0.050 | tr_acc 52.9% | val_loss 0.9933 | val_acc 59.5% *
    [tiny_s5:RealWorld-Waist] epoch 2 | act_loss 0.7522 | dom_loss
1.3188 | lambda 0.100 | tr_acc 68.1% | val_loss 0.6845 | val_acc 72.7% *
    [tiny_s5:RealWorld-Waist] epoch 4 | act_loss 0.6398 | dom_loss

```

```

1.2732 | lambda 0.197 | tr_acc 73.5% | val_loss 0.6853 | val_acc 74.2% *
[tiny_s5:RealWorld-Waist] epoch 7 | act_loss 0.5826 | dom_loss
1.2652 | lambda 0.336 | tr_acc 76.1% | val_loss 0.6687 | val_acc 74.7% *
[tiny_s5:RealWorld-Waist] epoch 8 | act_loss 0.5713 | dom_loss
1.2632 | lambda 0.380 | tr_acc 76.8% | val_loss 0.5833 | val_acc 77.9% *
[tiny_s5:RealWorld-Waist] epoch 20 | act_loss 0.5153 | dom_loss
1.2853 | lambda 0.762 | tr_acc 79.2% | val_loss 0.7617 | val_acc 67.7%
[tiny_s5:RealWorld-Waist] epoch 22 | act_loss 0.5184 | dom_loss
1.2877 | lambda 0.800 | tr_acc 78.9% | val_loss 0.5626 | val_acc 78.9% *
[tiny_s5:RealWorld-Waist] epoch 24 | act_loss 0.5166 | dom_loss
1.2892 | lambda 0.834 | tr_acc 79.2% | val_loss 0.5859 | val_acc 79.1% *
[tiny_s5:RealWorld-Waist] epoch 40 | act_loss 0.4992 | dom_loss
1.2977 | lambda 0.964 | tr_acc 80.4% | val_loss 0.5801 | val_acc 78.5%
[tiny_s5:RealWorld-Waist] early stop at epoch 44
[tiny_s5:RealWorld-Waist] src-val 79.10% | TARGET-TEST 74.31% |
saved ckpt + npz
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features each)...)
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[tiny_s5:UCI] trainable params: 4,731
[tiny_s5:UCI] epoch 1 | act_loss 1.0329 | dom_loss 1.4332 | lambda 0.050 | tr_acc 55.8% | val_loss 0.8274 | val_acc 70.3% *
[tiny_s5:UCI] epoch 2 | act_loss 0.7157 | dom_loss 1.3914 | lambda 0.100 | tr_acc 70.8% | val_loss 0.7086 | val_acc 73.3% *
[tiny_s5:UCI] epoch 3 | act_loss 0.6502 | dom_loss 1.3666 | lambda 0.149 | tr_acc 73.5% | val_loss 0.6991 | val_acc 73.7% *
[tiny_s5:UCI] epoch 5 | act_loss 0.6025 | dom_loss 1.3462 | lambda 0.245 | tr_acc 75.7% | val_loss 0.6342 | val_acc 76.1% *
[tiny_s5:UCI] epoch 12 | act_loss 0.5535 | dom_loss 1.3601 | lambda 0.537 | tr_acc 77.8% | val_loss 0.6295 | val_acc 76.4% *
[tiny_s5:UCI] epoch 14 | act_loss 0.5529 | dom_loss 1.3645 | lambda

```

```

da 0.604 | tr_acc 77.9% | val_loss 0.5982 | val_acc 76.7% *
    [tiny_s5:UCI] epoch 15 | act_loss 0.5493 | dom_loss 1.3661 | lamb
da 0.635 | tr_acc 77.9% | val_loss 0.6159 | val_acc 77.5% *
    [tiny_s5:UCI] epoch 20 | act_loss 0.5397 | dom_loss 1.3694 | lamb
da 0.762 | tr_acc 78.3% | val_loss 0.9637 | val_acc 62.5%
    [tiny_s5:UCI] epoch 26 | act_loss 0.5331 | dom_loss 1.3724 | lamb
da 0.862 | tr_acc 78.6% | val_loss 0.6152 | val_acc 77.9% *
    [tiny_s5:UCI] epoch 29 | act_loss 0.5319 | dom_loss 1.3730 | lamb
da 0.896 | tr_acc 78.8% | val_loss 0.5719 | val_acc 79.1% *
    [tiny_s5:UCI] epoch 40 | act_loss 0.5242 | dom_loss 1.3781 | lamb
da 0.964 | tr_acc 79.3% | val_loss 0.6277 | val_acc 78.6%
    [tiny_s5:UCI] epoch 41 | act_loss 0.5261 | dom_loss 1.3788 | lamb
da 0.967 | tr_acc 79.3% | val_loss 0.5721 | val_acc 80.0% *
    [tiny_s5:UCI] epoch 54 | act_loss 0.5157 | dom_loss 1.3805 | lamb
da 0.991 | tr_acc 79.7% | val_loss 0.5561 | val_acc 80.9% *
    [tiny_s5:UCI] epoch 60 | act_loss 0.5052 | dom_loss 1.3823 | lamb
da 0.995 | tr_acc 80.2% | val_loss 0.6619 | val_acc 70.3%
    [tiny_s5:UCI] epoch 67 | act_loss 0.4995 | dom_loss 1.3834 | lamb
da 0.998 | tr_acc 80.5% | val_loss 0.5714 | val_acc 81.1% *
    [tiny_s5:UCI] epoch 75 | act_loss 0.4908 | dom_loss 1.3830 | lamb
da 0.999 | tr_acc 81.0% | val_loss 0.5578 | val_acc 81.2% *
    [tiny_s5:UCI] epoch 80 | act_loss 0.4874 | dom_loss 1.3834 | lamb
da 0.999 | tr_acc 81.2% | val_loss 0.5761 | val_acc 80.8%
    [tiny_s5:UCI] epoch 92 | act_loss 0.4805 | dom_loss 1.3821 | lamb
da 1.000 | tr_acc 81.7% | val_loss 0.5575 | val_acc 81.2% *
    [tiny_s5:UCI] epoch 100 | act_loss 0.4776 | dom_loss 1.3818 | lamb
da 1.000 | tr_acc 81.5% | val_loss 0.5635 | val_acc 81.0%
    [tiny_s5:UCI] src-val 81.23% | TARGET-TEST 82.46% | saved ckpt +
npz
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:
(3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 28040 sensor graphs (6 nodes, 18 edges, 90 features each)
h)...
    built 5000/28040 graphs
    built 10000/28040 graphs
    built 15000/28040 graphs
    built 20000/28040 graphs
    built 25000/28040 graphs
    done: 28040 graphs built
    [tiny_s5:WISDM] trainable params: 4,731
    [tiny_s5:WISDM] epoch 1 | act_loss 1.0858 | dom_loss 1.3516 | lambda
mbda 0.050 | tr_acc 54.3% | val_loss 0.9016 | val_acc 63.0% *
    [tiny_s5:WISDM] epoch 2 | act_loss 0.7164 | dom_loss 1.3025 | lambda
mbda 0.100 | tr_acc 71.4% | val_loss 0.7826 | val_acc 71.9% *
    [tiny_s5:WISDM] epoch 4 | act_loss 0.6133 | dom_loss 1.3061 | lambda
mbda 0.197 | tr_acc 75.5% | val_loss 0.7256 | val_acc 73.5% *

```

```

[tiny_s5:WISDM] epoch 6 | act_loss 0.5813 | dom_loss 1.3091 | la
mbda 0.291 | tr_acc 76.9% | val_loss 0.6699 | val_acc 75.3% *
[tiny_s5:WISDM] epoch 7 | act_loss 0.5745 | dom_loss 1.3078 | la
mbda 0.336 | tr_acc 77.4% | val_loss 0.7203 | val_acc 75.7% *
[tiny_s5:WISDM] epoch 12 | act_loss 0.5406 | dom_loss 1.3186 | la
mbda 0.537 | tr_acc 79.0% | val_loss 0.6834 | val_acc 76.4% *
[tiny_s5:WISDM] epoch 13 | act_loss 0.5396 | dom_loss 1.3231 | la
mbda 0.572 | tr_acc 79.0% | val_loss 0.6943 | val_acc 77.3% *
[tiny_s5:WISDM] epoch 19 | act_loss 0.5172 | dom_loss 1.3292 | la
mbda 0.740 | tr_acc 79.8% | val_loss 0.6162 | val_acc 78.3% *
[tiny_s5:WISDM] epoch 20 | act_loss 0.5161 | dom_loss 1.3292 | la
mbda 0.762 | tr_acc 79.9% | val_loss 0.7729 | val_acc 72.2%
[tiny_s5:WISDM] epoch 27 | act_loss 0.5013 | dom_loss 1.3335 | la
mbda 0.874 | tr_acc 80.8% | val_loss 0.6364 | val_acc 79.2% *
[tiny_s5:WISDM] epoch 29 | act_loss 0.4941 | dom_loss 1.3338 | la
mbda 0.896 | tr_acc 81.1% | val_loss 0.5788 | val_acc 80.8% *
[tiny_s5:WISDM] epoch 40 | act_loss 0.4805 | dom_loss 1.3366 | la
mbda 0.964 | tr_acc 82.0% | val_loss 0.6099 | val_acc 77.6%
[tiny_s5:WISDM] epoch 47 | act_loss 0.4756 | dom_loss 1.3372 | la
mbda 0.982 | tr_acc 82.3% | val_loss 0.5968 | val_acc 81.0% *
[tiny_s5:WISDM] epoch 48 | act_loss 0.4710 | dom_loss 1.3354 | la
mbda 0.984 | tr_acc 82.5% | val_loss 0.5795 | val_acc 81.8% *
[tiny_s5:WISDM] epoch 54 | act_loss 0.4620 | dom_loss 1.3362 | la
mbda 0.991 | tr_acc 82.8% | val_loss 0.5660 | val_acc 82.5% *
[tiny_s5:WISDM] epoch 60 | act_loss 0.4563 | dom_loss 1.3357 | la
mbda 0.995 | tr_acc 83.0% | val_loss 0.6184 | val_acc 79.6%
[tiny_s5:WISDM] early stop at epoch 74
[tiny_s5:WISDM] src-val 82.49% | TARGET-TEST 70.85% | saved ckpt
+ npz

```

7. Aggregate — 5-seed accuracy report + Wilcoxon + GenPerByte

Fuses each seed's GNN probs with the matching `base_s{seed}` ConvNet ($w=0.6$), reports mean $\pm$ std, per-fold seed-means, GenPerByte with CI, and the paired Wilcoxon (no_gz vs full). Writes

`results/p24_seed_report_5.json`. Safe to run before all seeds finish — arms with missing seeds simply report a smaller n .

In [10]: `python -m p24.seed_report_5 --p21_sweep "$P21_SWEEP"`

```
=====
P24 - CANONICAL 5-SEED report (fused w=0.6, frozen ConvNet per seed)
seeds requested: [1, 2, 3, 5, 42] | canonical headline uses {1,
2,3,5,42}
Reference: P21 winner fused 77.96 ± 0.21% (5-seed)
=====
```

arm nPerByte	n	fused mean±std	GNN-only	Δ vs full	Ge
full (P21 winner)	5	77.96±0.21%	74.41% (reference)		0.
0035 accel_only	5	76.85±0.40%	68.29%	-1.11pp	0.
0035 gyro_only	5	73.14±1.19%	65.79%	-4.81pp	0.
0033 aw40	5	76.89±0.62%	72.78%	-1.06pp	0.
0035 aw30	5	75.73±0.22%	70.35%	-2.23pp	0.
0034 no_gz	5	77.97±0.75%	72.75%	+0.02pp	0.
0035 tiny	5	76.97±0.74%	72.07%	-0.99pp	0.
0205					

Per-fold fused seed-means:

fold	full (P21 win	accel_only	gyro_only	
aw40	aw30	no_gz	tiny	
KuHar		67.50%	62.64%	63.06%
65.00%	64.17%	65.83%	63.47%	
MotionSense		87.06%	87.14%	80.15%
85.20%	82.20%	87.87%	85.07%	
RealWorld-Thigh		74.93%	76.53%	70.75%
74.11%	72.51%	75.76%	75.09%	
RealWorld-Waist		76.49%	75.69%	70.23%
74.24%	73.56%	76.36%	76.68%	
UCI		86.38%	82.78%	83.86%
87.62%	87.25%	84.78%	87.04%	
WISDM		75.38%	76.30%	70.81%
75.19%	74.71%	77.23%	74.46%	

Paired Wilcoxon signed-rank across the 6 LOD0 folds (fused seed-means):

no_gz vs full (P21 winner): mean Δ +0.02pp | wilcoxon p=1.000 (n=6 folds – read effect size first)

saved /content/drive/MyDrive/Thesis/S0_Experiments/DAGHAR/P20_GraphRevival_DG/P24_S0_Compression/results/p24_seed_report_5.json

8. Aggregate — 5-seed RED/GS edge-budget table

Per-seed inference on the `full_s*` checkpoints, fused against `base_s{seed}` ; reports `policy×budget` seed `mean±std` and the paired Wilcoxon for `pearson-9` vs `RED-9` ("structure » random"). Writes `results/p24_edge_budget_multiseed.json` .

```
In [11]: python -m p24.edge_budget_multiseed --data_path "$DATA_PATH" --p21_sweep "$P21_SWEEP"
```

```

[edge_budget] seed 1 ...
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/35396 graphs
    built 10000/35396 graphs
    built 15000/35396 graphs
    built 20000/35396 graphs
    built 25000/35396 graphs
    built 30000/35396 graphs
    built 35000/35396 graphs
    done: 35396 graphs built
    [edge_budget s1:KuHar] done
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
    [edge_budget s1:MotionSense] done
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa

```

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irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/26450 graphs
  built 10000/26450 graphs
  built 15000/26450 graphs
  built 20000/26450 graphs
  built 25000/26450 graphs
  done: 26450 graphs built
  [edge_budget s1:RealWorld-Thigh] done
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/26456 graphs
  built 10000/26456 graphs
  built 15000/26456 graphs
  built 20000/26456 graphs
  built 25000/26456 graphs
  done: 26456 graphs built
  [edge_budget s1:RealWorld-Waist] done
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 34368 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/34368 graphs
  built 10000/34368 graphs
  built 15000/34368 graphs
  built 20000/34368 graphs
  built 25000/34368 graphs
  built 30000/34368 graphs
  done: 34368 graphs built

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[edge_budget s1:UCI] done
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 28040 sensor graphs (6 nodes, 18 edges, 90 features each)...)
    built 5000/28040 graphs
    built 10000/28040 graphs
    built 15000/28040 graphs
    built 20000/28040 graphs
    built 25000/28040 graphs
    done: 28040 graphs built
[edge_budget s1:WISDM] done
[edge_budget] seed 2 ...
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 35396 sensor graphs (6 nodes, 18 edges, 90 features each)...)
    built 5000/35396 graphs
    built 10000/35396 graphs
    built 15000/35396 graphs
    built 20000/35396 graphs
    built 25000/35396 graphs
    built 30000/35396 graphs
    built 35000/35396 graphs
    done: 35396 graphs built
[edge_budget s2:KuHar] done
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs:

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irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/33230 graphs
  built 10000/33230 graphs
  built 15000/33230 graphs
  built 20000/33230 graphs
  built 25000/33230 graphs
  built 30000/33230 graphs
  done: 33230 graphs built
  [edge_budget s2:MotionSense] done
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/26450 graphs
  built 10000/26450 graphs
  built 15000/26450 graphs
  built 20000/26450 graphs
  built 25000/26450 graphs
  done: 26450 graphs built
  [edge_budget s2:RealWorld-Thigh] done
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
  [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
  [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
  [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
  feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
  building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
  built 5000/26456 graphs
  built 10000/26456 graphs
  built 15000/26456 graphs
  built 20000/26456 graphs
  built 25000/26456 graphs
  done: 26456 graphs built

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[edge_budget s2:RealWorld-Waist] done
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 34368 sensor graphs (6 nodes, 18 edges, 90 features each)...)
    built 5000/34368 graphs
    built 10000/34368 graphs
    built 15000/34368 graphs
    built 20000/34368 graphs
    built 25000/34368 graphs
    built 30000/34368 graphs
    done: 34368 graphs built
    [edge_budget s2:UCI] done
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 28040 sensor graphs (6 nodes, 18 edges, 90 features each)...)
    built 5000/28040 graphs
    built 10000/28040 graphs
    built 15000/28040 graphs
    built 20000/28040 graphs
    built 25000/28040 graphs
    done: 28040 graphs built
    [edge_budget s2:WISDM] done
[edge_budget] seed 3 ...
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pairs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%

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[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[edge_budget s3:KuHar] done
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs
built 30000/33230 graphs
done: 33230 graphs built
[edge_budget s3:MotionSense] done
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs

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done: 26450 graphs built
[edge_budget s3:RealWorld-Thigh] done
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[edge_budget s3:RealWorld-Waist] done
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[edge_budget s3:UCI] done
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%

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[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[edge_budget s3:WISDM] done
[edge_budget] seed 5 ...
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs
built 35000/35396 graphs
done: 35396 graphs built
[edge_budget s5:KuHar] done
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/33230 graphs
built 10000/33230 graphs
built 15000/33230 graphs
built 20000/33230 graphs
built 25000/33230 graphs

```

```

built 30000/33230 graphs
done: 33230 graphs built
[edge_budget s5:MotionSense] done
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26450 graphs
built 10000/26450 graphs
built 15000/26450 graphs
built 20000/26450 graphs
built 25000/26450 graphs
done: 26450 graphs built
[edge_budget s5:RealWorld-Thigh] done
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[edge_budget s5:RealWorld-Waist] done
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%

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[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[edge_budget s5:UCI] done
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built
[edge_budget s5:WISDM] done
[edge_budget] seed 42 ...
[LOD0] target = KuHar
[LOD0] sources = ['MotionSense', 'RealWorld-Thigh', 'RealWorld-Waist',
'UCI', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=35396 val=5412 test=144
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (3, 4):100%, (3, 5):100%, (1, 2):100%, (1, 4):100%, (0, 2):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (0, 2):100%, (4, 5):100%, (3, 5):100%, (1, 4):100%, (2, 5):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%, (1, 4):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 35396 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/35396 graphs
built 10000/35396 graphs
built 15000/35396 graphs
built 20000/35396 graphs
built 25000/35396 graphs
built 30000/35396 graphs

```

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    built 35000/35396 graphs
    done: 35396 graphs built
    [edge_budget s42:KuHar] done
[LOD0] target = MotionSense
[LOD0] sources = ['KuHar', 'RealWorld-Thigh', 'RealWorld-Waist', 'UC
I', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=33230 val=5424 test=1062
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (3, 4):100%, (1, 2):100%, (4, 5):100%, (0, 1):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 33230 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/33230 graphs
    built 10000/33230 graphs
    built 15000/33230 graphs
    built 20000/33230 graphs
    built 25000/33230 graphs
    built 30000/33230 graphs
    done: 33230 graphs built
    [edge_budget s42:MotionSense] done
[LOD0] target = RealWorld-Thigh
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Waist', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26450 val=3990 test=2898
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
    [pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (1, 2):100%, (0, 1):100%, (3, 4):100%, (0, 3):100%
    feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
    building 26450 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
    built 5000/26450 graphs
    built 10000/26450 graphs
    built 15000/26450 graphs
    built 20000/26450 graphs
    built 25000/26450 graphs
    done: 26450 graphs built
    [edge_budget s42:RealWorld-Thigh] done
[LOD0] target = RealWorld-Waist
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'UCI', 'W
ISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=26456 val=3990 test=2592
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
    [pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
    [pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa

```

```

irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (0, 2):100%, (2, 5):100%, (1, 2):100%, (0, 1):100%, (3, 5):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 26456 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/26456 graphs
built 10000/26456 graphs
built 15000/26456 graphs
built 20000/26456 graphs
built 25000/26456 graphs
done: 26456 graphs built
[edge_budget s42:RealWorld-Waist] done
[LOD0] target = UCI
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'WISDM']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=34368 val=5504 test=690
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (4, 5):100%, (1, 2):100%, (0, 1):100%, (0, 3):100%, (0, 2):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 34368 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/34368 graphs
built 10000/34368 graphs
built 15000/34368 graphs
built 20000/34368 graphs
built 25000/34368 graphs
built 30000/34368 graphs
done: 34368 graphs built
[edge_budget s42:UCI] done
[LOD0] target = WISDM
[LOD0] sources = ['KuHar', 'MotionSense', 'RealWorld-Thigh', 'RealWorl
d-Waist', 'UCI']
[LOD0] feature_mode = hybrid normalize_mode = per_channel
[LOD0] samples: train=28040 val=4900 test=2604
[LOD0] P21 edge_mode=pearson select=topk topk=9 tau=0.3 alpha=0.5
[pearson train] edges/window: mean 9.0 min 9 max 9 | most-kept
pairs: (0, 2):100%, (3, 5):100%, (1, 4):100%, (4, 5):100%, (2, 5):100%
[pearson val] edges/window: mean 9.0 min 9 max 9 | most-kept pa
irs: (1, 2):100%, (3, 5):100%, (4, 5):100%, (0, 3):100%, (0, 1):100%
[pearson test] edges/window: mean 9.0 min 9 max 9 | most-kept p
airs: (3, 5):100%, (4, 5):100%, (3, 4):100%, (0, 3):100%, (0, 1):100%
feature_mode=hybrid, feat_dim=90, normalize_mode=per_channel
building 28040 sensor graphs (6 nodes, 18 edges, 90 features eac
h)...
built 5000/28040 graphs
built 10000/28040 graphs
built 15000/28040 graphs
built 20000/28040 graphs
built 25000/28040 graphs
done: 28040 graphs built

```

[edge_budget s42:WISDM] done

```
=====
=====
P24 - 5-SEED EDGE-BUDGET STUDY (RED/GS vs Pearson; frozen 'full' per
seed)
seeds used: [1, 2, 3, 5, 42]    budget = undirected edges/window (ful
l=15; trained on pearson-9)
=====
=====
```

policy	k	GNN mean±std	fused mean±std
pearson	9	74.84±0.53%	78.07±0.54%
pearson	6	63.77±1.99%	72.05±1.28%
pearson	3	51.62±3.18%	67.42±3.90%
RED	9	59.40±3.30%	69.82±1.05%
RED	6	52.69±4.82%	66.37±3.39%
RED	3	43.41±4.19%	62.71±4.57%
GS-cross	9	35.37±5.88%	48.91±5.99%
GS-cross	6	35.86±5.83%	48.79±6.24%
GS-cross	3	35.83±4.97%	49.66±6.24%
GS-intra	9	74.84±0.53%	78.07±0.54%
GS-intra	6	63.56±3.20%	71.73±3.43%
GS-intra	3	52.84±3.71%	67.42±4.82%

```
=====
Paired Wilcoxon (fused, 6 folds): pearson-9 vs RED-9    mean Δ +8.25pp
| wilcoxon p=0.031

saved /content/drive/MyDrive/Thesis/SO_Experiments/DAGHAR/P20_GraphR
evival_DG/P24_SO_Compression/results/p24_edge_budget_multiseed.json
=====
=====
```

9. Done — what you now have

- results/p24_seed_report_5.json — 5-seed fused/GNN mean±std for all thin + headline arms, per-fold means, GenPerByte+CI, Wilcoxon.
- results/p24_edge_budget_multiseed.json — 5-seed RED/GS/Pearson table + pearson-9-vs-RED-9 Wilcoxon.

These promote the P24 arms to "confirmed" and feed directly into thesis **Chapter 7** and

Thesis_Final_Structure_and_Remaining_Work.md §2 items 1, 2, 4. The remaining §2 items (combined-axis Pareto point §2.3, and the novelty-bar finalization §2.5) are handled elsewhere and don't depend on these runs.